Econometrics II

Estimation Methods and Applications

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1 Part 0 – Introduction to Econometrics

1.1 Role of Economic Theory

Economic theory makes (often competing) statements on the relationships among economic variables. Such statements are typically qualitative in nature: they describe directions of effects (positive or negative) rather than precise numerical values.

Example (Macroeconomics). Keynes (1936) states that people tend to increase their consumption when income rises, but not one–for–one. If we denote income by inc and consumption by c, we can write this informally as

$$c = f(\text{inc}, \ldots),$$

where income is the *explanatory* (independent) variable and consumption is the *dependent* (explained) variable.

Example (Microeconomics). In a standard demand setting, the quantity demanded of a good q depends negatively on its own price p. We write

$$q = f(p, \ldots),$$

with price as explanatory variable and quantity demanded as the dependent variable.

1.2 Role of Econometrics

Econometrics is the application of statistical and mathematical methods to economic data. A classical definition (Maddala, 1992) is:

> "The application of statistical and mathematical methods to the analysis of economic data, with a purpose of giving empirical content to economic theories and verifying them or refuting them."

This emphasizes three ingredients:

- **Economic theories**: provide qualitative predictions and structural relationships.
- **Economic data**: observations from the real world (time series, cross sections, panels).
- **Statistical and mathematical methods**: tools to estimate models and test hypotheses.

1.3 Econometric Modelling

An econometric analysis typically follows a sequence of steps:

1. Start from one or more **economic theories**.
2. Collect appropriate **economic data**.
3. **Specify** an econometric model (variables, functional form, error term).
4. **Estimate** the parameters of the model using sample data.
5. Perform **mis-specification tests** and, if needed, re-specify the model.
6. Carry out **hypothesis testing** (statistical inference).
7. Use the model for **forecasting** (prediction) and **policy analysis** (control).

1.4 Economic Data

Economic data can be classified along several dimensions.

1.4.1 4.1 Time series, cross-section, and panel data

Time series data consist of repeated observations on the same unit (country, firm, household, worker) over time, e.g. quarterly GDP of the EU from 1980.1 to 1999.4. There is a natural time ordering.

Cross-section data are observations on many units at a given point in time, e.g. GDP in 1980.1 for each EU country. There is no natural ordering of the units.

Panel data (or longitudinal data) combine both dimensions: many units observed over multiple time periods, e.g. GDP for each EU country from 1980.1 to 1999.4.

1.4.2 4.2 Continuous vs. discrete variables

A **continuous** variable can in principle take any value in an interval of the real line, such as income, consumption, or money holdings. A **discrete** variable takes only countable values, e.g. number of patents, income class, or unemployment status.

1.4.3 4.3 Levels of measurement

Variables can also be classified by their measurement scale:

- **Nominal**: categories with no natural order (e.g. gender, country). In econometrics, these are typically used as dummy variables (0/1).
- **Ordinal**: categories with a natural order but without meaningful distances between categories (e.g. education level: primary, secondary, tertiary). They can be treated as ordered categorical variables or recoded into dummies.
- **Interval**: numeric scale with equal units but an arbitrary zero (e.g. temperature in °C, calendar years). Differences are meaningful, but ratios are not.
- **Ratio**: numeric scale with equal units and a meaningful zero (e.g. income, age, number of children). Both differences and ratios make sense, and log transformations are meaningful.

Nominal and ordinal are qualitative, while interval and ratio are quantitative. Econometric models often recode nominal/ordinal variables into dummies.

This classification matters because some statistical concepts (like the mean or variance) are meaningful only for certain measurement levels. For example:

- For interval and ratio variables, mean, median, mode, variance, and covariance are all meaningful.
- For ordinal variables, median and mode are meaningful; a simple arithmetic mean may be hard to interpret.
- For nominal variables, only the mode is meaningful; mean or median do not make sense.

1.4.4 4.4 Data accuracy

Accurate econometric estimates require good-quality data. Several problems can arise:

- **Unobservable variables**: some theoretically relevant variables (e.g. expectations) cannot be directly observed.
- **Measurement error**: observable proxies may differ from the ideal theoretical variables (e.g. measured capital stock vs. "true" capital).
- **Inappropriate frequency**: data may be observed at a frequency that does not match the economic question (e.g. annual data when monthly responses to policy are relevant).
- **Selectivity bias**: in survey data, non-response may be systematic, so respondents are not a random sample from the population.

1.5 Specification of Econometric Models

Model specification involves several choices:

- **Selection of explanatory variables** (regressors).
- **Functional form** of the relationship.
- **Assumptions on explanatory variables** (treated as fixed or random).
- **Assumptions on the error term** and its distribution.

1.5.1 5.1 Choice of regressors

In a demand for oranges example, quantity q is expected to depend on the price of oranges p and consumer income x. These are *key* regressors. Quantity may also depend on prices of substitute goods (grapefruits, lemons, apples, ...) and other characteristics (diet habits, advertising, ...). These are *secondary* regressors. The number of possible secondary regressors can be large; some may be unobservable or measured with error.

1.5.2 5.2 Functional form

If we focus on price and income as relevant regressors for orange demand, we may start from

$$q_i = f(p_i, x_i), \quad i = 1, \ldots, n.$$

To make this usable in empirical work, we impose a functional form such as:

- **Linear:**
$$q_i = \beta_1 + \beta_2 p_i + \beta_3 x_i,$$
- **Log-linear:**
$$q_i = \beta_1 p_i^{\beta_2} x_i^{\beta_3},$$
- **Non-linear additive:**
$$q_i = \beta_1 + p_i^{\beta_2} + x_i^{\beta_3}.$$

1.5.3 5.3 Explanatory variables and experiments

In many natural sciences, the researcher controls the regressors in a designed experiment (e.g. fertilizer and water inputs in agriculture). In that case, regressors are often treated as deterministic. In economics, data are rarely generated by controlled experiments; instead, income, wealth, consumption, prices, etc. are observed. Thus both dependent and explanatory variables are usually treated as random variables.

1.5.4 5.4 Error term

With a linear functional form, we might write

$$q_i = \beta_1 + \beta_2 p_i + \beta_3 x_i.$$

This would imply an exact relationship between demand and the two regressors, which is unrealistic. Many influences are omitted, variables are imperfectly measured, functional form might be misspecified, and human behaviour is intrinsically random. To account for this, we introduce an error term:

$$q_i = \beta_1 + \beta_2 p_i + \beta_3 x_i + \varepsilon_i.$$

The error term is not meant to "fix" a wrong model (e.g. omitted key regressors); rather, mis-specification tests often use residuals to diagnose model failures. Different econometric models in the course are distinguished by assumptions on the conditional distribution of the error vector ε given the data matrix X:

$$E[\varepsilon \mid X] = ?, \qquad Var[\varepsilon \mid X] = ?, \qquad \varepsilon \mid X \sim ?.$$

1.6 Estimation of Econometric Models

Consider a generic linear model

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i.$$

The parameters $\beta_1, \beta_2, \beta_3$ are unknown and must be estimated from sample data. Common estimation methods are:

- **Ordinary Least Squares** (OLS) and **Generalized Least Squares** (GLS),
- **Maximum Likelihood** (ML),
- **Generalized Method of Moments** (GMM).

These methods produce sample estimates b_1, b_2, b_3 that we hope are "good" proxies for the true parameters. The development of appropriate estimators was a central topic in econometric theory in the 1950s and 1960s.

1.7 Mis-specification Testing

Before using parameter estimates for theory testing, forecasting, or policy analysis, it is essential to check whether the model specification is adequate. Estimation methods deliver good estimates only if the assumptions used in specifying the model are not rejected by the data. Typical questions:

- Are relevant variables omitted? Are irrelevant variables included?
- Is the chosen functional form appropriate?
- Are assumptions on the conditional distribution of disturbances compatible with the data?

1.8 Hypothesis Testing

Within a correctly specified model, we can test economic theories. In the orange demand example

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i,$$

economic theory suggests that demand is a decreasing function of price ($\beta_2 < 0$) and typically an increasing function of income ($\beta_3 > 0$). These inequalities define testable hypotheses on model parameters.

Traditional hypothesis testing works well when theories are nested within the same model, but is less suitable when we need to choose between non-nested theories. Such questions lead to model selection.

1.9 Forecasting and Policy Analysis

Econometric models are used mainly for two purposes:

- **Forecasting (prediction)**: e.g. predicting inflation, unemployment, or sales.
- **Policy analysis (control)**: e.g. assessing the impact of a minimum wage increase on employment.

1.9.1 9.1 Lucas Critique

Robert Lucas (1976) argued that models based only on past correlations may break down when policy rules change. The reason is that people adjust expectations and behaviour in response to policy, so the historical relationships captured by the model may no longer hold. For reliable policy evaluation, we need structural models based on deep parameters (preferences, technology, constraints), not just reduced-form correlations.

1.9.2 9.2 Alternative approaches to policy evaluation

Modern empirical economics often uses experimental and quasi-experimental methods:

- **Randomized Controlled Trials (RCTs)**: treatment is randomly assigned to units, creating comparable treatment and control groups and eliminating selection bias under ideal conditions. For example, a job-training program can be evaluated by randomizing access.
- **Difference-in-Differences (DiD)**: compares changes over time in treated vs. untreated groups, relying on a "parallel trends" assumption in the absence of treatment. For instance, one can compare employment in regions that increase the minimum wage with those that do not.

These strategies allow causal inference without fully specified structural models, but they still rely on strong identification assumptions and research design.

2 Section 1 – Probability Refresher

This section recalls key probability concepts needed for econometrics: random variables and their distributions, moments, independence, and large-sample tools such as the Law of Large Numbers (LLN) and the Central Limit Theorem (CLT).

2.1 Random variables and distributions

A random variable Y can be **discrete** or **continuous**.

If Y is discrete, it takes values in a countable set (e.g. $\{0, 1, 2, \dots\}$). Its distribution is described by the probability mass function (pmf)

$$p_Y(y) = P(Y = y), \quad \sum_y p_Y(y) = 1.$$

The expectation and variance are

$$E[Y] = \sum_y y\, p_Y(y), \qquad Var(Y) = \sum_y (y - E[Y])^2 p_Y(y).$$

If Y is continuous, it takes values in intervals of $\mathbb{R}$. Its distribution is described by the probability density function (pdf) $f_Y(y) \geq 0$ such that

$$\int_{-\infty}^{\infty} f_Y(y)\, dy = 1.$$

For continuous Y, $P(Y = y) = 0$; probabilities are defined over intervals. Expectation and variance are

$$E[Y] = \int y f_Y(y)\, dy, \qquad Var(Y) = \int (y - E[Y])^2 f_Y(y)\, dy.$$

In both cases, the cumulative distribution function (CDF) is

$$F_Y(y) = P(Y \leq y),$$

which is nondecreasing and right-continuous.

2.2 Moments of a distribution

The **mean** is $E[Y]$.
The **variance** is

$$Var(Y) = E[(Y - E[Y])^2] = E[Y^2] - (E[Y])^2.$$

The **standard deviation** is $\sqrt{Var(Y)}$, and has the same units as Y.

Higher moments describe shape:

- **Skewness** (asymmetry) is $E[((Y - \mu)/\sigma)^3]$, where $\mu = E[Y]$ and $\sigma = \sqrt{Var(Y)}$.
- **Kurtosis** (tails and peakedness) is $E[((Y - \mu)/\sigma)^4]$.

2.3 Two random variables: covariance and correlation

Let (X, Z) be a pair of random variables. The **covariance** is

$$Cov(X, Z) = E[(X - EX)(Z - EZ)].$$

The **correlation** is

$$Corr(X, Z) = \frac{Cov(X, Z)}{\sqrt{Var(X)\, Var(Z)}} \in [-1, 1].$$

If X and Z are independent, then (provided second moments exist) $Cov(X, Z) = 0$. The converse is not true in general, except under special distributions such as multivariate normal, where zero covariance implies independence.

Linearity properties:

- $E[aX + bZ + c] = aEX + bEZ + c$.
- $Var(aX + bZ) = a^2 Var(X) + b^2 Var(Z) + 2ab\, Cov(X, Z)$.

2.4 Conditional distributions and moments

The **conditional distribution** of Y given $X = x$ describes the behaviour of Y when X takes a specific value. From it we obtain:

- the conditional mean $E[Y \mid X = x]$,
- the conditional variance $Var(Y \mid X = x)$.

Conditional means are central both for prediction and for defining regression functions.

2.5 Independence vs. uncorrelatedness

Two variables X and Z are **independent** if their joint distribution factorizes as

$$F_{X,Z}(x, z) = F_X(x) F_Z(z),$$

or, for densities/pmfs, $f_{X,Z}(x, z) = f_X(x) f_Z(z)$. They are **uncorrelated** if $Cov(X, Z) = 0$.

Independence implies uncorrelatedness (if second moments exist), but the reverse is not true in general. In multivariate normal families, however, zero covariance does imply independence.

2.6 Common probability distributions

Discrete:

- **Bernoulli**(p): $Y \in \{0, 1\}$, $P(Y = 1) = p$, $P(Y = 0) = 1 - p$. Mean $E[Y] = p$, variance $Var(Y) = p(1 - p)$.
- **Binomial**(n, p): $Y \in \{0, \dots, n\}$, $P(Y = y) = \binom{n}{y} p^y (1 - p)^{n-y}$. Mean $E[Y] = np$, variance $Var(Y) = np(1 - p)$.
- **Poisson**(λ): $Y \in \{0, 1, 2, \dots\}$, $P(Y = y) = e^{-\lambda} \lambda^y / y!$. Mean and variance are both λ.

Continuous:

- **Normal** $N(\mu, \sigma^2)$: density

$$f(y) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(y - \mu)^2}{2\sigma^2} \right).$$

Other common continuous distributions can be defined using standard normal variables $Z, Z_1, \dots, Z_k \sim$ *i.i.d.* $N(0, 1)$:

- **Chi-square** with k degrees of freedom: $\chi^2_k = \sum_{i=1}^{k} Z_i^2$.
- **Student-t** with ν degrees of freedom: if $V \sim \chi^2_\nu$ independently of Z, then

$$T = \frac{Z}{\sqrt{V/\nu}} \sim t_\nu.$$

- **F** distribution with (d_1, d_2) degrees of freedom: if $U_1 \sim \chi^2_{d_1}$ and $U_2 \sim \chi^2_{d_2}$ independent, then

$$F = \frac{U_1/d_1}{U_2/d_2} \sim F_{d_1, d_2}.$$

2.7 Large-sample behaviour of test distributions

As degrees of freedom grow:

- $t_\nu \Rightarrow N(0, 1)$ as $\nu \to \infty$; critical values approximate standard normal.
- χ^2_k has mean k and variance $2k$; standardized $(\chi^2_k - k)/\sqrt{2k} \Rightarrow N(0, 1)$ as $k \to \infty$.
- F_{d_1, d_2} becomes concentrated around 1 as d_1, d_2 grow; large-df critical values are close to 1.

If only the denominator degrees of freedom $d_2 \to \infty$ with d_1 fixed, then

$$F_{d_1, d_2} \Rightarrow \frac{\chi^2_{d_1}}{d_1},$$

and upper-tail critical values satisfy approximately

$$F_{d_1, d_2; 1-\alpha} \approx \frac{\chi^2_{d_1; 1-\alpha}}{d_1}$$

for large d_2. Intuitively, U_2/d_2 converges in probability to 1 by LLN, so F behaves like the scaled chi-square in the numerator.

2.8 Random sampling and the sample mean

Under simple random sampling from a population, the observations $\{Y_i\}_{i=1}^n$ are independent and identically distributed (i.i.d.) with the population distribution. Before data are observed, they are random; once observed, we treat the realized sample $(y_1, \ldots, y_n)$ as fixed.

The sample mean is

$$\bar{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i.$$

If $E[Y_i] = \mu$ and $Var(Y_i) = \sigma^2 \in (0, \infty)$, then

$$E[\bar{Y}_n] = \mu, \qquad Var(\bar{Y}_n) = \frac{\sigma^2}{n}, \qquad SD(\bar{Y}_n) = \frac{\sigma}{\sqrt{n}}.$$

If the Y_i are normal, then $\bar{Y}_n \sim N(\mu, \sigma^2/n)$ exactly; otherwise, CLT gives approximate normality for large n.

2.9 Laws of Large Numbers (LLN)

Let $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^n Y_i$ and suppose $E[Y_i] = \mu$.

The **Weak Law of Large Numbers (WLLN)** states that if Y_i are i.i.d. with $E|Y_i| < \infty$, then

$$\bar{Y}_n \xrightarrow{p} \mu,$$

i.e., for large n, the sample mean is close to μ with high probability.

The **Strong Law of Large Numbers (SLLN)** states that, under similar conditions, $\bar{Y}_n \to \mu$ almost surely, meaning that for almost every infinite sequence (sample path), the sample mean converges to μ.

Finite variance ($Var(Y) < \infty$) implies WLLN (e.g. via Chebyshev's inequality) but is stronger than needed; LLN can still hold with infinite variance (e.g. some Pareto distributions with tail index between 1 and 2).

2.10 Probability limits and consistency

The notation $\operatorname{plim}_{n\to\infty} \hat{\theta}_n = \theta_0$ means that $\hat{\theta}_n$ converges in probability to θ_0:

$$\lim_{n\to\infty} P(|\hat{\theta}_n - \theta_0| > \varepsilon) = 0 \quad \forall\, \varepsilon > 0.$$

An estimator is **(weakly) consistent** if its probability limit equals the true parameter.

Some useful algebra of probability limits: if $\operatorname{plim} a_n = a$ and $\operatorname{plim} b_n = b$, then

- $\operatorname{plim}(a_n + b_n) = a + b$, $\operatorname{plim}(a_n b_n) = ab$,
- if $b \neq 0$, $\operatorname{plim}(a_n / b_n) = a/b$,
- if g is continuous at θ_0 and $\operatorname{plim} \hat{\theta}_n = \theta_0$, then $\operatorname{plim} g(\hat{\theta}_n) = g(\theta_0)$.

By WLLN, for example, $\operatorname{plim} \bar{Y}_n = \mu$.

2.11 Central Limit Theorem (CLT)

If Y_i are i.i.d. with $E[Y_i] = \mu$ and $Var(Y_i) = \sigma^2 \in (0, \infty)$, then

$$\sqrt{n}(\bar{Y}_n - \mu) \Rightarrow N(0, \sigma^2),$$

or, in standardized form,

$$\frac{\sqrt{n}(\bar{Y}_n - \mu)}{\sigma} \Rightarrow N(0, 1).$$

The CLT justifies normal approximations for sample averages and many estimators, and underlies standard errors, t-statistics, and Wald tests. Versions of CLT also hold under weaker dependence and heteroskedasticity assumptions (Lindeberg–Feller CLT) with suitable regularity.

3 Section 2 – Statistical Inference

Statistical inference links probability theory to empirical analysis. The goal is to learn about unknown population quantities (parameters, functions) using sample data, by constructing estimators, hypothesis tests, and confidence intervals. Large-sample tools such as LLN and CLT play a key role.

3.1 From probability to inference

Suppose the population is described by a parameter θ (or vector of parameters). We observe a sample $Y_1, \ldots, Y_n$ generated according to a model. Our aim is to:

- construct **estimators** of θ,
- test **hypotheses** about θ,
- quantify **uncertainty** via standard errors and confidence intervals.

The sampling distribution implied by the model (possibly approximated using asymptotics: LLN and CLT) is the basis for inference.

3.2 Point estimators

A **point estimator** of θ is a function of the sample:

$$\hat{\theta}_n = g(Y_1, \ldots, Y_n).$$

The estimator is random because it depends on the random sample.

Typical examples:

- **Sample mean**: $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^n Y_i$.
- **Sample variance**: $s^2 = \frac{1}{n-1}\sum_{i=1}^n (Y_i - \bar{Y}_n)^2$.

Once data $y_1, \ldots, y_n$ are observed, the estimator produces a numerical **estimate** $\hat{\theta}_n(y_1, \ldots, y_n)$.

3.3 Properties of point estimators

Let $\hat{\theta}_n$ estimate a scalar parameter θ_0. Important properties:

- **Unbiasedness**: $E[\hat{\theta}_n] = \theta_0$.
- **Consistency**: $\hat{\theta}_n \xrightarrow{p} \theta_0$ as $n \to \infty$.
- **Asymptotic distribution**: often $\sqrt{n}(\hat{\theta}_n - \theta_0) \Rightarrow N(0, \sigma^2)$, a normal limit with some asymptotic variance σ^2.
- **Efficiency**: among a class of estimators, an efficient estimator has the smallest variance (in finite samples) or the smallest asymptotic variance.

3.4 P-values, hypothesis tests, and confidence intervals

Suppose we test a null hypothesis $H_0 \colon \theta = \theta_0$ against an alternative H_1. We construct a **test statistic** T from the sample, often of the form

$$T = \frac{\hat{\theta} - \theta_0}{\widehat{se}(\hat{\theta})},$$

where $\widehat{se}(\hat{\theta})$ is an estimate of the standard error of $\hat{\theta}$.

The **P-value** is the probability, under H_0, of obtaining a value of the test statistic at least as extreme as the observed one T_{obs}. For a two-sided test,

$$p_{\text{two-sided}} = P_{H_0}(|T| \geq |T_{\text{obs}}|).$$

If T is approximately standard normal, we can approximate $p \approx 2\Phi(-|T_{\text{obs}}|)$. If T follows a Student-t distribution with given degrees of freedom, or another distribution, we use the corresponding CDF.

A test at significance level α can be conducted in two equivalent ways:

- **P-value rule**: reject H_0 if $p < \alpha$.
- **Critical value rule**: for a two-sided test with symmetric distribution, reject if $|T_{\text{obs}}| \geq c_{1-\alpha/2}$, where $c_{1-\alpha/2}$ is the appropriate quantile (e.g. $z_{1-\alpha/2}$ for normal, $t_{df,1-\alpha/2}$ for t-distributions).

For one-sided alternatives (e.g. $H_1 \colon \theta > \theta_0$), we consider upper-tail probabilities $P_{H_0}(T \geq T_{\text{obs}})$ and critical values $c_{1-\alpha}$.

3.5 Type I and Type II errors, power

A **Type I error** occurs when we reject a true null hypothesis H_0. Its probability is the significance level α.

A **Type II error** occurs when we fail to reject H_0 even though it is false. The probability of Type II error is often denoted β. The **power** of a test is $1 - \beta$, the probability of correctly rejecting a false null.

Power increases when:

- the true effect size (distance between true parameter and null value) is larger,
- the estimator is more precise (lower variance / smaller standard error),
- the sample size n is larger.

3.6 Confidence intervals

A $(1-\alpha)$ **confidence interval** (CI) for θ_0 has the generic form

$$\theta_0 \in \left[\hat{\theta} \pm c_{1-\alpha/2}\, \widehat{se}(\hat{\theta})\right],$$

where $c_{1-\alpha/2}$ is the appropriate critical value (from a normal or t distribution, etc.). The interpretation is that if we repeated the sampling procedure many times and constructed the CI each time, approximately a fraction $(1-\alpha)$ of the intervals would contain the true parameter.

3.7 Testing workflow

To test $H_0 : \theta = \theta_0$ vs. H_1 (two-sided by default):

1. Choose a statistic T_n whose distribution under H_0 is known or approximable.
2. Compute the observed value T_{obs}.
3. Compute the P-value $p = P_{H_0}(|T_n| \geq |T_{\text{obs}}|)$.
4. Reject H_0 if $p < \alpha$, or equivalently if $|T_{\text{obs}}|$ exceeds the critical value.

3.8 Example: estimating the population mean

Suppose $Y_1, \ldots, Y_n$ are i.i.d. with $E[Y_i] = \mu$ and $Var(Y_i) = \sigma^2$. Consider estimating μ under quadratic loss (least squares). For any constant m, the sum of squared deviations is

$$\sum_{i=1}^{n} (Y_i - m)^2.$$

Differentiating with respect to m and setting the derivative to zero gives

$$\frac{d}{dm} \sum_{i=1}^{n} (Y_i - m)^2 = -2 \sum_{i=1}^{n} (Y_i - m) = 0$$

so $\sum_{i=1}^{n} Y_i = nm$, hence

$$m = \frac{1}{n} \sum_{i=1}^{n} Y_i = \bar{Y}_n.$$

Thus the sample mean is the least squares estimator of μ. It can be shown that $\bar{Y}_n$ is unbiased and, among linear unbiased estimators, has the smallest variance.

3.9 Sampling distribution of the sample mean

As seen in the probability refresher, if $Y_1, \ldots, Y_n$ are i.i.d. with mean μ and variance $\sigma^2 \in (0, \infty)$, the sample mean

$$\bar{Y}_n = \frac{1}{n} \sum_{i=1}^{n} Y_i$$

satisfies

$$E[\bar{Y}_n] = \mu, \qquad Var(\bar{Y}_n) = \frac{\sigma^2}{n}, \qquad SD(\bar{Y}_n) = \frac{\sigma}{\sqrt{n}}.$$

By the CLT,

$$\sqrt{n}(\bar{Y}_n - \mu) \Rightarrow N(0, \sigma^2),$$

so $\bar{Y}_n$ is approximately normal for large n, even if the underlying Y_i are not normal.

3.10 Estimating the variance for the mean

In practice, σ^2 is unknown and must be estimated. The **sample variance** is

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y}_n)^2.$$

This estimator is unbiased for σ^2, i.e. $E[s^2] = \sigma^2$, and is consistent: $s^2 \xrightarrow{p} \sigma^2$.
An estimator for the variance and standard error of the mean is then

$$\widehat{Var}(\bar{Y}_n) = \frac{s^2}{n}, \qquad \widehat{se}(\bar{Y}_n) = \frac{s}{\sqrt{n}}.$$

3.11 Testing hypotheses about the mean

Consider testing

$$H_0 \colon \mu = \mu_0 \quad \text{vs.} \quad H_1 \colon \mu \neq \mu_0$$

based on the i.i.d. sample $Y_1, \ldots, Y_n$. Define

$$\bar{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y}_n)^2.$$

A natural test statistic is

$$T = \frac{\bar{Y}_n - \mu_0}{s/\sqrt{n}}.$$

If we additionally assume $Y_i \sim N(\mu, \sigma^2)$, then:

- $\frac{\bar{Y}_n - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$,
- $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$,
- $\bar{Y}_n$ and s^2 are independent.

Under $H_0 \colon \mu = \mu_0$, this implies

$$T = \frac{\bar{Y}_n - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

exactly. For a two-sided test at level α, we reject H_0 if $|T| \geq t_{n-1,1-\alpha/2}$, and the P-value is $p = 2P(t_{n-1} \geq |T_{\text{obs}}|)$.
Even without normality, a large-sample approximation is available: by CLT,

$$\sqrt{n}(\bar{Y}_n - \mu) \Rightarrow N(0, \sigma^2),$$

and since $s \xrightarrow{p} \sigma$ (Slutsky's theorem),

$$T = \frac{\bar{Y}_n - \mu_0}{s/\sqrt{n}} d N(0,1).$$

For large n, we thus approximate the distribution of T by the standard normal and reject if $|T| \geq z_{1-\alpha/2}$, with P-value $p \approx 2\Phi(-|T_{\text{obs}}|)$.

3.12 CLT example: Bernoulli mean (sample proportion)

If $Y_i \sim Bernoulli(p)$ i.i.d., then $\mu = p$ and $\sigma^2 = p(1-p)$. The sample mean

$$\hat{p} = \bar{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i$$

is the sample proportion of successes. By CLT,

$$\sqrt{n}(\hat{p} - p) \Rightarrow N\big(0,\ p(1-p)\big).$$

For small sample sizes (e.g. $n = 1, 2, 3$), the exact sampling distribution of $\hat{p}$ is discrete and can be worked out from the Bernoulli probabilities. As n grows (e.g. to 10 or 100), the distribution of $\hat{p}$ becomes more concentrated around p and more closely approximated by a normal distribution.

4 Part I - The Classical Linear Regression Model (CLRM)

4.1 Set 1: The Classical Linear Regression Model (CLRM) and the OLS Estimator

4.1.1 What is a Regression Model?

Regression provides a framework to model the relationship between a dependent variable y and one or more explanatory variables x, using observed data.

In the simplest case with one regressor and an intercept, we write:

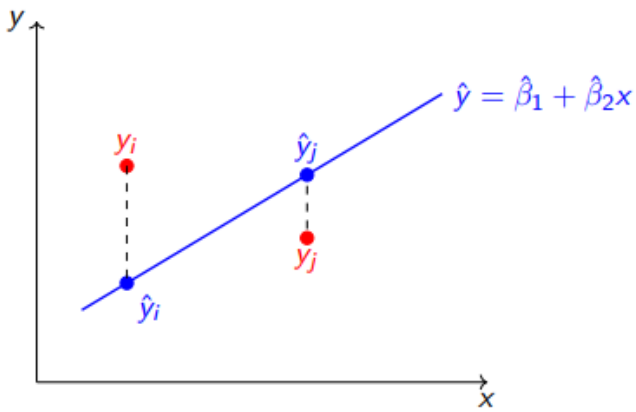
$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i, \qquad i = 1, \ldots, n.$$

Here:

- β_1 is the intercept,
- β_2 is the slope measuring the effect of x on y,
- ε_i is the error term capturing the part of y not explained by the included regressors. It reflects omitted factors, random noise, and possible measurement errors.

A key idea: **the regression line provides the best linear approximation to the data**, typically obtained by minimizing the sum of squared residuals.

4.1.2 Graphical and Geometric Intuition



- Point on the left: observed y_i above the line $\Rightarrow$ positive residual.
- Point on the right: observed y_j below the line $\Rightarrow$ negative residual.

For each observation x_i, the fitted value is:

$$\hat{y}_i = \hat{\beta}_1 + \hat{\beta}_2 x_i.$$

The residual is:

$$e_i = y_i - \hat{y}_i.$$

The OLS estimator chooses $(\hat{\beta}_1, \hat{\beta}_2)$ to minimize:

$$\sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

Geometrically, residuals represent vertical distances from each data point to the regression line. OLS selects the line that is "as close as possible" to all the points, in the sense of least squares.

4.1.3 Regression as a Statistical Device

Regression can be used descriptively:

- It summarizes how y is associated with x in the sample.
- The coefficient $\hat{\beta}_2$ gives the average change in y when x increases by one unit.

- It allows prediction of y given x, without requiring causal interpretation.

Important: Regression always captures an *association*. By itself, it does not establish causality.

4.1.4 From Description to Causality

To interpret β_2 as a causal effect of x on y, the following assumptions (for a single regressor) are needed:

1. **Linearity:** $y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$. The effect of x on y is linear on average.
2. **Exogeneity:** No omitted variables correlated with x:
$$[E[\varepsilon_i \mid x_i] = 0.$$
3. **Random sampling:** Observations are independently drawn from the population.
4. **Variation in x:** we need differences in x to identify its effect.
5. **Homoskedasticity (optional):** $\text{Var}(\varepsilon_i \mid x_i) = \sigma^2$. The noise in y has constant variance across x.

With credible exogeneity and no omitted confounders, regression can be interpreted causally.

4.1.5 Matrix Notation and the CLRM

Let the dependent variable be collected in:
$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}.$$

For each regressor x_k, define:
$$\mathbf{x}_k = \begin{bmatrix} x_{1k} \\ x_{2k} \\ \vdots \\ x_{nk} \end{bmatrix}, \qquad k = 1, \ldots, K.$$

Stack the regressors into the $n \times K$ matrix:
$$\begin{bmatrix} x_{11} & x_{12} & \ldots & x_{1K} \\ x_{21} & x_{22} & \ldots & x_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \ldots & x_{nK} \end{bmatrix} = \begin{bmatrix} \cdots & x_{1k} & \cdots \\ \cdots & x_{2k} & \cdots \\ \vdots & \vdots & \ddots \\ \cdots & x_{nk} & \cdots \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1' \\ \mathbf{x}_2' \\ \vdots \\ \mathbf{x}_n' \end{bmatrix} = \mathbf{X}$$

The matrix $\mathbf{X}$ is an $n \times K$ data matrix containing the n sample observations on the K independent variables. Usually, the vector $\mathbf{x}_1$ is assumed to be a column of ones, representing the constant term.

Notice the difference in notation between x_{ik}, which is a scalar and is written in plain text, and vectors and matrices such as $\mathbf{x}_1$ or $\mathbf{X}$, which are written in bold.

Let the error vector be:
$$\varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

The regression model becomes:
$$y = X\beta + \varepsilon.$$

4.1.6 CLRM Assumption 1: Linearity

Observed data are generated by the following linear model (with respect to the parameters):
$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_K x_{iK} + \varepsilon_i = x_i'\beta + \varepsilon_i, \qquad i = 1, 2, \ldots, n.$$

The K unknown parameters of the model can be collected into a column vector:
$$\beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_K \end{bmatrix}.$$

The model can then be rewritten in compact form:

$$y = x_1\beta_1 + x_2\beta_2 + \cdots + x_K\beta_K + \varepsilon,$$

or, more concisely,

$$y = X\beta + \varepsilon.$$

4.1.7 CLRM Assumption 2: Strict Exogeneity

The expected value of each error term ε_i, conditional on all observations, is equal to zero:

$$E[\varepsilon_i \mid X] = 0, \qquad i = 1, 2, \ldots, n.$$

In compact form:

$$E[\varepsilon \mid X] = 0.$$

Loosely speaking, this means that knowing X does not give us information about the distribution of the error terms.

Let X and Z be continuous random variables. Three concepts are relevant:

Statistical Independence:

$$f_{X,Z}(x, z) = f_X(x) f_Z(z),$$

that is, the joint density factorizes into the marginal densities.

Mean Independence:

$$E[Z \mid X] = E[Z].$$

Uncorrelation:

$$E[XZ] = E[X]\,E[Z].$$

$$\text{Independence} \;\Rightarrow\; \text{Mean Independence} \;\Rightarrow\; \text{Uncorrelation}.$$

- Independence is the strongest condition: it implies full factorization of the joint density $f_{X,Z}$.
- Mean independence means that X does not influence the mean of Z (although it may affect its variance).
- Uncorrelation is the weakest condition: it means zero covariance, i.e. no linear association (a nonlinear relationship may still exist).
- The implications do not go in the opposite direction.

Mean independence implies:

$$E[X \mid Z] = E[X].$$

Then,

$$E[ZX] = E[\,E[ZX \mid Z]\,] = E[\,Z\,E[X \mid Z]\,] = E[\,Z\,E[X]\,] = E[Z]\,E[X],$$

where the Law of Iterated Expectations has been applied.

Since the covariance is defined as

$$\mathrm{Cov}(Z, X) = E[ZX] - E[Z]E[X],$$

we obtain:

$$\mathrm{Cov}(Z, X) = 0.$$

The unconditional mean of each error term is also zero. Using the Law of Total Expectations:

$$E[\varepsilon_i] = E_X[\,E[\varepsilon_i \mid X]\,] = 0, \qquad i = 1, 2, \ldots, n.$$

Mean independence implies zero covariance (and therefore zero correlation) between regressors and error terms:

$$\mathrm{Cov}(x_{jk},\, \varepsilon_i) = E[x_{jk}\varepsilon_i] - E[x_{jk}]\,E[\varepsilon_i] = 0, \qquad i, j = 1, \ldots, n,\; k = 1, \ldots, K.$$

Strict exogeneity implies that regressors are uncorrelated not only with the error from the same observation, but also with error terms from other observations.

4.1.8 CLRM Assumption 3: No Perfect Multicollinearity

The rank of the $n \times K$ matrix $\mathbf{X}$ is equal to K with probability one. This implies that $\mathbf{X}$ has full column rank: the columns of $\mathbf{X}$ are linearly independent, and therefore there must be at least K observations ($n \geq K$).

If $\mathbf{X}$ does not have full column rank, then the matrix $X'X$ is singular and not invertible.

Example 1: Constant and "Year of Attendance" (always 4)

$$X = \begin{bmatrix} 1 & 4 \\ 1 & 4 \\ 1 & 4 \end{bmatrix}, \qquad X'X = \begin{bmatrix} 3 & 12 \\ 12 & 48 \end{bmatrix}, \qquad \det(X'X) = 0.$$

Since "Year of attendance" is constant across all three observations, it is perfectly collinear with the intercept, generating perfect multicollinearity.

Example 2: Lecture Attendance (yes/no, 1 or 0) and Econometrics Mark

$$X = \begin{bmatrix} 1 & 30 \\ 1 & 30 \\ 0 & 0 \end{bmatrix}, \qquad X'X = \begin{bmatrix} 2 & 60 \\ 60 & 1800 \end{bmatrix}, \qquad \det(X'X) = 2 \cdot 1800 - 60^2 = 0.$$

Here, Mark $=$ Attendance $\times$ 30, meaning the second variable is an exact linear transformation of the first (when nonzero). This again creates perfect multicollinearity.

Takeaway. If one regressor is a linear combination of others, the matrix $X'X$ becomes singular and therefore non-invertible. This prevents the computation of the OLS estimator $(X'X)^{-1}X'y$. (Recall how to compute the inverse of a 2×2 matrix.)

4.1.9 CLRM Assumption 4: Spherical Errors

A variance–covariance matrix is called *spherical*, or *isotropic*, if it is proportional to the identity matrix:

$$V = \lambda I,$$

that is, it is diagonal and all diagonal elements are equal.

4.1.10 4a. Homoskedasticity Assumption

The conditional second moment of each error term ε_i is constant:

$$E[\varepsilon_i^2 \mid X] = \sigma^2, \qquad i = 1, \ldots, n.$$

4.1.11 4b. No-Correlation Assumption

The conditional second cross-moment between ε_i and ε_j is zero for all $i \neq j$:

$$E[\varepsilon_i \varepsilon_j \mid X] = 0, \qquad i \neq j.$$

4.1.12 Equivalent Formulation

Since:

$$\mathrm{Var}(\varepsilon_i \mid X) = E[\varepsilon_i^2 \mid X] - E[\varepsilon_i \mid X]^2 = E[\varepsilon_i^2 \mid X] = \sigma^2,$$

and

$$\mathrm{Cov}(\varepsilon_i, \varepsilon_j \mid X) = E[\varepsilon_i \varepsilon_j \mid X] - E[\varepsilon_i \mid X]\, E[\varepsilon_j \mid X] = E[\varepsilon_i \varepsilon_j \mid X] = 0,$$

these two assumptions can be expressed compactly as:

$$E[\varepsilon \varepsilon' \mid X] = \mathrm{Var}(\varepsilon \mid X) = \sigma^2 I,$$

where I is the $n \times n$ identity matrix.

4.1.13 Least Squares Estimation

The parameters of the linear regression model, β, and the common variance of the error terms, σ^2, are unknown quantities. Using the available sample data, estimation methods provide estimates of these unknown parameters.

Although we do not observe the vector of error terms,

$$\varepsilon = y - X\beta,$$

we can compute the vector of *residuals* implied by any hypothetical value $\tilde{\beta}$ as:

$$\tilde{\varepsilon} = y - X\tilde{\beta}.$$

The least squares criterion chooses the value of $\tilde{\beta}$ that minimizes the sum of squared residuals.

4.1.14 Residual Sum of Squares

Given any coefficient vector $\tilde{\beta}$, the objective function is:

$$S(\tilde{\beta}) = (y - X\tilde{\beta})'(y - X\tilde{\beta}).$$

Note that:

$$(y - X\tilde{\beta})'(y - X\tilde{\beta}) = \sum_i (y_i - x_i'\tilde{\beta})^2 = \sum_i \tilde{\varepsilon}_i^2.$$

4.1.15 First Order Conditions

The gradient of the objective function with respect to $\tilde{\beta}$ is:

$$\frac{\partial S(\tilde{\beta})}{\partial \tilde{\beta}} = -2X'y + 2X'X\tilde{\beta} = 0.$$

Let b denote the minimizing value. Then b satisfies the *normal equations*:

$$X'Xb = X'y.$$

If the matrix $X'X$ is invertible (which is guaranteed when X has full column rank), the solution is:

$$b = (X'X)^{-1}X'y.$$

The least squares residuals are:

$$e = y - Xb.$$

To verify that this solution minimizes the sum of squares, note that the Hessian is:

$$\frac{\partial^2 S(b)}{\partial b\, \partial b'} = 2X'X,$$

which must be positive definite (and is so if X has full column rank).

4.1.16 Properties of the Normal Equations

From the normal equations:

$$X'Xb = X'y$$

we obtain:

$$X'(y - Xb) = X'e = 0.$$

Therefore, for every column x_k of X:

$$x_k'e = 0,$$

and if the first column of X is a column of ones:

$$x_1'e = \mathbf{i}'e = 0.$$

This shows that *the OLS residuals sum to zero*, provided that the model includes a constant.

4.1.17 Example: The Case $K = 2$

When $K = 2$, the regression model becomes:

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i, \qquad i = 1, \ldots, n.$$

We minimize the residual sum of squares:

$$S(\tilde{\beta}_1, \tilde{\beta}_2) = \sum_{i=1}^{N} (y_i - \tilde{\beta}_1 - \tilde{\beta}_2 x_i)^2.$$

The first order conditions are:

$$\frac{\partial S(\tilde{\beta}_1, \tilde{\beta}_2)}{\partial \tilde{\beta}_1} = -2 \sum_{i=1}^{N} (y_i - \tilde{\beta}_1 - \tilde{\beta}_2 x_i) = 0,$$

$$\frac{\partial S(\tilde{\beta}_1, \tilde{\beta}_2)}{\partial \tilde{\beta}_2} = -2 \sum_{i=1}^{N} x_i (y_i - \tilde{\beta}_1 - \tilde{\beta}_2 x_i) = 0.$$

4.1.18 Explicit Solutions

Solving these two equations yields:

$$b_1 = \frac{1}{N} \sum_{i=1}^{N} y_i - b_2 \frac{1}{N} \sum_{i=1}^{N} x_i = \bar{y} - b_2 \bar{x},$$

and

$$b_2 = \frac{\sum_{i=1}^{N} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N} (x_i - \bar{x})^2}.$$

Thus, b_2 is the ratio of the sample covariance of x and y to the sample variance of x. The intercept b_1 is chosen to ensure that the residuals average to zero.

4.1.19 An Alternative Expression for the OLS Estimator

The OLS estimator:

$$b = (X'X)^{-1} X'y$$

can be rewritten as:

$$b = \left(X'X \frac{1}{n} \right)^{-1} X'y \frac{1}{n} = S_{XX}^{-1} S_{XY},$$

where

$$S_{XX} = \left(X'X \frac{1}{n} \right) = \frac{1}{n} \sum_i x_i x_i',$$

and

$$S_{XY} = X'y \frac{1}{n} = \frac{1}{n} \sum_i x_i y_i.$$

Intuition: S_{XX} and S_{XY} can be interpreted as sample averages of $x_i x_i'$ and $x_i y_i$ respectively. This form is fundamental in large-sample theory.

4.1.20 More Concepts and Algebra

Vector of fitted values

$$\hat{y} = Xb$$

Vector of residuals

$$e = y - Xb = y - \hat{y}$$

4.1.21 Two Useful Matrices

Projection matrix

$$P = X(X'X)^{-1}X'$$

Annihilator matrix

$$M = I - P$$

Both P and M are $n \times n$, symmetric, and idempotent. That is:

$$PX = X, \qquad MX = 0.$$

Definition. A matrix A is *idempotent* if $A^2 = A$. If A is also symmetric, then $A'A = A$.

4.1.22 Estimate of the Variance of the Error Term

The OLS estimate of σ^2, denoted s^2, is the sum of squared residuals divided by $n - K$:

$$s^2 = \frac{e'e}{n-K}.$$

Standard Error of the Regression (SER) The square root of s^2, denoted s, is called the standard error of the regression.

4.1.23 Goodness of Fit and Analysis of Variance

The least squares estimator minimizes the residual sum of squares. However, minimization alone does not indicate whether the residual variation is large or small.

To address this, we compare the RSS to a benchmark: the total variation in the dependent variable around its sample mean, the *total sum of squares* (TSS).

4.1.24 Derivation of the Coefficient of Determination

The OLS decomposition is:

$$y = \hat{y} + e$$

Subtracting $\bar{y}$ from both sides:

$$y - \bar{y} = \hat{y} - \bar{y} + e.$$

Using the mean-centering matrix $M^0 = I - \frac{1}{n}ii'$, we have:

$$M^0 y = M^0 \hat{y} + e = M^0 Xb + e.$$

The total sum of squares is:

$$(M^0 y)'(M^0 y) = (M^0 Xb + e)'(M^0 Xb + e).$$

Provided the model contains an intercept, this becomes:

$$y'M^0 y = b'X'M^0 Xb + e'e.$$

4.1.25 The TSS Decomposition

The total sum of squares can be written as the sum of:

$$TSS = ESS + RSS$$

The coefficient of determination is:

$$R^2 = \frac{ESS}{TSS} = \frac{b'X'M^0Xb}{y'M^0y} = 1 - \frac{e'e}{y'M^0y}$$

4.1.26 Problems with the Use of the Coefficient of Determination

The R^2 statistic never decreases when additional explanatory variables are added to the regression. For this reason, alternative measures are used, such as the adjusted R^2 (adjusted for degrees of freedom):

$$\bar{R}^2 = 1 - \frac{e'e/(n-K)}{y'M^0y/(n-1)}$$

$$\hat{y} = y - e \quad \bar{\hat{y}} = \frac{1}{n}\sum \hat{y} = \frac{1}{n}\sum y_i - \frac{1}{n}\sum e_i$$

This adjusted measure can decrease if the newly added regressor contributes little explanatory power.

If the model does not include a constant, the coefficient of determination is not bounded between zero and one and can even become negative.

4.2 Set 2: Statistical Properties of the OLS Estimator

4.2.1 Sampling Error

The quantity $b - \beta$ is called *sampling error*.

$$\begin{aligned} b - \beta &= (X'X)^{-1}X'y - \beta \\ &= (X'X)^{-1}X'(X\beta + \varepsilon) - \beta \\ &= (X'X)^{-1}(X'X)\beta + (X'X)^{-1}X'\varepsilon - \beta \\ &= \beta + (X'X)^{-1}X'\varepsilon - \beta \\ &= (X'X)^{-1}X'\varepsilon \end{aligned}$$

4.2.2 Finite Sample Properties of the OLS Estimator

Unbiasedness (proof)

$$E[b \mid X] = \beta \quad \Longleftrightarrow \quad E[b - \beta \mid X] = 0$$

Proof.

$$\begin{aligned} E[b - \beta \mid X] &= E[(X'X)^{-1}X'\varepsilon \mid X] \\ &= (X'X)^{-1}X'E[\varepsilon \mid X] = 0 \end{aligned}$$

Thus:

$$E[b] = E_X[E[b \mid X]] = \beta$$

4.2.3 Unbiasedness (proof)

4.2.4 Expression for the variance (proof)

$$Var[b \mid X] = \sigma^2(X'X)^{-1}$$

Proof.

$$\begin{aligned} Var[b \mid X] &= Var[b - \beta \mid X] \\ &= Var[(X'X)^{-1}X'\varepsilon \mid X] \\ &= (X'X)^{-1}X'Var[\varepsilon \mid X]X(X'X)^{-1} \\ &= (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} \\ &= \sigma^2(X'X)^{-1} \end{aligned}$$

4.2.5 Uncorrelatedness (proof)

$$Cov[b, e \mid X] = 0$$

Proof.

$$\begin{aligned} Cov[b, e \mid X] &= E[(b - E[b \mid X])(e - E[e \mid X])' \mid X] \\ &= E[(X'X)^{-1}X'\varepsilon(M\varepsilon)' \mid X] \\ &= E[(X'X)^{-1}X'\varepsilon\varepsilon'M \mid X] \\ &= (X'X)^{-1}X'E[\varepsilon\varepsilon' \mid X]M \\ &= \sigma^2(X'X)^{-1}X'M = 0 \end{aligned}$$

since $X'M = (MX)' = 0$.

4.2.6 Estimator of the Variance

It can be shown that

$$s^2 = \frac{e'e}{n - K}$$

is an unbiased estimator of σ^2.

4.2.7 Estimate of Var[b $\mid X$]

With s^2 we can estimate the covariance matrix as:

$$\widehat{Var}[b \mid X] = s^2(X'X)^{-1}$$

Finally, the square root of the kth diagonal element gives the standard error:

$$s_{b_k} = \left[s^2(X'X)^{-1}\right]_{kk}^{1/2}$$

4.2.8 Hypothesis Testing under Normality

4.2.9 Assumption 5: Normality of the error terms

The distribution of ε conditional on X is jointly normal.
Since the mean and the variance of $\varepsilon \mid X$ are already specified in assumptions 2 (strict exogeneity) and 4 (spherical error variance), we can write:

$$\varepsilon \mid X \sim N(0, \sigma^2 I)$$

The distribution of ε conditional on X does not depend on X. ε and X are therefore independent. As a consequence the marginal (or unconditional) distribution of ε is the same as the conditional distribution on X.

4.2.10 Distribution of the OLS estimator under normality

Finally, since a linear function of random variables that are jointly normally distributed is itself normally distributed, we can write:

$$b - \beta \mid X \sim N\left(0, \sigma^2(X'X)^{-1}\right)$$

or, analogously,

$$b \mid X \sim N\left(\beta, \sigma^2(X'X)^{-1}\right)$$

and for each element of the vector $b - \beta$:

$$b_k - \beta_k \mid X \sim N\left(0, \sigma^2(X'X)^{-1}_{kk}\right)$$

or, analogously,

$$b_k \mid X \sim N\left(\beta_k, \sigma^2(X'X)^{-1}_{kk}\right)$$

4.2.11 Testing Hypotheses on Individual Regression Parameters

The null hypothesis we consider is about the k-th parameter:

$$H_0 : \beta_k = \beta_k^*$$

Let S^{kk} be the k-th diagonal element of $(X'X)^{-1}$. Under the null, the distribution of the statistic

$$z_k = \frac{b_k - \beta_k^*}{\sqrt{\sigma^2 S^{kk}}}$$

conditional on X is:

$$z_k \mid X \sim$$

Since the distribution of $z_k \mid X$ does not depend on X, the marginal distribution of z_k is the same as the distribution of $z_k \mid X$.

In principle, we could therefore use z_k to test hypotheses or form confidence intervals about each individual element β_k. The problem here is that σ^2 is unknown. What we observe is s^2. By using s^2 instead of σ^2 we can derive a statistic to use in place of z_k.

Let us define the quantity:

$$q = \frac{(n-K)s^2}{\sigma^2} = \frac{e'e}{\sigma^2} = \frac{\varepsilon' M \varepsilon}{\sigma^2} = \left(\frac{\varepsilon}{\sigma}\right)' M \left(\frac{\varepsilon}{\sigma}\right)$$

It can be proved that:

$q \mid X$ has a χ^2 distribution with $n-K$ degrees of freedom, where $n-K$ is the rank of M.

The two random variables z_k and q are independent conditional on X.

Regression packages typically report the following t–value, to test the null that coefficient k is equal to zero:

$$t_k = \frac{b_k}{se(b_k)}$$

If the hypothesis is rejected, that is if the computed t_k is greater than 1.96, we say that b_k is statistically different from zero at the 5% level. P–values and confidence intervals are also typically reported.

4.2.12 A joint test of significance of regressions coefficients

Let's consider the null that J of the K coefficients are equal to zero (with $J < K$):

$$H_0 : \beta_{K-J+1} = \cdots = \beta_K = 0$$

The alternative hypothesis is that *at least one* of those J coefficients is not equal to zero. One easy way to test this hypothesis is to compare the sum of squared residuals of the restricted model (with the restrictions imposed, call it RSS_0) with that of the full model (RSS_1). If the null is correct, the difference between the two should be "small". It can be shown that under the null it holds that:

$$\frac{RSS_0 - RSS_1}{\sigma^2} \sim \chi_J^2$$

And using the fact that $(n-K)s^2/\sigma^2$ is distributed as a χ^2 with $(n-K)$ degrees of freedom, we have:

$$F = \frac{(RSS_0 - RSS_1)/J}{RSS_1/(n-K)}$$

Under the null, F has an F distribution with $(J, n-K)$ degrees of freedom. Using the definition of R^2, we can write:

$$F = \frac{(R_1^2 - R_0^2)/J}{1 - R_1^2}(n-K)$$

Hence the test can be interpreted as testing whether the increase in R^2 moving from the restricted model to the more general model is significant.

A special case of this test, typically provided by computer packages, is a test of a joint significance of all regressors, excluding the constant. In this case, the F statistic can be written using only the unrestricted model as:

$$\frac{R^2/(K-1)}{(1-R^2)/(n-K)}$$

Proof: it follows from the previous test, noticing that in the restricted model, i.e. in a model that only includes the constant, $R^2 = 0$.

4.2.13 Testing Linear Hypotheses – The General Case

The null hypothesis is a set of J linear restrictions on the parameters:

$$\begin{aligned} R_{11}\beta_1 + R_{12}\beta_2 + \cdots + R_{1K}\beta_K &= r_1 \\ R_{21}\beta_1 + R_{22}\beta_2 + \cdots + R_{2K}\beta_K &= r_2 \\ &\vdots \\ R_{J1}\beta_1 + R_{J2}\beta_2 + \cdots + R_{JK}\beta_K &= r_J \end{aligned}$$

or, in compact form:

$$H_0 : R\beta = r$$

where R and r are known and specified by the hypothesis. R has J rows and K columns. Assume that $J < K$ and that the rows are linearly independent. Therefore R has full row rank. With the restrictions imposed, there are only $K - J$ free parameters.

4.2.14 Linear restriction example: $R\beta = r$

Assume $K = 4$. We want to test the joint null

$$H_0 : \beta_2 = \beta_3 \quad \text{and} \quad \beta_4 = 0.$$

This can be written as $R\beta = r$ with $m = 2$ restrictions:

$$R = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad r = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

4.2.15 Unrestricted model

$$y = X\beta + \varepsilon$$

Set of linear restrictions

$$H_0 : R\beta = r$$

The unrestricted sample estimate of $R\beta$ is Rb. Furthermore, since

$$Var[b \mid X] = \sigma^2 (X'X)^{-1}$$

we can write

$$Var[Rb \mid X] = \sigma^2 R(X'X)^{-1}R'.$$

This statistic

$$(Rb - R\beta)' \left[\sigma^2 R(X'X)^{-1}R'\right]^{-1} (Rb - R\beta)$$

conditional on X has a known distribution

$$w \mid X \sim \chi^2(J)$$

To test linear hypotheses we look for a statistic that has a known distribution under the null. This is simply

$$w = (Rb - r)' \left[\sigma^2 R(X'X)^{-1}R'\right]^{-1}(Rb - r)$$

which, conditional on X, is obviously:

$$w \mid X \sim \chi^2(J)$$

Again, since the conditional distribution on X does not depend on X, w and X are independently distributed and the marginal distribution of w is the same as the conditional distribution of w.

In principle, we could therefore use w to test linear hypotheses. The problem here is that σ^2 is unknown. What we observe is s^2. With the purpose of eliminating σ^2 we can define this additional statistic, known as the F-statistic:

$$F = \frac{w/J}{q/(n-K)}$$

Under the null, the distribution of the F-statistic conditional on X is

$$F \mid X \sim F(J, n-K)$$

In fact, it is the ratio of two independent χ^2 variables, each divided by its respective degrees of freedom.

Finally, we can rewrite the F-statistic as:

$$F = \frac{(Rb - r)' \left[s^2 R(X'X)^{-1}R'\right]^{-1}(Rb - r)}{J}$$

If F is large then the null hypothesis is rejected. The intuition is straightforward: the null hypothesis is not rejected when Rb is not "distant" from r and/or when the estimated variance of Rb is "large".

4.2.16 Large Sample Properties of the OLS Estimator

In the CLRM with normally distributed errors we have been able to derive the finite sample distribution (that is the distribution for any given sample size) of the OLS estimator and of its associated test statistics (t and F).

However, in order to obtain these finite sample results we have been forced to make very strong assumptions:

- 1. linearity of the regression equation;
- 2. strict exogeneity of the regressors;
- 3. normality of the error terms.

Indeed, the finite sample theory breaks down if only one of these assumptions is violated.

It is therefore necessary to resort to large sample (or asymptotic) theory, that is, theory which applies to the case in which the sample size is infinitely large.

However, since infinite samples are obviously not available, by applying large sample theory to finite samples we will only be able to derive an approximation of the distribution of the estimator and of its associated statistics.

4.2.17 Consistency of the OLS Estimator

The OLS estimator is consistent under weaker conditions than the CLRM assumptions.

To see this let's write the OLS estimator as

$$b = \left(\frac{1}{n}\sum_i x_i x_i'\right)^{-1}\left(\frac{1}{n}\sum_i x_i y_i\right)$$

This expression can be rewritten as

$$\begin{aligned} b &= \left(\frac{1}{n}\sum_i x_i x_i'\right)^{-1}\left(\frac{1}{n}\sum_i x_i (x_i'\beta + \varepsilon_i)\right) \\ &= \beta + \left(\frac{1}{n}\sum_i x_i x_i'\right)^{-1}\left(\frac{1}{n}\sum_i x_i \varepsilon_i\right) \end{aligned}$$

Consistency requires that

$$p\lim(b - \beta) = p\lim\left[\left(\frac{1}{n}\sum_i x_i x_i'\right)^{-1}\left(\frac{1}{n}\sum_i x_i \varepsilon_i\right)\right] = 0$$

4.2.18 Assumption 6

$\left(\frac{1}{n}\sum_i x_i x_i'\right)$ converges in probability to a finite non-singular matrix Σ_{XX}.

The non-singularity of Σ_{XX} requires that there is no multicollinearity in the limit. This is a technical regularity condition that holds in most empirical applications.

Under weak conditions sample averages converge to the corresponding population means (Law of Large Numbers). Under assumption 6 we have therefore

$$p\lim(b-\beta) = \Sigma_{XX}^{-1} E[x_i\varepsilon_i]$$

where Σ_{XX} is the population mean of $\left(\frac{1}{n}\sum_i x_i x_i'\right)$ and $E[x_i\varepsilon_i]$ is the population mean of $\left(\frac{1}{n}\sum_i x_i\varepsilon_i\right)$.

4.2.19 Assumption 7

$$E[x_i\varepsilon_i] = 0$$

Assumption 7, which is needed for consistency, is weaker than assumption 2 which is needed for unbiasedness.

It only states that the error terms have zero mean and are uncorrelated (as opposed to mean independent) with any of the explanatory variables.

In fact, given assumption 7, we can write:

$$p\lim(b-\beta) = \Sigma_{XX}^{-1} 0 = 0$$

4.2.20 Asymptotic Normality of the OLS Estimator

Even under the assumptions of the CLRM (Assumptions 1–4), the exact distribution of the OLS estimator cannot be derived without assuming normality (Assumption 5).

We can however derive an asymptotic approximation based on the asymptotic distribution. For the OLS estimator it can be shown that under the conditions of the CLRM augmented by assumptions 6 and 7:

$$\sqrt{n}(b-\beta) \xrightarrow{d} N(0, \sigma^2\Sigma_{XX}^{-1})$$

In words, $\sqrt{n}(b-\beta)$ converges in distribution to a normally distributed random vector with 0 mean and $\sigma^2\Sigma_{XX}^{-1}$ variance. This results in an application of the Central Limit Theorem.

In practice this result can be used to approximate the distribution of b as:

$$b \overset{a}{\sim} N\left(\beta, \frac{\sigma^2\Sigma_{XX}^{-1}}{n}\right)$$

where $\overset{a}{\sim}$ means "is approximatively distributed as".

Since both σ^2 and Σ_{XX} are unknown, the asymptotic approximation of the variance has to be estimated. A consistent estimator is:

$$\frac{s^2\left(\frac{1}{n}\sum_i x_i x_i'\right)^{-1}}{n} = s^2\left(\sum_i x_i x_i'\right)^{-1}$$

4.2.21 Appendix A: Gauss-Markov Theorem (proof)

Expression for the OLS estimator b

$$A = (X'X)^{-1}X'$$
$$b = Ay$$

Expression for the alternative linear estimator b0

$$b_0 = Cy = (D+A)y$$
$$= Dy + Ay$$
$$= D(X\beta+\varepsilon) + b$$
$$= DX\beta + D\varepsilon + b$$

Condition for the unbiasedness of b0

$$\begin{aligned} E[b_0 \mid X] &= DX\beta + DE[\varepsilon \mid X] + E[b \mid X] \\ &= DX\beta + \beta \\ &= DX\beta + \beta \end{aligned}$$

Condition for unbiasedeness is therefore

$$DX = 0$$

It follows that

$$b_0 = D\varepsilon + b$$

and

$$\begin{aligned} b_0 - \beta &= D\varepsilon + (b - \beta) \\ &= (D + A)\varepsilon \end{aligned}$$

BLUE definition and variance comparison in matrix sense The OLS estimator b is BLUE (Best Linear Unbiased Estimator) if for any unbiased estimator b_0 that is linear in y

$$Var[b_0 \mid X] \geq Var[b \mid X]$$

in matrix sense. We say that $Var[b_0 \mid X] \geq Var[b \mid X]$ in matrix sense if

$$Var[b_0 \mid X] - Var[b \mid X]$$

is positive semidefinite, that is if

$$c' \left[Var[b_0 \mid X] - Var[b \mid X] \right] c \geq 0$$

where c is a generic conformable vector.

Derivation of Var[b0 ∣ X] = $\sigma^2[DD' + (X'X)^{-1}] \geq \sigma^2(X'X)^{-1}$

$$\begin{aligned} Var[b_0 \mid X] &= Var[b_0 - \beta \mid X] = Var[(D + A)\varepsilon \mid X] \\ &= (D + A)Var[\varepsilon \mid X](D + A)' \\ &= \sigma^2(D + A)(D' + A') \\ &= \sigma^2(DD' + AD' + DA' + AA') \end{aligned}$$

and since

$$\begin{aligned} DA' &= DX(X'X)^{-1} = 0 \\ AA' &= (X'X)^{-1}(X'X)(X'X)^{-1} = (X'X)^{-1} \end{aligned}$$

we obtain our final result

$$Var[b_0 \mid X] = \sigma^2[DD' + (X'X)^{-1}] \geq \sigma^2(X'X)^{-1}$$

since DD' is positive semidefinite.

4.2.22 Appendix B: Unbiasedness of s^2 (proof)

Expression for E[e'e | X]

$$\begin{aligned} E[e'e \mid X] &= E[e'Me \mid X] \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} m_{ij} E(\varepsilon_i \varepsilon_j \mid X) \\ &= \sum_{i=1}^{n} m_{ii}\sigma^2 = \sigma^2 \sum_{i=1}^{n} m_{ii} \\ &= \sigma^2 \,\text{trace}(M) \end{aligned}$$

Computing trace(M)

$$\begin{aligned} \text{trace}(M) &= \text{trace}(I_n - P) \\ &= \text{trace}(I_n) - \text{trace}(P) \\ &= n - \text{trace}(P) \end{aligned}$$

and, since

$$\begin{aligned} \text{trace}(P) &= \text{trace}[X(X'X)^{-1}X'] \\ &= \text{trace}[X'X(X'X)^{-1}] \\ &= \text{trace}(I_k) = K \end{aligned}$$

we obtain

$$E[e'e \mid X] = (n-K)\sigma^2$$

Result E[e'e | X] = (n - K)σ^2

Bias of (1/n)e'e and unbiasedness of $s^2 = (1/(n-K))e'e$ This proves that the "natural" estimator is biased (even if the bias becomes smaller as the sample size increases) since:

$$E\left[\frac{1}{n}e'e \mid X\right] = \frac{n-K}{n}\sigma^2$$

s^2 is instead unbiased. In fact

$$E\left[\frac{1}{n-K}e'e \mid X\right] = \frac{n-K}{n-K}\sigma^2 = \sigma^2$$

4.3 Set 3: Model Specification: Interpretation and Misspecification

4.3.1 Interpreting the model

Consider the model

$$y_i = x_i'\beta + \varepsilon_i$$

with

$$E(\varepsilon_i \mid x_i) = 0.$$

With this assumption, the model can be interpreted as describing the conditional expected value of y_i given values for the explanatory variables x_i.

It follows that we can interpret the β coefficients as the expected change in y_i if x_{ik} changes by one unit but all other variables in x_i do not change (ceteris paribus condition):

$$\frac{\partial E(y_i \mid x_i)}{\partial x_{ik}} = \beta_k.$$

4.3.2 Non-linear regressors

- Depending on the specification, marginal effects can vary over the observations.
- If $x_i'\beta$ includes, for example, both $\text{age}_i\beta_2 + \text{age}_i^2\beta_3$, then the marginal effect of age will depend on age:

$$\frac{\partial E(y_i \mid x_i)}{\partial \text{age}_i} = \beta_2 + 2\,\text{age}_i\,\beta_3$$

4.3.3 Units and scaling

- In the linear model, β_k is also called **level effect**: a one-unit increase in x_k changes y by β_k units, ceteris paribus.
- Units matter: if x_k is in thousands, β_k is the effect of a \$1,000 change.
- Rescaling an explanatory variable x_k by c rescales its coefficient by $1/c$; t-statistics are invariant.
- Standardizing x_k (z-score) gives the effect of a one-SD increase in x_k on y.
- Always map back to original units when explaining your results.

4.3.4 Dummy variables (indicator regressors)

Let $D_i \in \{0, 1\}$. In the regression

$$y_i = \beta_0 + \beta_D D_i + \gamma' x_i + u_i,$$
$$E[y_i \mid D_i = 1, x] - E[y_i \mid D_i = 0, x] = \beta_D.$$

- β_D is the difference in conditional means between $D = 1$ and the baseline $D = 0$.
- With c categories, include $c - 1$ dummies to avoid the "dummy trap"; coefficients are relative to the baseline (omitted category).

4.3.5 Interactions: allowing effects to differ

Interactions $D \times X$ allow the slope on X to differ by group.

$$y_i = \beta_0 + \beta_1 x_i + \beta_D D_i + \beta_{XD}(x_i D_i) + u_i.$$

- Slope for $D = 0$: β_1; slope for $D = 1$: $\beta_1 + \beta_{XD}$.
- β_{XD} is the difference in slopes.

4.3.6 Other non-linear regressor: logarithms of variables (1/2)

Why are (natural) logarithms of variables used?

- **Linearization of nonlinear relations:** Many economic relations are multiplicative, e.g. $y = Ax^\beta e^u$. Taking logs yields a linear form suitable for OLS estimation.
- **Elasticity interpretation:** In a log–log model

$$\ln(y_i) = \beta_0 + \beta_1 \ln(x_i) + v_i,$$

  the slope β_1 measures the elasticity: a 1% increase in x is associated with an approximate β_1% change in y.
- **Semi-elasticities:** In a model where only y is logged,

$$\ln(y_i) = \beta_0 + \beta_1 x_i + v_i,$$

  $100\beta_1$% measures the percentage change in y for a one-unit change in x.

4.3.7 Other non-linear regressor: logarithms of variables (2/2)

- **Reducing skewness and heteroskedasticity:** Many variables (income, sales, firm size) are highly skewed. Log transformation compresses large values and stabilizes variance.
- **Comparability and interpretation:** Logs express effects in relative rather than absolute terms, often more meaningful in economics.
- **Summary:** Taking logs makes models:
  - Easier to interpret (elasticities, semi-elasticities);
  - More suitable for OLS (linear in parameters, homoskedastic errors);
  - More robust to scale and outliers.

4.3.8 A note on elasticities (log–log model)

Formally, the elasticity of one variable y with respect to another variable x is the percentage change in y relative to the percentage change in x:

$$\frac{dy}{dx}\frac{x}{y} = \frac{d\ln y}{d\ln x}$$

In a regression model, y depends on both x and on an error term. It is conventional to evaluate the elasticity as the percentage change not of y but of the predicted component of y, that is of the conditional mean $\mathrm{E}(Y|X)$ with respect to X, which is

$$\frac{d\,\mathrm{E}(y_i|x_i)}{dx_i}\frac{x_i}{\mathrm{E}(y_i|x_i)} = \frac{d\ln \mathrm{E}(y_i|x_i)}{d\ln x_i}.$$

Can we recover this elasticity from a regression model? We can *approximate* it with the so-called log–log linear model.

4.3.9 The log–log linear model

- The log–log linear model is a linear regression model in which all variables are expressed in (natural) logarithms (except for dummy variables). For example, with one regressor:

$$\ln(y_i) = \beta_0 + \beta_1 \ln(x_i) + v_i,$$

- In this case

$$\frac{d\,\mathrm{E}(y_i|x_i)}{dx_i}\frac{x_i}{\mathrm{E}(y_i|x_i)} = \frac{d\ln \mathrm{E}(y_i|x_i)}{d\ln x_i} \approx \frac{d\,\mathrm{E}(\ln y_i|\ln x_i)}{d\ln x_i} = \beta_1$$

- which is only approximately true because $\mathrm{E}(\ln y_i|\ln x_i) \neq \ln \mathrm{E}(y_i|x_i)$.
- Hence, we can say that in the log–log–linear model the elasticity (i.e. the relative change in y_i due to a relative change in x_i) is *approximately* constant.

4.3.10 The log–log model: elasticities (exact vs. approx.)

Model (with K variables):

$$\ln y_i = \alpha + \ln(x_i)'\beta + v_i, \qquad \mathrm{E}[v_i|x_i] = 0.$$

Exact (geometric-mean / "typical" Y) elasticity:

$$g(x_i) \equiv \exp\{\mathrm{E}[\ln y_i|x_i]\} = e^{\alpha}\prod_k x_{ik}^{\beta_k} \quad \Rightarrow \quad \frac{\partial \ln g(x_i)}{\partial \ln x_{ik}} = \beta_k \quad \text{(constant, exact)}.$$

Arithmetic mean and the "smearing factor":

$$\mathrm{E}[y_i|x_i] = e^{\alpha}\prod_k x_{ik}^{\beta_k}\frac{\mathrm{E}[e^{v_i}|x_i]}{S(x_i)} \quad \Rightarrow \quad \frac{\partial \ln \mathrm{E}[y_i|x_i]}{\partial \ln x_{ik}} = \beta_k + \frac{\partial \ln S(x_i)}{\partial \ln x_{ik}}.$$

If $S(x_i)$ is (approximately) constant (e.g. log-error spread doesn't vary with x_i), this equals β_k (constant); otherwise it can vary with x_i.

4.3.11 Reminder: arithmetic vs geometric mean (incl. lognormal)

Arithmetic mean (AM): $E[Y \mid X]$ (additive center; totals/budgets).
Geometric mean (GM): $\exp\{E[\ln Y \mid X]\}$ (multiplicative "typical" level; robust to right-tail).

AM–GM inequality: AM $\geq$ GM (equality iff all values equal).

Lognormal special case: If $Y \mid X \sim \text{Lognormal}(\mu(X), \sigma^2(X))$,

$$GM(Y \mid X) = e^{\mu(X)}, \qquad AM(Y \mid X) = e^{\mu(X)+\sigma^2(X)/2}$$

The gap widens with log-variance $\sigma^2(X)$.

4.3.12 Summarizing the log–log model: exact finite changes and small-change approximations

Consider the simple model:

$$\ln(y) = \beta_0 + \beta_1 \ln(x) + v$$

Exact finite change: consider an increase in x from x_0 to $x_1 = x_0(1+\tau)$,

$$\mathrm{E}[\ln y_1|x] - \mathrm{E}[\ln y_0|x] = \beta_1(\ln x_1 - \ln x_0)$$

Exponentiating $\mathrm{E}[\ln y_1|x]$ we obtain the geometric mean $g(x_1)$

$$\frac{g(x_1)}{g(x_0)} = \left(\frac{x_1}{x_0}\right)^{\beta_1} = (1+\tau)^{\beta_1} \quad \Rightarrow \quad \%\Delta g = 100\left((1+\tau)^{\beta_1} - 1\right)\%.$$

Small-change (derivative) approximation:

$$d\ln y = \beta_1 d\ln x = \frac{\Delta y}{y} \approx \beta_1 \frac{\Delta x}{x} \quad \Rightarrow \quad \%\Delta y \approx \beta_1 \times \%\Delta x.$$

In words: a 1% increase in x changes y by approximately β_1%.

4.3.13 Log-linear: ln Y on levels (semi-elasticity)

Model: $\ln y = \beta_0 + \beta_1 x + v$.

Exact finite change: consider an increase in one unit in x:

$$E[\ln y_1 \mid x] - E[\ln y_0 \mid x] = \beta_1$$

Exponentiating $E[\ln y \mid x]$ we obtain the geometric mean $g(x)$:

$$\frac{g(x_1)}{g(x_0)} = (e)^{\beta_1} \quad \Rightarrow \quad \%\Delta g = 100\,(\exp(\beta_1) - 1)\,\%.$$

In general, the exact percent change of (the geometric mean of) y for some Δx:

$$100(\exp(\beta_1 \Delta X) - 1)\%.$$

A one-unit increase in x changes y by approximately $100\beta_1$%.

4.3.14 Linear-log: y on ln x

Model: $\mathrm{y} = \beta_0 + \beta_1 \ln x + v$.

Exact finite change: for a change $x_0 \to x_1 = x_0(1+\tau)$ is simply:

$$\Delta y = \beta_1 \ln(1+\tau).$$

A 1% increase in x $(\Delta \ln X = 0.01)$changes y by approximately $\beta_1/100$ units.

4.3.15 Dummy in log outcome

Model: $\ln y = \beta_0 + \beta_D D + \cdots + v$.

Effect on $\ln y$ is β_D;

The exact percent difference on y is $100 \times (\exp(\beta_D) - 1)\%$;

Can approximate $100\,\beta_D$% if effect is small.

Specification	Coefficient	Interpretation (ceteris paribus)
$\ln Y$ on $\ln X$	β	Elasticity: β (constant). Exact finite change: $\%\Delta Y = 100((1+\tau)^{\beta} - 1)\%$. Approx: $+1\%$ in $X \Rightarrow \beta\%$ in Y.
$\ln Y$ on X	β	Semi-elasticity: β (constant). Exact: $\%\Delta Y = 100(\exp(\beta\,\Delta X) - 1)\%$. Approx: $+1$ unit in $X \Rightarrow 100\,\beta\%$ in Y.
Y on $\ln X$	β	Exact: $\Delta Y = \beta \ln(1+\tau)$ for a τ-change. Approx: $+1\%$ in $X \Rightarrow \beta/100$ units in Y.
$\ln Y$ on D	β_D	Exact: $\%\Delta Y = 100(\exp(\beta_D) - 1)\%$. Approx: $D = 1 \Rightarrow 100\,\beta_D\%$.

4.3.16 Partitioned regression

To understand the meaning of coefficients in a multiple regression model, we can use the partitioned regression result, which shows how the estimated effect of one group of variables changes once the influence of other variables is taken into account. Consider a correctly specified regression and define two separate sets of variables, X_1 and X_2:

$$y = X\beta + \varepsilon = X_1\beta_1 + X_2\beta_2 + \varepsilon$$

We are interested in the coefficients relative to the second subset of variables. What is the algebraic solution for b_2? We can write the first order conditions:

$$\begin{bmatrix} X_1'X_1 & X_1'X_2 \\ X_2'X_1 & X_2'X_2 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} X_1'y \\ X_2'y \end{bmatrix}$$

We could obtain the solution applying the partitioned inverse matrix or solve the above system, first for b_1:

$$b_1 = (X_1'X_1)^{-1}X_1'y - (X_1'X_1)^{-1}X_1'X_2b_2 = (X_1'X_1)^{-1}X_1'(y - X_2b_2)$$

This solution states that b_1 is the set of coefficients in the regression of y on X_1, minus a correction vector. Suppose that $X_1'X_2 = 0$. Then b_1 is simply the coefficient vector in the regression of y on X_1:

$$b_1 = (X_1'X_1)^{-1}X_1'y$$

Hence, if in the linear least squares regression the two sets of variables are orthogonal, then the separate coefficient vectors can be obtained by separate regressions of y on X_1 and y on X_2 alone.

Now, in the general case in which the two sets of regressors are *not* orthogonal, we can substitute the expression of b_1 into the second set of first order conditions and get:

$$X_2'X_1(X_1'X_1)^{-1}X_1'y - X_2'X_1(X_1'X_1)^{-1}X_1'X_2b_2 + X_2'X_2b_2 = X_2'y$$

Collecting terms:

$$X_2'[I - X_1(X_1'X_1)^{-1}X_1']X_2b_2 = X_2'[I - X_1(X_1'X_1)^{-1}X_1']y$$

$$X_2'M_1X_2b_2 = X_2'M_1y$$

Hence:

$$b_2 = (X_2'M_1X_2)^{-1}X_2'M_1y$$

- As an example, consider a regression of earnings on a constant, age, and education. The coefficient on education can be obtained by first regressing earnings on age and education on age, and then running a simple regression using the residuals from these two regressions. In other words, the *ceteris paribus* effect of education on earnings is the effect of education on earnings after removing (or "partialling out") the effect of age.
- In a time series setting, introducing a time trend in a regression or "detrending" the data by running a regression of the individual series on using the residuals, gives the same results on the coefficients of interest.
- As a further (fun!) application, consider the case where X_1 is just a column of ones: applying the procedure of the partitioned regression we can see that including a constant in a regression is equivalent to running the regression on the demeaned data (i.e. transforming the data into deviations from their mean).

4.3.17 Omission of Relevant Variables

So far our analysis (estimation, hypothesis testing) has assumed that the correct specification of the model is known to be

$$y = X\beta + \varepsilon$$

Let us now consider the possibility of having two (nested) competing models

$$\text{Model 1} : y = X\beta + Z\gamma + v$$

and

$$\text{Model 2} : y = X\beta + \varepsilon$$

Both models can be interpreted as expected values for y conditional on X and Z. Model 2 implicitly assumes that Z is irrelevant ($\gamma = 0$). Model 2 is said to be nested in Model 1.

Let us assume that Model 1 is the true model and that we are interested in estimating β. The relevant question to be addressed is whether the OLS estimator based on Model 2

$$b_2 = (X'X)^{-1}(X'y)$$

is an unbiased estimator for β.

The expression in the previous slide can be written as

$$b_2 = (X'X)^{-1}(X'(X\beta + Z\gamma + v))$$
$$= \beta + (X'X)^{-1}X'Z\gamma + (X'X)^{-1}X'v$$

This allows us to write

$$E[b_2 \mid X, Z] = \beta + (X'X)^{-1}X'Z\gamma$$

b_2 is an unbiased estimator of β only if $X'Z = 0$ (an unlikely event) or $\gamma = 0$ (which is false if Model 1 is the true model).

4.3.18 Inclusion of Irrelevant Variables

Suppose now that the true model is Model 2

$$\text{Model 2: } y = X\beta + \varepsilon$$

and we estimate Model 1 instead

$$\text{Model 1: } y = X\beta + Z\gamma + v$$

The relevant question to be addressed is whether the OLS estimator based on Model 1 is an unbiased estimator of β. What we are doing in this case is simply failing to impose the restriction

$$\gamma = 0$$

$$E[b, g \mid X] = \begin{bmatrix}\beta \\ \gamma\end{bmatrix} = \begin{bmatrix}\beta \\ 0\end{bmatrix}$$

$$E\left[\frac{\hat{v}'\hat{v}}{n - K_1 - K_2}\right] = \sigma^2$$

However, the price one has to pay for not imposing the appropriate restriction is the reduced precision of estimates. In fact the covariance matrix in the short regression is never larger than the covariance matrix obtained in the presence of superfluous variables.

4.3.19 Multicollinearity

- In general, explanatory variables may be mutually correlated
- However, if correlation between two variables is too high, this may lead to problems
- Technically, these problems arise because the matrix $X'X$ is close to being not invertible: this may lead to unreliable estimates of an unexpected sign or magnitude with high standard errors
- The intuition is that if two variables are highly correlated (e.g. age and experience in the labor market) it may be hard for the model to identify the individual impact of these two variables
- This is situation is referred to as *multicollinearity*: an approximate linear relationship exists among two (or more) explanatory variables
- When the linear relationship is perfect (as opposed to approximate) then we have *perfect multicollinearity*, in which case the OLS estimator is not defined (the matrix $X'X$ is not invertible)
- Consider the general expression for the variance of the OLS estimator:

$$\text{Var}[b \mid X] = \sigma^2(X'X)^{-1}$$

- For a single coefficient β_k this can be written as:

$$\text{Var}[b_k \mid X] = \frac{\sigma^2}{1 - R_k^2}\frac{1}{N}\left(\frac{1}{N}\sum_{i=1}^{N}(x_{ik} - \bar{x}_k)^2\right)^{-1}$$

for $k = 2, \ldots, K$.

- The term R_k^2 denotes the squared multiple correlation coefficient between x_{ik} and all the other explanatory variables
- In other words, it is the R^2 from regressing x_{ik} on the remaining explanatory variables, including the constant
- If R_k^2 is close to 1, x_{ik} can be closely approximated by a linear combination of the other regressors, and the variance of β_k will be large.
- However, if there is enough variation in x_{ik}, the sample is sufficiently large and the variance of the error term is sufficiently small, a large value of R_k^2 need not cause problems.
- The variance inflation factor is used to detect multicollinearity:

$$\text{VIF}[b_k] = \frac{1}{1 - R_k^2}$$

4.4 Set 4: Heteroskedasticity and Autocorrelation

4.4.1 The Generalized Linear Regression Model

Assumption 1: linearity

$$y = X\beta + \varepsilon$$

Assumption 2: strict exogeneity

$$E[\varepsilon \mid X] = 0$$

Assumption 3: no (perfect) multicollinearity

$$\Pr[\text{rank}(X) = K] = 1$$

Assumption 4b: non-spherical error variance

$$E[\varepsilon\varepsilon' \mid X] = \text{Var}[\varepsilon \mid X] = \sigma^2\Omega$$

where Ω is assumed to be symmetric and positive definite. For simplicity we assume that Ω is known and that σ^2 is an unknown positive scalar.
The covariance matrix $E[\varepsilon\varepsilon' \mid X]$ is therefore known up to the multiplicative constant σ^2.

4.4.2 The Ordinary Least Squares Estimator (OLS)

Unbiasedness
The OLS estimator is still unbiased in the Generalized Linear Regression Model (GLRM). In fact,

$$b_{\text{OLS}} = \beta + (X'X)^{-1}X'\varepsilon$$

and therefore

$$E(b_{\text{OLS}} - \beta \mid X) = (X'X)^{-1}X'E(\varepsilon \mid X) = 0$$

Expression for the variance
The variance of the OLS estimator is

$$\text{Var}[b_{\text{OLS}} \mid X] = \text{Var}[b_{\text{OLS}} - \beta \mid X] = \text{Var}[(X'X)^{-1}X'\varepsilon \mid X]$$

$$= (X'X)^{-1}X' \,\text{Var}[\varepsilon \mid X]X(X'X)^{-1}$$

$$= \sigma^2(X'X)^{-1}X'\Omega X(X'X)^{-1}$$

Note that in the special case $\Omega = I$, we have

$$\text{Var}[b_{\text{OLS}} \mid X] = \sigma^2(X'X)^{-1}.$$

4.4.3 Conditional distribution of the OLS estimator

Assume that $\varepsilon \mid X$ is jointly normal. Since a linear function of normally distributed variables is normally distributed, we obtain

$$b_{\text{OLS}} - \beta \mid X \sim N\Big(0,\ \sigma^2 (X'X)^{-1}(X'\Omega X)(X'X)^{-1}\Big)$$

or equivalently

$$b_{\text{OLS}} \mid X \sim N\Big(\beta,\ \sigma^2 (X'X)^{-1}(X'\Omega X)(X'X)^{-1}\Big)\,.$$

4.4.4 The Gauss-Markov Theorem

Under the assumption that Ω is known, it can be proved that:

1. there exists a BLUE estimator for the GLRM, and
2. this estimator is **not** the OLS estimator.

4.4.5 The Generalized Least Squares (GLS) Estimator

Intuition

By exploiting the fact that Ω is by construction symmetric and positive definite, y, X and ε can be transformed so that the transformed model satisfies all the assumptions of the CLRM.

The OLS estimator of this transformed model is called the Generalized Least Squares (GLS) Estimator.

The GLS estimator is BLUE in the context of the GLRM.

Since Ω is a symmetric and positive definite matrix, there exists a non singular $n \times n$ matrix P such that:

$$\Omega^{-1} = P'P$$

and therefore

$$\Omega = (P'P)^{-1} = P^{-1}P^{-1'}$$
$$P\Omega P' = PP^{-1}P^{-1'}P' = I$$

It follows that

$$E[P\varepsilon \mid X] = PE[\varepsilon \mid X] = 0$$

and

$$\text{Var}[P\varepsilon \mid X] = P\,\text{Var}[\varepsilon \mid X]P' = \sigma^2 P\Omega P' = \sigma^2 I$$

In words, $P\varepsilon$ satisfies the conditions of the Gauss–Markov Theorem.

In order to derive the GLS estimator, pre-multiply the GLRM by P:

$$Py = (PX)\beta + P\varepsilon$$

The model can therefore be transformed as:

$$y^* = X^*\beta + \varepsilon^*$$

where $Py = y^*$, $PX = X^*$ and $P\varepsilon = \varepsilon^*$.

The GLS estimator is simply the OLS estimator of the transformed model

$$b_{GLS} = (X^{*'}X^*)^{-1}X^{*'}y^* = (X'P'PX)^{-1}X'P'Py = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y$$

Obviously

$$E(b_{GLS} - \beta \mid X) = (X^{*'}X^*)^{-1}X^{*'}E(\varepsilon^* \mid X) = 0$$

and

$$\text{Var}(b_{GLS} \mid X) = \text{Var}(b_{GLS} - \beta \mid X) = \text{Var}\left[(X^{*'}X^*)^{-1}X^{*'}\varepsilon^* \mid X\right]$$

$$= (X^{*'}X^*)^{-1}X^{*'}\ \mathrm{Var}(\varepsilon^* \mid X)\, X^*(X^{*'}X^*)^{-1}$$

$$= \sigma^2\,(X^{*'}X^*)^{-1}X^{*'}X^*(X^{*'}X^*)^{-1}$$

$$= \sigma^2\,(X^{*'}X^*)^{-1} = \sigma^2\,(X'\Omega^{-1}X)^{-1}$$

The expression above can be estimated by replacing σ^2 with its unbiased estimator:

$$\hat{\sigma}^2 \;=\; \frac{\hat{\varepsilon}^{*'}\hat{\varepsilon}^*}{n-K}$$

where $\hat{\varepsilon}^*$ is the vector of residuals of the transformed model.

4.4.6 Heteroskedasticity and the GLS estimator

Assume

$$E[\varepsilon_i^2 \mid X] = E[\varepsilon_i^2 \mid x_i] = \sigma^2 h_i^2, \qquad i = 1, 2, \ldots, n$$

where the h_i^2s are known and strictly positive. If we make the assumption of no-correlation, assumption 4b can be written as

$$E[\varepsilon\varepsilon' \mid X] = \mathrm{Var}[\varepsilon \mid X] = \sigma^2\,\mathrm{Diag}(h_i^2) = \sigma^2\Omega$$

where $\mathrm{Diag}(h_i^2)$ is a diagonal matrix with $h_1^2, h_2^2, \ldots, h_n^2$ on the main diagonal. In such a case

$$P = \mathrm{Diag}(h_i^{-1})$$

and therefore the transformed model can be specialized as:

$$y_i^* = x_i^{*'}\beta + \varepsilon_i^*$$

which is the same as

$$\frac{y_i}{h_i} = \frac{x_i'}{h_i}\beta + \frac{\varepsilon_i}{h_i}$$

The GLS estimator can therefore be written as

$$b_{GLS} = \left(\sum_i h_i^{-2} x_i x_i'\right)^{-1}\left(\sum_i h_i^{-2} x_i y_i\right)$$

This GLS estimator is a special case of the general GLS estimator in the equation of b_{GLS}. It is commonly known as the Weighted Least Squares (WLS) estimator since each observation is weighted by a factor proportional to the inverse of the error variance.

In most cases the h_i^2s in (20) are unknown and therefore the GLS estimator is unfeasible.

The obvious solution seems to replace the unknown h_i^2s with biased or consistent estimates and hope that this does not affect the properties of the modified estimator.

This is not simple, however, since there would be $n + K$ parameters and only n observations to estimate them.

The solution to this problem is to specify the unknown variances as a function of observables and of a small number of parameters. The corresponding estimator

$$b_{FGLS} = \left(\sum_i \hat{h}_i^{-2} x_i x_i'\right)^{-1}\left(\sum_i \hat{h}_i^{-2} x_i y_i\right)$$

is known as "Feasible" GLS (FGLS) or "Estimated" GLS (EGLS).

It can be shown that the GLS and the FGLS estimators are asymptotically equivalent and therefore converge to the same asymptotic distribution.

Unfortunately, the FGLS estimator does not share the small sample properties of the GLS estimator and therefore we cannot say that the FGLS estimator is BLUE. This happens to be the case since $\hat{h}_i^2$ is usually a non-linear function of y_i's and therefore is not a linear estimator.

4.4.7 OLS and heteroskedasticity

The GLS estimator is efficient but it is unfeasible in practice since it requires Ω to be known.

The FGLS estimator is feasible but: i) its sample properties are unknown, ii) more importantly, it requires us to specify a function for the variances.

A third, more common, strategy is to use the OLS estimator and estimate the appropriate covariance matrix:

$$\begin{aligned}\text{Var}(b_{\text{OLS}} \mid X) &= \sigma^2 (X'X)^{-1}(X'\Omega X)(X'X)^{-1} \\ &= (X'X)^{-1}\left[X' \,\text{Diag}(\sigma_i^2) X\right](X'X)^{-1}.\end{aligned}$$

At first sight it seems that to estimate this covariance matrix we need to estimate all σ_{ij}^2. However in an important paper White (1980) argues that only a consistent estimator of the $K \times K$ matrix

$$\Sigma = \frac{1}{n} X' \text{Diag}(\sigma_i^2) X = \frac{1}{n} \sum_i \sigma_i^2 x_i x_i'$$

is required. A consistent estimator is given by

$$S = \frac{1}{n} \sum_i e_i^2 x_i x_i'$$

where e_i is the usual OLS residual.

Therefore

$$\begin{aligned}\text{Var}(\boldsymbol{b}_{\text{OLS}} \mid X) &= (X'X)^{-1} \sum_i e_i^2 x_i x_i' (X'X)^{-1} \\ &= \left(\sum_i x_i x_i'\right)^{-1} \sum_i e_i^2 x_i x_i' \left(\sum_i x_i x_i'\right)^{-1}\end{aligned}$$

Standard errors retrieved from S (after) are defined as **heteroskedasticity-consistent standard errors** or, simply, White standard errors.

4.4.8 Autocorrelation and the GLS estimator

In the presence of autocorrelation, in general we can write:

$$E[ee'|X] = \text{Var}[\varepsilon|X] = \sigma_\varepsilon^2 \Omega$$

with

$$\Omega = \begin{pmatrix} 1 & \rho_1 & \cdots & \rho_{T-2} & \rho_{T-1} \\ \rho_1 & 1 & \cdots & \cdots & \rho_{T-2} \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ \rho_{T-2} & \cdots & \cdots & 1 & \rho_1 \\ \rho_{T-1} & \rho_{T-2} & \cdots & \rho_1 & 1 \end{pmatrix}$$

and

$$\rho_s = \text{Corr}(\varepsilon_t, \varepsilon_{t-s}) = \frac{\text{Cov}(\varepsilon_t, \varepsilon_{t-s})}{\sqrt{\text{Var}(\varepsilon_t)\text{Var}(\varepsilon_{t-s})}}$$

where ρ_s denotes the autocorrelation between ε_t and ε_{t-s}.

- One could obtain a feasible GLS estimator by estimating the $n(n+1)/2$ unknown elements in Ω
- In practice, it is possible to impose some structure to the error term. For example, if it follows a simple AR(1) or MA(1) process, than it can be shown that the Ω matrix depends on just one parameter, that could be easily estimated (more on this in the second part of the course!)
- Alternatively, one can still apply OLS adjusting the standard errors to this more general case...

4.4.9 OLS and autocorrelation

As for the heteroskedasticity case, we can still apply the OLS approach while adjusting its standard errors. This last approach is particularly useful when the correlation between ε_t and ε_{t-s} can be thought as virtually zero after H lags. In fact if we are willing to assume that

$$E(\varepsilon_t \varepsilon_{t-s}) = 0$$

$$s = H, H+1, \dots$$

a consistent estimator of the variance of the OLS estimator is given by

$$\begin{aligned} \text{Var}(\boldsymbol{b}_{\text{OLS}} \mid X) &= (X'X)^{-1} TS (X'X)^{-1} \\ &= \left(\sum_t x_t x_t' \right)^{-1} TS \left(\sum_i x_i x_i' \right)^{-1} \end{aligned}$$

where

$$S = \frac{1}{T} \sum_{t=1}^{T} e_t^2 x_t x_t' + \frac{1}{T} \sum_{j=1}^{H-1} w_j \sum_{s=j+1}^{T} e_s e_{s-j} (x_s x_s' - x_{s-j} x_s')$$

Standard errors based on (62) are referred to as **heteroskedasticity-and-autocorrelation consistent standard errors** (HAC) or simply Newey-West standard errors.

4.5 Set 5: SUMMARY: Finite and large sample properties of OLS

4.5.1 CLRM Assumptions (Finite-Sample Framework)

Model:

$$y = X\beta + \varepsilon.$$

1. **Linearity in parameters:**
$$y_i = x_i' \beta + \varepsilon_i.$$
2. **Strict exogeneity:**
$$E[\varepsilon \mid X] = 0 \quad \text{(equivalently, } E[\varepsilon_i \mid X] = 0 \text{ for all } i\text{)}.$$
3. **No perfect multicollinearity:**
$$\text{rank}(X) = K.$$
4. **Spherical errors:**
   - **Homoskedasticity:**
   $$E[\varepsilon_i^2 \mid X] = \sigma^2.$$
   - **No autocorrelation:**
   $$E[\varepsilon_i \varepsilon_j \mid X] = 0 \quad \text{for } i \neq j.$$
5. **Normality (for exact finite-sample inference):**
$$\varepsilon \mid X \sim N(0, \sigma^2 I_n).$$

4.5.2 OLS: Finite-Sample Properties

Let $\hat{\beta}$ be the OLS estimator.

- **Unbiasedness**
$$E[\hat{\beta} \mid X] = \beta.$$
- **Variance**
$$\text{Var}(\hat{\beta} \mid X) = \sigma^2 (X'X)^{-1}.$$
- **Gauss–Markov Theorem** Under CLRM assumptions 1–4, $\hat{\beta}$ is BLUE (Best Linear Unbiased Estimator).
- **Estimator of σ^2** Let $e = y - X\hat{\beta}$ be the residual vector. Then
$$s^2 = \frac{e'e}{n-K}$$
is an unbiased estimator of σ^2.

4.5.3 Finite-Sample Inference (with Normality)

Under normality (CLRM 5):

$$\hat{\beta} \mid X \sim N(\beta,\ \sigma^2(X'X)^{-1}).$$

t-test for a single coefficient For the null $H_0\colon \beta_k = \beta_k^*$, define

$$t_k = \frac{\hat{\beta}_k - \beta_k^*}{s\sqrt{(X'X)^{-1}_{kk}}} \ \sim\ t_{n-K},$$

where $(X'X)^{-1}_{kk}$ is the k-th diagonal element of $(X'X)^{-1}$.

F-test for joint linear restrictions For the null $H_0\colon R\beta = r$ with J linear restrictions (rows of R), the F-statistic is

$$F = \frac{(\,R\hat{\beta} - r\,)'\left[\,R\,s^2(X'X)^{-1}R'\,\right]^{-1}(\,R\hat{\beta} - r\,)}{J} \ \sim\ F(J, n-K).$$

4.5.4 Large-Sample Assumptions for the OLS Estimator (Homoskedastic Case)

Consider again the linear model

$$y = X\beta + \varepsilon. \tag{1}$$

Assume:

(1) Linearity: observed data are generated by (1).

(2b) Orthogonality (per observation):

$$E[x_i\varepsilon_i] = 0 \quad \text{for each } i.$$

This is weaker than strict exogeneity $E[\varepsilon \mid X] = 0$ because it allows $E[x_j\varepsilon_i] \neq 0$ for $j \neq i$.

(3b) No multicollinearity in the limit:

$$\operatorname{plim}\left(\frac{1}{n}X'X\right) = E[x_i x_i'] = \Sigma_{XX},$$

where Σ_{XX} is finite and non-singular.

(4) i.i.d. errors with homoskedasticity:

$$\varepsilon_i \sim i.i.d.\ (0, \sigma^2).$$

Assumption (4) implies both homoskedasticity and no-correlation of the errors.

Under (1), (2b), (3b), (4) it can be shown that:

(a) **Consistency**

$$\hat{\beta}_{\text{OLS}} \xrightarrow{p} \beta.$$

(b) **Asymptotic normality**

$$\sqrt{n}\,(\hat{\beta} - \beta) \xrightarrow{d} N(0,\ \sigma^2\Sigma_{XX}^{-1}).$$

(c) **Approximate finite-sample distribution**

$$\hat{\beta} \ \overset{a}{\sim}\ N\left(\beta,\ \frac{\sigma^2}{n}\Sigma_{XX}^{-1}\right).$$

(d) **Consistent estimator of the asymptotic variance** A consistent estimator of $\frac{\sigma^2}{n}\Sigma_{XX}^{-1}$ is

$$s^2\left(\sum_i x_i x_i'\right)^{-1} = s^2(X'X)^{-1}.$$

4.5.5 Asymptotic tests (homoskedastic case)

(e) For the null

$$H_0 \colon \beta_k = \beta_k^*,$$

the statistic

$$t_k = \frac{\hat{\beta}_k - \beta_k^*}{\sqrt{\left(s^2 (X'X)^{-1}\right)_{kk}}} \xrightarrow{d} N(0,1).$$

(f) For the null

$$H_0 \colon R\beta = r,$$

the Wald statistic

$$w = (R\hat{\beta} - r)' \left[s^2 R (X'X)^{-1} R' \right]^{-1} (R\hat{\beta} - r) \xrightarrow{d} \chi^2(J).$$

4.5.6 Large-Sample Properties under Heteroskedasticity

Now consider the more general case where homoskedasticity is not imposed.
Assume:

(1) Linearity:

$$y = X\beta + \varepsilon.$$

(2b) Orthogonality (per observation):

$$E[x_i \varepsilon_i] = 0 \quad \text{for each } i.$$

(3b) No multicollinearity in the limit:

$$\operatorname{plim} \left(\frac{1}{n} X'X \right) = E[x_i x_i'] = \Sigma_{XX},$$

with Σ_{XX} finite and non-singular.

(4c) Zero mean and no serial correlation of errors:

$$E[\varepsilon_i] = 0, \qquad E[\varepsilon_i \varepsilon_j] = 0 \quad \text{for } i \neq j.$$

Assumption (4c) allows for heteroskedasticity (variance may depend on i).

Define

$$\Sigma = E[\varepsilon_i^2 x_i x_i'].$$

Under (1), (2b), (3b), (4c):

(aa) **Consistency**

$$\hat{\beta}_{\text{OLS}} \xrightarrow{p} \beta.$$

(bb) **Asymptotic normality**

$$\sqrt{n}\,(\hat{\beta} - \beta) \xrightarrow{d} N\big(0,\ \Sigma_{XX}^{-1} \Sigma \Sigma_{XX}^{-1}\big).$$

(cc) **Approximate finite-sample distribution**

$$\hat{\beta} \overset{a}{\sim} N\left(\beta,\ \frac{1}{n} \Sigma_{XX}^{-1} \Sigma \Sigma_{XX}^{-1} \right).$$

4.5.7 Heteroskedasticity-robust variance estimator

Let

$$S_{XX} = \frac{1}{n} \sum_i x_i x_i', \qquad S = \frac{1}{n} \sum_i e_i^2 x_i x_i',$$

with e_i the OLS residuals. A consistent estimator for $\frac{1}{n} \Sigma_{XX}^{-1} \Sigma \Sigma_{XX}^{-1}$ is

$$\left(\sum_i x_i x_i' \right)^{-1} \left(\sum_i e_i^2 x_i x_i' \right) \left(\sum_i x_i x_i' \right)^{-1}.$$

(dd) The matrix

$$\widehat{\text{Var}}(\hat{\beta}) = \left(\sum_i x_i x_i' \right)^{-1} \left(\sum_i e_i^2 x_i x_i' \right) \left(\sum_i x_i x_i' \right)^{-1}$$

is the heteroskedasticity-consistent (White) variance estimator.

4.5.8 Asymptotic tests (heteroskedastic-robust)

(ee) For the null

$$H_0\colon \beta_k = \beta_k^*,$$

define

$$t_k = \frac{\hat{\beta}_k - \beta_k^*}{s_k}, \qquad s_k^2 = \left[\left(\sum_i x_i x_i' \right)^{-1} \left(\sum_i e_i^2 x_i x_i' \right) \left(\sum_i x_i x_i' \right)^{-1} \right]_{kk}.$$

Then

$$t_k \xrightarrow{d} N(0,1).$$

(ff) For the null

$$H_0\colon R\beta = r,$$

the robust Wald statistic

$$w = n\,(R\hat{\beta} - r)' \left[\, RS_{XX}^{-1} S S_{XX}^{-1} R' \,\right]^{-1} (R\hat{\beta} - r) \xrightarrow{d} \chi^2(J).$$

5 Part II - Endogeneity and Instrumental Variables

5.1 Set 1: Endogeneity and Instrumental Variables (IV): Introduction

5.1.1 Violations of the contemporaneous orthogonality assumption

In many applications of interest there are good (economic and/or statistical) reasons which do not allow us to impose the orthogonality condition

$$E[x_i \varepsilon_i] = 0.$$

In such a case we can no longer argue that the OLS estimator is consistent or unbiased and we therefore need to rely on alternative estimation techniques.

Some examples include:

1. the presence of a lagged dependent variable and autocorrelation in the error term;
2. measurement errors in the regressors;
3. omitted variables;
4. simultaneity (or reverse causality).

5.1.2 Orthogonality and consistency with a lagged dependent variable

Condition for OLS consistency:

$$E[x_i \varepsilon_i] = 0.$$

This ensures

$$\frac{1}{n} X'\varepsilon \xrightarrow{p} 0 \quad \Rightarrow \quad \hat{\beta}_{\text{OLS}} \xrightarrow{p} \beta.$$

Example (AR(1) model).

$$y_t = \rho y_{t-1} + u_t, \qquad E[u_t] = 0, \quad E[u_t u_s] = 0 \ (t \neq s).$$

Here the orthogonality condition is

$$E[y_{t-1} u_t] = 0.$$

If it holds, OLS is consistent for ρ.

Important. In this example strict exogeneity does *not* hold, hence OLS is not unbiased.

5.1.3 Lagged dependent variable with correlated errors

Consider the model

$$y_t = \rho y_{t-1} + u_t, \qquad u_t = \varphi u_{t-1} + e_t,$$

where $E[e_t] = 0$ and e_t is white noise.

Compute the key moment:

$$\begin{aligned} E[y_{t-1}u_t] &= E\big[y_{t-1}(\varphi u_{t-1} + e_t)\big] \\ &= \varphi E[y_{t-1}u_{t-1}] + E[y_{t-1}e_t] \end{aligned}$$

and $E[y_{t-1}e_t] = 0$ by assumption.

Since $E[y_{t-1}u_{t-1}] \neq 0$ when $\rho \neq 0$, we get

$$E[y_{t-1}u_t] \neq 0.$$

Implications:

- the regressor y_{t-1} is correlated with the current error u_t;
- the orthogonality condition $E[y_{t-1}u_t] = 0$ fails;
- therefore OLS is inconsistent for ρ;
- a different estimator is needed (e.g. Maximum Likelihood).

5.1.4 Measurement error

Suppose that the model of interest is

$$y_i = \beta_1 + \beta_2 w_i + v_i, \tag{2}$$

with

$$E(v_i \mid w_i) = 0, \qquad Var(v_i \mid w_i) = \sigma_v^2.$$

Suppose also that the "true" explanatory variable w_i cannot be measured accurately and denote the "measured" explanatory variable by x_i.

5.1.5 Assumptions on the measurement error

By construction, the measured variable x_i is equal to the sum of the true variable w_i and a measurement error u_i:

$$x_i = w_i + u_i. \tag{3}$$

Assume:

- $E(u_i \mid w_i) = 0$: measurement error is mean independent of w_i;
- $E(u_i^2 \mid w_i) = \sigma_u^2$: measurement error has constant conditional variance;
- $E(u_i \mid v_i) = 0$: measurement error is mean independent of v_i.

5.1.6 Substitution and composite error

Replacing $w_i = x_i - u_i$ from (3) into (2):

$$y_i = \beta_1 + \beta_2(x_i - u_i) + v_i,$$

so

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i, \qquad \varepsilon_i = v_i - \beta_2 u_i.$$

From this it follows immediately that

$$E(x_i \varepsilon_i) \neq 0.$$

Is x_i independent of $\varepsilon_i = v_i - \beta_2 u_i$? No, because $x_i = w_i + u_i$ and the measurement error u_i enters ε_i.

5.1.7 Illustrating the inconsistency of OLS

The OLS slope estimator in the regression of y_i on x_i is

$$\hat{b}_2 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}.$$

We can rewrite it as

$$\hat{b}_2 = \beta_2 + \frac{\sum_i (x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon})}{\sum_i (x_i - \bar{x})^2},$$

where we have used

$$y_i - \bar{y} = \beta_2(x_i - \bar{x}) + (\varepsilon_i - \bar{\varepsilon}).$$

5.1.8 Large-sample limit (LLN)

As the sample size N increases to infinity, sample moments converge to their corresponding population moments (Law of Large Numbers). Thus we can write

$$\text{plim}(\hat{b}_2) = \beta_2 + \frac{\text{plim } \frac{1}{N}\sum_i (x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon})}{\text{plim } \frac{1}{N}\sum_i (x_i - \bar{x})^2}. \tag{4}$$

This becomes

$$\text{plim}(\hat{b}_2) = \beta_2 + \frac{E(x_i \varepsilon_i)}{Var(x_i)}.$$

The OLS estimator $\hat{b}_2$ is therefore biased towards zero; this is often called *attenuation bias*.

5.1.9 Cross-moment and variance terms

The second term in (4) is not equal to zero. In fact:

$$\begin{aligned} E(x_i \varepsilon_i) &= E\big[(w_i + u_i)(v_i - \beta_2 u_i)\big] \\ &= -\beta_2 \sigma_u^2, \end{aligned}$$

and

$$Var(x_i) = Var(w_i + u_i) = \sigma_w^2 + \sigma_u^2,$$

where $Var(w_i) = \sigma_w^2$.

5.1.10 Asymptotic bias

Hence we can rewrite $\text{plim}(\hat{b}_2)$ as:

$$\text{plim}(\hat{b}_2) = \beta_2 - \beta_2 \frac{\sigma_u^2}{\sigma_w^2 + \sigma_u^2}.$$

The OLS estimator of β_2 is inconsistent if $\sigma_u^2 > 0$, and its asymptotic bias is an increasing function of the "noise-to-signal" ratio σ_u^2/σ_w^2.

Exercise. What happens if y is measured with error (so that we observe $y = y^* + u$)?

5.1.11 Endogeneity and omitted variable bias

We have already seen that the OLS estimator is biased if a relevant variable is omitted from the model.

Omitted variable bias also arises if there are unobservable omitted factors in the model that are correlated with one or more explanatory variables.

Example (wage equation).

$$y_i = x_{1i}' \beta_1 + x_{2i} \beta_2 + \mu_i \gamma + v_i,$$

where:

- y_i is individual log wage,
- x_{1i} is a vector of individual characteristics,
- x_{2i} is years of schooling,
- μ_i is individual ability.

It is likely that individual ability μ_i correlates with years of schooling x_{2i}. However, individual ability is typically not observed.

The econometrician then estimates

$$y_i = x_{1i}' \beta_1 + x_{2i} \beta_2 + \varepsilon_i,$$

which can be written in compact form as

$$y_i = x_i' \beta + \varepsilon_i, \qquad \varepsilon_i = \mu_i \gamma + v_i.$$

The error term in the estimated regression, ε_i, is correlated with years of schooling x_{2i}. Hence the OLS estimator for β is biased and inconsistent.

This bias is of particular concern if we wish to interpret the model coefficients as causal effects of the included variables.

5.1.12 Simultaneity and reverse causality

Another form of the endogeneity problem is *reverse causality*. This refers to the possibility that not only x_i has an impact on y_i, but at the same time y_i has an impact on x_i.

A situation of reverse causality naturally emerges when y and x are simultaneously determined.

5.1.13 Wright (1928): Demand and supply for butter

Historical context

- Philip G. Wright (1928), *The Tariff on Animal and Vegetable Oils*, provides one of the first clear discussions of the problem of simultaneous causation between price and quantity.
- Wright noticed that estimating a demand or supply curve by OLS was biased, because both P and Q are jointly determined by market equilibrium.

The butter market Define:

$$Q^d = \ln(\text{demanded quantity}), \quad Q^s = \ln(\text{supplied quantity}), \quad P = \ln(\text{price}).$$

Demand:

$$Q^d = \beta_0 + \beta_1 P + u_d$$

Supply:

$$Q^s = \gamma_0 + \gamma_1 P + u_s$$

Equilibrium: $Q^d = Q^s = Q$, jointly determining P and Q.

In equilibrium, both P and Q depend on the demand and supply shocks u_d and u_s.

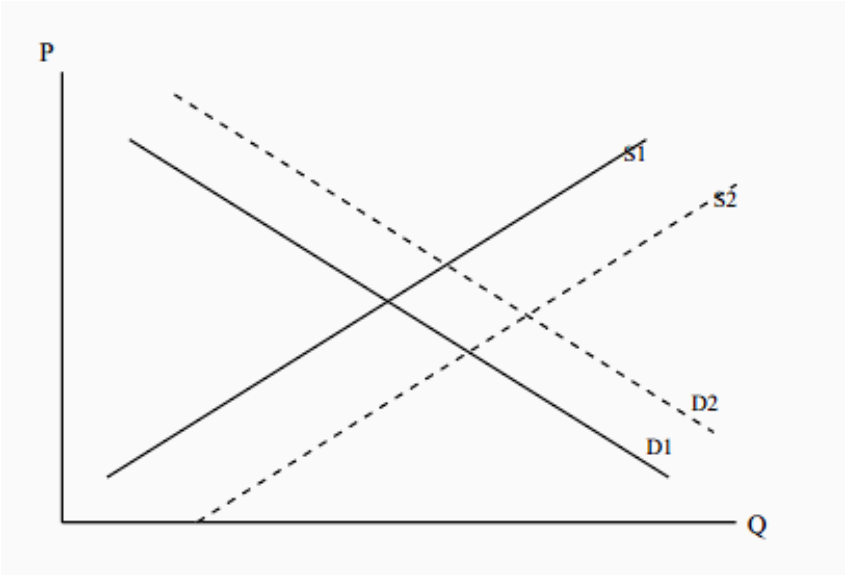


Figure 1: Graphical intuition for Wright's instrumental variable idea.

Wright's instrumental variable idea **Key insight.** To identify one relation (say, demand), we need a variable that shifts the other relation (supply) but not demand.

- Wright's instrument: weather and harvest conditions (affecting butter production costs) as instruments.
- These variables affect supply, but not consumers' tastes or demand.

Example. Consider rainfall and construct a variable that captures movements in the price due to rainfall (supply factors) alone:

$$P = \alpha_0 + \alpha_1 \,\text{Rainfall} + u_p.$$

Use the fitted value $\hat{P}$ as a regressor in the demand equation:

$$Q = \beta_0 + \beta_1 \hat{P} + u_d,$$

where $\hat{P}$ is the part of (log) price that is explained by rainfall. Then β_1 identifies the elasticity of demand.

5.2 Set 2: Endogeneity and Instrumental Variables (IV) Estimator

5.2.1 Why OLS Can Be Inconsistent

OLS relies on the orthogonality condition

$$E[x_i \varepsilon_i] = 0.$$

If this fails, OLS no longer isolates the causal effect of x_i on y_i.

Reasons why this condition can fail:

- **Omitted variable bias**: an unobserved factor affects both x_i and y_i.
- **Simultaneity**: x_i and y_i influence each other (e.g. price and quantity).
- **Measurement error**: x_i is measured with noise correlated with ε_i.
- More generally, the explanatory variable is correlated with unobserved shocks that also affect y_i.

In all these cases OLS attributes spurious correlation to x_i.

5.2.2 OLS: Causality vs. Prediction (a digression)

- If
$$E[\varepsilon_i \mid X_i] = 0,$$
then $\hat{\beta}_{OLS}$ is unbiased and consistent, and β measures the *causal* effect of X on Y.
- If instead $E[\varepsilon_i \mid X_i] \neq 0$:
  - OLS is biased and inconsistent, so no causal interpretation is valid.
  - But OLS still minimizes
$$E\big[(Y - X'\beta)^2\big].$$

Implication:

- OLS remains the **Best Linear Predictor** (BLP) of Y from X.
- It is useful for prediction and descriptive analysis.
- It is not valid for policy evaluation or counterfactual analysis when exogeneity fails.

5.2.3 Causality vs. prediction

	Causality	**Prediction**
Requires $E[\varepsilon \mid X] = 0$	yes	no
OLS unbiased/consistent?	yes	not required
Interpretation	effect of X on Y	forecast of Y given X
Criterion	exogeneity	mean squared error

5.2.4 Why OLS Still Works for Prediction

Even if $\mathrm{Cov}(X, \varepsilon) \neq 0$, OLS still finds the line

$$\min_{\beta} E\big[(Y - X'\beta)^2\big].$$

- OLS is the **best linear fit** for the observed data.
- It predicts Y from X with the smallest possible mean squared error (in-sample).
- However, the slope $\hat{\beta}_{OLS}$ mixes causal and spurious correlation.

Use OLS when:

- the goal is to forecast Y given X;
- the relationship between Y and X is stable over time.

5.2.5 The IV estimator

If one or more explanatory variables are endogenous, so that (5) fails, the OLS estimator is biased and inconsistent.

OLS orthogonality condition:

$$E[x_i \varepsilon_i] = 0. \tag{5}$$

Suppose, for simplicity, that x_i includes only two explanatory variables. Then (5) can be written as:

$$E[x_{1i} \varepsilon_i] = 0, \qquad E[x_{2i} \varepsilon_i] = 0. \tag{6}$$

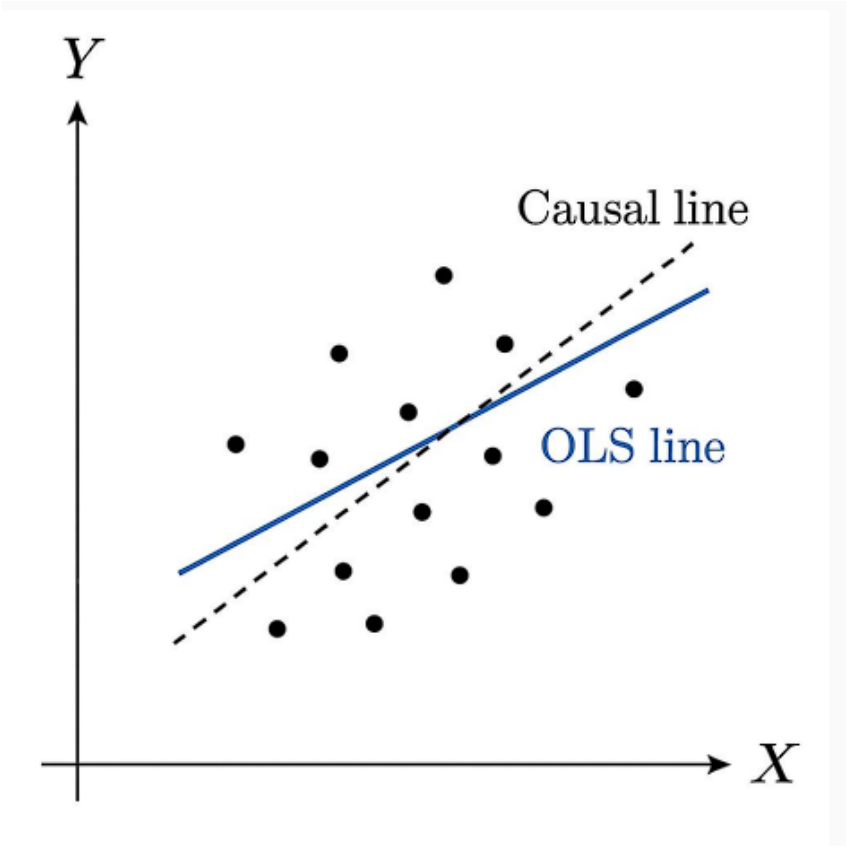


Figure 2: OLS fits data well for prediction, but the slope need not be causal.

These are conditions on the mixed second moments of x_i and ε_i.

Equivalently,

$$E\left[x_{1i}(y_i - \beta_1 x_{1i} - \beta_2 x_{2i})\right] = 0, \tag{7}$$
$$E\left[x_{2i}(y_i - \beta_1 x_{1i} - \beta_2 x_{2i})\right] = 0. \tag{8}$$

5.2.6 General rule (identification via moment conditions)

A model is statistically identified if the number of moment conditions is greater or equal to the number of parameters to be estimated. If a model is not statistically identified any estimator is necessarily inconsistent.

5.2.7 OLS as solution to sample moment conditions

The OLS estimator can be interpreted as the solution of the system where the population moment conditions in (7)–(8) are imposed on the corresponding sample moments:

$$\frac{1}{n}\sum_i x_{1i}(y_i - b_{1,OLS}x_{1i} - b_{2,OLS}x_{2i}) = 0, \tag{9}$$
$$\frac{1}{n}\sum_i x_{2i}(y_i - b_{1,OLS}x_{1i} - b_{2,OLS}x_{2i}) = 0. \tag{10}$$

These are equivalent to the first-order conditions from minimizing the sum of squared residuals.

5.2.8 Instrumental variable and new moment condition

Suppose now that the condition for x_{2i} in (6) is violated. The model is no longer statistically identified.

To identify the model, we need at least one additional moment condition, typically derived from an *instrumental variable*.

An **instrumental variable** z_i is a variable that:

- is correlated with the endogenous regressor x_{2i};

- is not correlated with the error term ε_i.

If such a variable exists, we can replace the problematic condition with

$$E\left[z_i(y_i - \beta_1 x_{1i} - \beta_2 x_{2i})\right] = 0, \tag{11}$$

which, together with (7), allows statistical identification of the model.

The IV estimator $(b_{1,IV}, b_{2,IV})$ is then defined by the sample counterparts:

$$\frac{1}{n}\sum_i x_{1i}(y_i - b_{1,IV}x_{1i} - b_{2,IV}x_{2i}) = 0, \tag{12}$$

$$\frac{1}{n}\sum_i z_i(y_i - b_{1,IV}x_{1i} - b_{2,IV}x_{2i}) = 0. \tag{13}$$

5.2.9 Matrix notation for IV

Let:

- X be the matrix containing all (endogenous and exogenous) explanatory variables (here x_1 and x_2);
- Z be the matrix including all instruments: exogenous regressors and external instruments (here x_1 and z).

The first-order conditions can be written as:

$$Z'y - Z'Xb_{IV} = 0.$$

The general solution is

$$b_{IV} = (Z'X)^{-1}Z'y = \left(\frac{1}{n}\sum_i z_i x_i'\right)^{-1}\left(\frac{1}{n}\sum_i z_i y_i\right), \tag{14}$$

where x_i is the vector of explanatory variables and z_i is the vector of instruments.
In the simple IV case, the dimensions of X and Z coincide (one instrument per endogenous variable).

5.2.10 Properties of the IV estimator

Assume:

1. Observed data are generated by the linear model

$$y = X\beta + \varepsilon. \tag{15}$$

2. Error terms and instruments are independent for each i:

$$\varepsilon_i \text{ and } z_i \text{ are independent (for each } i). \tag{16}$$

   This implies $E[z_i\varepsilon_i] = 0$.

Invertibility (relevance and nonsingularity).

$$\operatorname{plim}\left(\frac{1}{n}Z'X\right) = E[z_i x_i'] = \Sigma_{ZX}, \tag{17}$$

$$\operatorname{plim}\left(\frac{1}{n}Z'Z\right) = E[z_i z_i'] = \Sigma_{ZZ}, \tag{18}$$

where Σ_{ZX} and Σ_{ZZ} are both finite and invertible matrices.
A necessary condition for (17) is that the instrument z_i is correlated with x_{2i} and not a linear function of x_{1i}. In this case we say that the instrument is *relevant*.

Assume also:

3. Error terms are identically and independently distributed with mean 0 and variance σ^2:

$$\varepsilon_i \sim iid(0, \sigma^2). \tag{19}$$

   This implies both homoskedasticity and no-correlation.

5.2.11 Asymptotic distribution

Under the above assumptions it can be shown that:

(a) **Consistency**

$$b_{IV} \xrightarrow{p} \beta. \tag{20}$$

(b) **Asymptotic normality**

$$\sqrt{n}\,(b_{IV} - \beta) \xrightarrow{d} N\Big(0,\ \sigma^2(\Sigma_{XZ}\Sigma_{ZZ}^{-1}\Sigma_{ZX})^{-1}\Big), \tag{21}$$

where $\Sigma_{XZ} = \Sigma_{ZX}'$.

(c) **Approximate finite-sample distribution**

$$b_{IV} \overset{a}{\sim} N\left(\beta,\ \frac{\sigma^2}{n}(\Sigma_{XZ}\Sigma_{ZZ}^{-1}\Sigma_{ZX})^{-1}\right). \tag{22}$$

5.2.12 Estimating the asymptotic variance

A consistent estimator for $\dfrac{\sigma^2}{n}(\Sigma_{XZ}\Sigma_{ZZ}^{-1}\Sigma_{ZX})^{-1}$ is

$$\widehat{\text{Avar}}(b_{IV}) = \hat{\sigma}^2\big(S_{XZ}S_{ZZ}^{-1}S_{ZX}\big)^{-1}\frac{1}{n}, \tag{23}$$

that is

$$\begin{aligned}\widehat{\text{Avar}}(b_{IV}) &= \hat{\sigma}^2\left[\left(\frac{1}{n}\sum_i x_i z_i'\right)\left(\frac{1}{n}\sum_i z_i z_i'\right)^{-1}\left(\frac{1}{n}\sum_i z_i x_i'\right)\right]^{-1}\frac{1}{n}\\ &= \hat{\sigma}^2\left[\left(\sum_i x_i z_i'\right)\left(\sum_i z_i z_i'\right)^{-1}\left(\sum_i z_i x_i'\right)\right]^{-1},\end{aligned}$$

where

$$S_{XZ} = \frac{1}{n}\sum_i x_i z_i', \qquad S_{ZZ} = \frac{1}{n}\sum_i z_i z_i', \qquad S_{ZX} = S_{XZ}'.$$

Here

$$\hat{\sigma}^2 = \frac{\sum_i (y_i - x_i' b_{IV})^2}{n}$$

is a consistent estimator of σ^2.

5.2.13 Inference with IV

Under the null

$$H_0 : \beta_k = \beta_k^*, \tag{24}$$

the IV t-statistic

$$t_k = \frac{b_k - \beta_k^*}{\sqrt{\widehat{\text{Avar}}(b_{IV})_{kk}}} \xrightarrow{d} N(0,1), \tag{25}$$

where the denominator is the square root of the kk-th element of $\widehat{\text{Avar}}(b_{IV})$.

Under the joint null

$$H_0 : R\beta = r, \tag{26}$$

the Wald statistic

$$w = (Rb_{IV} - r)'\big[\,R\,\widehat{\text{Avar}}(b_{IV})\,R'\,\big]^{-1}(Rb_{IV} - r) \xrightarrow{d} \chi^2(J). \tag{27}$$

Summing up, in large samples the IV estimator is approximately normally distributed, so the usual inference procedures (based on the normal and chi-square distributions) can be applied.

5.3 Set 3: Generalized Method of Moments (GMM) Estimators

5.3.1 Intuition Behind GMM Estimation

Many estimators in econometrics are based on moment conditions, that is, statements that are true in the population:

$$E[m_i(\beta)] = 0,$$

where $m_i(\beta)$ is a vector of functions of the data and parameters.

The Generalized Method of Moments (GMM) chooses the parameter vector β that makes the sample moments as close to zero as possible:

$$\hat{\beta}_{GMM} = \arg\min_{\beta} \; g_n(\beta)' W_n g_n(\beta),$$

where

$$g_n(\beta) = \frac{1}{n} \sum_i m_i(\beta),$$

and W_n is a positive definite weighting matrix.

The moment conditions need not be linear in the parameters.

- In IV estimation, $m_i(\beta) = z_i(y_i - x_i'\beta)$ is linear in β.
- In nonlinear models (e.g. nonlinear IV, Probit, dynamic panels) the same principle applies with nonlinear $m_i(\beta)$.

GMM therefore provides a unifying framework that includes OLS and IV as special cases, but also allows for nonlinear models and more instruments than parameters.

In what follows, we will illustrate the GMM estimator mainly using moment conditions that are linear in the parameters. This will allow us to show more easily how many popular estimators are, in fact, special cases of the GMM estimator.

5.3.2 The GMM estimator

Assume there are R instruments available in the vector z_i which correspond to the following R moment conditions:

$$E[z_i \varepsilon_i] = 0. \tag{28}$$

These conditions can be written as:

$$E\left[z_i(y_i - x_i'\beta)\right] = 0. \tag{29}$$

The corresponding R sample moments are:

$$g_n(\beta) = \frac{1}{n} \sum_i z_i(y_i - x_i'\beta). \tag{30}$$

5.3.3 Identification

- If $R < K$ the model is *not identified* and no consistent estimator can be found.
- If $R = K$ the model is *exactly identified* and we are back to the IV case, where β can be solved from the sample moment conditions in (30).
- If $R > K$ the model is *overidentified*. In this case it is not possible to solve for an estimate of β by simply replacing (29) with its sample counterparts, because there are more equations than unknowns.

Instead of dropping instruments, one can choose β in such a way that the R sample moments in (30) are as close to zero as possible.

5.3.4 The GMM Objective Function

This is done by minimizing the following quadratic form:

$$\min_{\beta} Q_n(\beta, W_n) = g_n(\beta)' W_n g_n(\beta), \tag{31}$$

that is:

$$Q_n(\beta, W_n) = \left(\frac{1}{n} \sum_i z_i(y_i - x_i'\beta) \right)' W_n \left(\frac{1}{n} \sum_i z_i(y_i - x_i'\beta) \right),$$

or, in matrix notation:

$$Q_n(\beta, W_n) = \left(\frac{1}{n}Z'(y - X\beta)\right)' W_n \left(\frac{1}{n}Z'(y - X\beta)\right),$$

where Z is the $n \times R$ matrix of instruments.

The weighting matrix W_n is a positive definite symmetric matrix satisfying

$$W_n \xrightarrow{p} W,$$

where W is itself positive definite and symmetric.

5.3.5 Expanding the GMM Objective Function

Using matrix notation, the GMM objective function can be written as:

$$Q_n(\beta, W_n) = \left(\frac{1}{n}Z'(y - X\beta)\right)' W_n \left(\frac{1}{n}Z'(y - X\beta)\right).$$

Expanding:

$$Q_n(\beta, W_n) = \frac{1}{n^2}\Big(y'ZW_nZ'y - y'ZW_nZ'X\beta - \beta'X'ZW_nZ'y + \beta'X'ZW_nZ'X\beta\Big),$$

which can also be written as

$$Q_n(\beta, W_n) = \frac{1}{n^2}\Big(y'ZW_nZ'y - 2y'ZW_nZ'X\beta + \beta'X'ZW_nZ'X\beta\Big).$$

5.3.6 First Order Conditions

Using the derivation rules

$$\frac{\partial a'\beta}{\partial \beta} = a, \qquad \frac{\partial \beta' A\beta}{\partial \beta} = 2A\beta,$$

the first order condition (FOC) for minimizing (31) gives:

$$\frac{\partial Q_n(\beta, W_n)}{\partial \beta} = -2X'ZW_nZ'y + 2X'ZW_nZ'X\hat{\beta}_{GMM} = 0,$$

which implies

$$X'ZW_nZ'y = X'ZW_nZ'X\hat{\beta}_{GMM}. \tag{32}$$

5.3.7 The GMM estimator formula

Equation (32) is a system with K equations and K unknowns. Provided that the matrix $X'Z$ is of rank K, the solution is given by:

$$\hat{\beta}_{GMM} = (X'ZW_nZ'X)^{-1}X'ZW_nZ'y. \tag{33}$$

5.3.8 The weighting matrix

In general, the GMM estimator $\hat{\beta}_{GMM}$ is a function of the weighting matrix W_n.

- The matrix W_n determines how much weight is given to each moment condition.
- Different weighting matrices lead to different consistent GMM estimators with generally different asymptotic covariance matrices.
- If W_n is chosen arbitrarily, the estimator is still consistent.
- However, some moment conditions are more informative (have lower variance) than others.
- To make the estimator efficient, we want to give more weight to precise moments and less to noisy ones.

5.3.9 The optimal weighting matrix

It can be shown that the *optimal* weighting matrix is proportional to the inverse of the covariance matrix of the sample moments:

$$\Sigma = E[\varepsilon_i^2 z_i z_i'], \tag{34}$$

which can be estimated consistently by:

$$S = \frac{1}{n}\sum_i \hat{\varepsilon}_i^2 z_i z_i', \tag{35}$$

where $\hat{\varepsilon}_i$ is the residual obtained in a first step where the weighting matrix W_n is set equal to the identity matrix.

5.3.10 The efficient GMM estimator

Using the optimal weighting matrix, the efficient GMM estimator is:

$$\hat{\beta}_{GMM}^{opt} = \left(X'ZS^{-1}Z'X\right)^{-1}X'ZS^{-1}Z'y. \tag{36}$$

Summarizing, using the optimal weighting matrix yields the *efficient* GMM estimator, which has the smallest asymptotic variance among all GMM estimators.

5.3.11 The GMM estimator properties: assumptions

Under the assumptions:

1. Observed data are generated by the linear model:

$$y = X\beta + \varepsilon. \tag{37}$$

2b. Error terms are orthogonal (and therefore uncorrelated) to instruments for each i:

$$E[z_i\varepsilon_i] = 0. \tag{38}$$

5.3.12 The GMM estimator properties: assumptions (cont.)

3b. Invertibility:

$$\operatorname{plim}\left(\frac{1}{n}\sum_i z_i x_i'\right) = E[z_i x_i'] = \Sigma_{ZX}, \tag{39}$$

$$\operatorname{plim}\left(\frac{1}{n}\sum_i \hat{\varepsilon}_i^2 z_i z_i'\right) = E[\varepsilon_i^2 z_i z_i'] = \Sigma, \tag{40}$$

where Σ_{ZX} and Σ are both finite and non-singular matrices.

4b. Error terms are uncorrelated with expectation zero:

$$E[\varepsilon_i] = 0, \qquad E[\varepsilon_i\varepsilon_j] = 0 \quad \text{for } i \neq j. \tag{41}$$

Assumption 4b accommodates heteroskedasticity.

5.3.13 The GMM estimator properties

It can be proved that:

(a) The GMM estimator converges in probability to the true parameter vector β:

$$\hat{\beta}_{GMM} \xrightarrow{p} \beta. \tag{42}$$

Note: in general, GMM estimators are consistent, but not unbiased in finite samples.

5.3.14 Asymptotic distribution

(bb) The GMM estimator is asymptotically distributed as:

$$\sqrt{n}\left(\hat{\beta}_{GMM} - \beta\right) \xrightarrow{d} N\Big(0,\ (\Sigma_{XZ}W\Sigma_{ZX})^{-1}\Sigma_{XZ}W\Sigma W\Sigma_{ZX}(\Sigma_{XZ}W\Sigma_{ZX})^{-1}\Big), \tag{43}$$

where $\Sigma_{XZ} = \Sigma_{ZX}'$.

(cc) The expression above simplifies to

$$\sqrt{n}\left(\hat{\beta}_{GMM}^{opt} - \beta\right) \xrightarrow{d} N\Big(0,\ (\Sigma_{XZ}\Sigma^{-1}\Sigma_{ZX})^{-1}\Big) \tag{44}$$

when the optimal weighting matrix is chosen.

5.3.15 Approximate finite-sample distribution

(cc) In finite samples it holds approximately that:

$$\hat{\beta}_{GMM} \overset{a}{\sim} N\Big(\beta,\ (\Sigma_{XZ}W\Sigma_{ZX})^{-1}\Sigma_{XZ}W\Sigma W\Sigma_{XZ}(\Sigma_{XZ}W\Sigma_{ZX})^{-1}\frac{1}{n}\Big), \tag{45}$$

and, if the weighting matrix is chosen optimally:

$$\hat{\beta}_{GMM}^{opt} \overset{a}{\sim} N\Big(\beta,\ (\Sigma_{XZ}\Sigma^{-1}\Sigma_{ZX})^{-1}\frac{1}{n}\Big). \tag{46}$$

5.3.16 Estimated covariance matrix

(dd) A consistent estimator for the covariance in (46) is given by:

$$\text{Avar}(\hat{\beta}_{GMM}) = (S_{XZ}W_nS_{ZX})^{-1}S_{XZ}W_nSW_nS_{ZX}(S_{XZ}W_nS_{ZX})^{-1}\frac{1}{n}, \tag{47}$$

which simplifies to:

$$\text{Avar}(\hat{\beta}^{opt}_{GMM}) = (S_{XZ}S^{-1}S_{ZX})^{-1}\frac{1}{n} \tag{48}$$

when the weighting matrix is chosen optimally.

Summing up, in large samples the GMM estimator is approximately normally distributed, so the usual inference procedures can be applied (using standard normal and chi-square distributed tests).

5.3.17 Special case: the IV estimator

Given the usual R moment conditions

$$E[z_i(y_i - x_i'\beta)] = 0, \tag{49}$$

the $X'Z$ matrix is square (and invertible) if $R = K$. In this case the choice of the weighting matrix becomes irrelevant. In fact, from (33):

$$\hat{\beta}_{GMM} = (X'ZW_nZ'X)^{-1}X'ZW_nZ'y \tag{50}$$

$$= (Z'X)^{-1}W_n^{-1}(X'Z)^{-1}X'ZW_nZ'y \tag{51}$$

$$= (Z'X)^{-1}Z'y = \hat{\beta}_{IV}. \tag{52}$$

The expression above makes it clear that the IV estimator $\hat{\beta}_{IV}$ can be seen as a special case of GMM estimation.

5.3.18 Special case: the OLS estimator

For consistency, OLS estimation requires:

$$E[x_i\varepsilon_i] = 0, \tag{53}$$

which can be rewritten as:

$$E[x_i(y_i - x_i'\beta)] = 0. \tag{54}$$

In this case, the GMM estimator becomes:

$$\hat{\beta}_{GMM} = (X'X)^{-1}W_n^{-1}(X'X)^{-1}X'XW_nX'y \tag{55}$$

$$= (X'X)^{-1}X'y = \hat{\beta}_{OLS}. \tag{56}$$

5.3.19 Special case: The Generalized IV estimator (GIVE)

If we assume homoskedastic error terms:

$$\varepsilon_i \sim iid(0, \sigma^2), \tag{57}$$

in addition to independence between ε_i and z_i for each i, then the asymptotic variance of the sample moments simplifies to:

$$\Sigma = \sigma^2\Sigma_{ZZ}, \tag{58}$$

which implies the optimal weighting matrix:

$$W_n^{opt} = \left(s^2\frac{1}{n}Z'Z\right)^{-1}, \tag{59}$$

where s^2 is a consistent estimator of σ^2.

Finally, we can write the efficient GMM estimator (from (36)) as:

$$\hat{\beta}_{GMM} = \left(X'Z(Z'Z)^{-1}Z'X\right)^{-1}X'Z(Z'Z)^{-1}Z'y = \hat{\beta}_{GIVE}.$$

$\hat{\beta}_{GIVE}$ is the Generalized Instrumental Variable Estimator. Its formula involves terms like

$$P_Z = Z(Z'Z)^{-1}Z',$$

which suggest an alternative (and equivalent) way to compute it.

5.3.20 The 2SLS (Two-Stage Least Squares) estimator

The GIVE estimator can also be obtained in two steps, both of which can be estimated by ordinary least squares.

- In the first step the reduced form is estimated by OLS. In practice, each variable in the x_i vector is regressed against all variables in the z_i vector.

The fitted values for a generic regressor x_k can be written as:

$$\hat{x}_k = Z(Z'Z)^{-1}Z'x_k = P_Z x_k,$$

and for the full set of explanatory variables X:

$$\hat{X} = Z(Z'Z)^{-1}Z'X = P_Z X.$$

- In the second step the original structural equation is estimated by OLS, replacing all endogenous variables on the right-hand side with their predicted values from the reduced form.

The OLS estimator in the second step is:

$$\hat{\beta}_{2SLS} = (\hat{X}'\hat{X})^{-1}\hat{X}'y = \left(X'Z(Z'Z)^{-1}Z'X\right)^{-1}X'Z(Z'Z)^{-1}Z'y = \hat{\beta}_{GIVE}. \tag{60}$$

5.3.21 Specification tests: the Sargan–Hansen test

If $R = K$, the following statistic:

$$\begin{aligned}\xi &= n\left(\frac{1}{n}\sum_i z_i(y_i - x_i'\hat{\beta}_{GMM}^{opt})\right)'\left(\frac{1}{n}\sum_i \hat{\varepsilon}_i^2 z_i z_i'\right)^{-1}\left(\frac{1}{n}\sum_i z_i(y_i - x_i'\hat{\beta}_{GMM}^{opt})\right) \\ &= n\left(\frac{1}{n}\sum_i z_i\hat{\varepsilon}_i\right)'\left(\frac{1}{n}\sum_i \hat{\varepsilon}_i^2 z_i z_i'\right)^{-1}\left(\frac{1}{n}\sum_i z_i\hat{\varepsilon}_i\right) \\ &= \left(\sum_i z_i\hat{\varepsilon}_i\right)'\left(\sum_i \hat{\varepsilon}_i^2 z_i z_i'\right)^{-1}\left(\sum_i z_i\hat{\varepsilon}_i\right)\end{aligned} \tag{61}$$

is equal to zero, since $\hat{\beta}_{GMM}^{opt}$ is the solution to the minimization problem in (31) when $R = K$.

If $R > K$, the statistic in (61) will in general be positive.
Under the null of instrument validity, it should be small since

$$\text{plim}\left(\frac{1}{n}\sum_i z_i\hat{\varepsilon}_i\right) = 0,$$

and therefore we can expect all elements in $\frac{1}{n}\sum_i z_i\hat{\varepsilon}_i$ to be close to zero.

It can be proved that under the null:

$$\xi \xrightarrow{d} \chi^2(R - K). \tag{62}$$

The associated test is commonly known as the **Sargan–Hansen test** or, alternatively, as the *overidentifying restrictions test*.

5.3.22 Specification issues: Weak Instruments

If the instrument(s) exhibit only weak correlation with the endogenous regressor(s), the properties of the IV estimator can be very poor.

In these cases, the normal distribution provides a very poor approximation of the true distribution of the IV estimator, even when the sample size is large. As a result:

- the IV estimator is biased towards the OLS estimator;
- its standard errors are misleading;
- hypothesis tests are unreliable.

5.3.23 Weak Instruments: simple IV example

Consider the case of a single regressor and a single instrument:

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i. \tag{63}$$

The IV estimator can be derived starting from:

$$\begin{aligned} \mathrm{Cov}(y_i, z_i) &= \mathrm{Cov}(\beta_1 + \beta_2 x_i + \varepsilon_i, z_i) && (64)\\ &= \mathrm{Cov}(\beta_1, z_i) + \beta_2 \mathrm{Cov}(x_i, z_i) + \mathrm{Cov}(\varepsilon_i, z_i) && (65)\\ &= \beta_2 \mathrm{Cov}(x_i, z_i), && (66) \end{aligned}$$

which gives

$$\beta_2 = \frac{\mathrm{Cov}(y_i, z_i)}{\mathrm{Cov}(x_i, z_i)}. \tag{67}$$

The IV estimator is obtained by replacing the population covariances with their sample counterparts:

$$\hat{\beta}_{2,IV} = \frac{\frac{1}{n}\sum_i (y_i - \bar{y})(z_i - \bar{z})}{\frac{1}{n}\sum_i (x_i - \bar{x})(z_i - \bar{z})}. \tag{68}$$

If the instrument is not correlated at all with the regressor in the population, then the denominator in (68) is equal to zero. In this case the IV estimator is inconsistent and the asymptotic distribution of $\hat{\beta}_{2,IV}$ deviates substantially from a normal distribution.

If the instrument is weak, the asymptotic normal distribution does not provide a good approximation in finite (even if very large) samples.

To test whether there is a weak instruments problem, it is useful to estimate the first stage regression and evaluate the explanatory power of the additional instruments that are not included in the equation of interest.

5.3.24 Weak Instruments: first-stage regression

Consider, for instance, the linear model with one endogenous regressor, x_{2i}:

$$y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i, \tag{69}$$

with

$$E(x_{1i}\varepsilon_i) = 0, \qquad E(z_i\varepsilon_i) = 0.$$

The appropriate first-stage equation is:

$$x_{2i} = \pi_1 x_{1i} + \pi_2 z_i + v_i. \tag{70}$$

If $\pi_2 = 0$, the instrument z is irrelevant. If π_2 is close to zero, the instrument is weak. In both cases the IV estimator is biased even when the sample is large.

5.3.25 Weak Instruments: first-stage F-statistic

The first-stage F-statistic tests the hypothesis that the instruments do not enter the first-stage regression.

Weak instruments imply a small first-stage F-statistic.

Rule-of-thumb:

- If the first-stage F-statistic is less than 10, then the set of instruments is weak.
- A first-stage $F > 10$ implies that the bias of IV, relative to OLS, is less than 10%.
- If $F < 10$, the relative bias exceeds 10%.

5.3.26 Weak Instruments: why this rule-of-thumb?

Starting from

$$\text{plim}(\hat{\beta}_{2,OLS}) = \beta_{2,OLS} \neq \beta_2, \tag{71}$$

it can be shown that the bias of the IV estimator is approximately equal to:

$$E(\hat{\beta}_{2,IV}) - \beta_2 = \frac{\hat{\beta}_{2,OLS} - \beta_2}{E(F) - 1}, \tag{72}$$

where F is the first-stage F-statistic.

If $E(F) = 10$, then

$$\frac{E(\hat{\beta}_{2,IV}) - \beta_2}{\hat{\beta}_{2,OLS} - \beta_2} \approx \frac{1}{9}, \tag{73}$$

which is considered acceptable in applications.

5.4 Set 4: Instrumental Variables – Examples

5.4.1 Acemoglu, Johnson and Robinson (2001)

The Royal Swedish Academy of Sciences awarded the 2024 Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel to Daron Acemoglu, Simon Johnson, and James A. Robinson "for studies of how institutions are formed and affect prosperity."

The seminal contribution in this field is the widely cited instrumental variable analysis by Acemoglu, Johnson, and Robinson (2001, henceforth AJR 2001), with further details published in 2012. In these papers, the authors study the effect of political institutions on economic performance.

The theory is that good institutions — such as rule of law and secure property rights — should result in higher long-term economic output compared to countries with weak institutions. To investigate this question, they focus on a sample of 64 former European colonies.[1]

The authors argue that modern political institutions were shaped by colonization. In particular, they distinguish between:

- *Extractive states*: established mainly to extract resources for the colonizer and not largely settled by Europeans. In these colonies, there was little incentive to set up good political institutions.
- *Settler colonies*: attracted large numbers of European migrants who sought to replicate institutions from their home countries, thereby creating incentives to establish strong political institutions.

Because the nature of institutions is highly persistent over time, these 19th-century foundations continue to shape modern institutional quality. Hence, the colonial legacy is predictive of both institutional development and modern economic performance.

To begin their analysis, AJR (2001) report an OLS regression of log GDP per capita in 1995 on a measure of political institutions called *risk*, which captures legal protection against expropriation (interpreted as "protection against expropriation risk" — the higher the value, the stronger the protection). This variable ranges from 0 to 10, with 0 representing the lowest protection and 10 the highest. For each country, they compute the average index value over 1985–1995 (mean 6.5, standard deviation 1.5). Their OLS estimates are:

$$\log(\text{GDP per capita}) = 4.69 + 0.52 \times \text{risk}, \tag{74}$$

with standard errors (in parentheses) equal to (0.42) and (0.06), respectively.

Interpretation.

Equation (74) is a log–level specification: the dependent variable is in logarithms, while the regressor (*risk*) is in levels. The coefficient 0.52 therefore measures the proportional change in GDP per capita associated with a one-unit change in the risk index.

- For a one-unit *increase* in risk, the predicted change in GDP per capita is

$$\%\Delta\text{GDP} = \big[\exp(0.52) - 1\big] \times 100 \approx +68.2\%.$$

- For a one-unit *decrease* in risk, the predicted change is

$$\%\Delta\text{GDP} = \big[\exp(-0.52) - 1\big] \times 100 \approx -40.6\%.$$

[1] Their paper, dataset and replication do-file are in the folder *Acemoglu, Johnson and Robinson (2001)* on the data section of the course website.

If we used the linear approximation $\%\Delta\text{GDP} \approx 100 \times 0.52 = 52\%$, the result would lie between the two exact values above. This is conceptually similar to an *arc elasticity* in economics, which measures a symmetric percentage change between two points.

Hence, a one-unit improvement in the protection against expropriation (*risk*) is associated with approximately a 68% higher GDP per capita, while a one-unit deterioration is associated with roughly a 41% lower GDP per capita.

The authors argue that the risk variable is endogenous, both because economic output may influence political institutions and because *risk* is likely measured with error. These issues induce least-squares bias in different directions, so the overall bias is ambiguous.

To address endogeneity, they seek an instrumental variable correlated with political institutions but not directly with economic performance. Their innovative suggestion was to use the mortality rate faced by potential European settlers in the 19th century. Colonies with high expected mortality were less attractive to settlers and therefore more likely to develop extractive institutions. Mortality rates are measured as annual deaths per 1,000 soldiers ("`logmort0`" in the dataset). The first-stage regression is:

$$\text{risk} = 9.37 - 0.61 \times \text{logmort0}, \tag{75}$$

with standard errors (0.61) and (0.13), respectively.

These estimates confirm that high historical mortality rates are associated with lower-quality modern institutions.

Using log(mortality) as an instrument for *risk*, they estimate the structural equation via 2SLS:

$$\log(\text{GDP per capita}) = 1.99 + 0.93 \times \text{risk}, \tag{76}$$

with standard errors (1.15) and (0.17), respectively.

This 2SLS coefficient is much higher than the OLS estimate, suggesting that a one-unit increase in the risk index corresponds to roughly a 150% higher GDP per capita.

These are simple regressions with only one explanatory variable. The authors also estimated more general models. Notably, traditional OLS regressions had found latitude and an Africa dummy to be significant predictors of output. However, once institutions were instrumented, these variables became statistically insignificant, indicating that their correlation with income operated through institutions.

To evaluate the specification, they apply the Stock–Yogo and endogeneity tests. From the reduced form (equation (75)), the instrument has a t-statistic of 4.8 (or $F = 23$), exceeding the Stock–Yogo critical value; hence, it is a strong instrument.

5.4.2 Card (1995)

In 2021, the Royal Swedish Academy of Sciences awarded the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel to David Card (who received one half of the prize) and Joshua D. Angrist and Guido W. Imbens (who shared the other half).

You will encounter many works by Card, Angrist, and Imbens in econometrics and labor economics courses, as their contributions have been hugely influential.

Among other topics, David Card studied the causal relationship between education and earnings, showing that school resources play a much larger role in students' future labor market success than previously thought.

To investigate the causal effect of education on earnings, Card (1995) proposed that *proximity to a college* reduces the cost of attendance and increases the likelihood of enrollment. College proximity does not directly affect a student's abilities, and thus should not have a direct effect on wages. These arguments suggest that college proximity can serve as an instrument for education in a wage regression.[2]

Card used data from the 1976 National Longitudinal Survey of Young Men (NLSYM). The dependent variable is the log of weekly earnings; regressors include:

- education (years of schooling),
- experience (years of work experience, computed as age minus education plus 6),
- experience squared divided by 100,
- dummies for being Black, residing in the South, and living in an urban area (SMSA).

Observations with missing wages were excluded, leaving 3,010 observations. The data are available in the file `Card1995` on the course website.

The OLS regression suggests that each additional year of education is associated with a 7% increase in earnings.

However, education is an individual choice likely correlated with unobserved characteristics (e.g. ability) that also affect earnings. Hence, the OLS estimate reflects a linear projection but is inconsistent for the structural parameter representing the causal effect of education on wages. Because ability, education, and wages are expected to be

[2]His paper, dataset and replication do-file are in the folder *Card 1995* on the data section of the course website.

positively correlated, OLS likely overstates the causal effect (upward bias). However, with multiple regressors and other endogeneity sources, the bias direction is not guaranteed.

To instrument for education, Card used indicators for whether an individual grew up near a college. We consider three measures from his dataset:

- **college**: grew up in the same county as a four-year college (used in column 2);
- **public**: grew up in the same county as a four-year public college (used in columns 3 and 4);
- **private**: grew up in the same county as a four-year private college (used in columns 3 and 4).

5.4.3 Table: Returns to education

The table below reports estimates of returns to education under different estimators (OLS, IV, 2SLS, GMM). Standard errors are reported in parentheses; $*$, $**$, and $***$ denote significance at the 10%, 5%, and 1% levels, respectively.

Table 1: Returns to education

	(1) OLS	(2) IV	(3) 2SLS	(4) GMM
Years of education	0.0740***	0.132***	0.161***	0.162***
	(0.00364)	(0.0485)	(0.0405)	(0.0405)
Experience	0.0836***	0.107***	0.119***	0.120***
	(0.00673)	(0.0211)	(0.0182)	(0.0182)
Squared Experience	-0.224***	-0.228***	-0.231***	-0.232***
	(0.0318)	(0.0346)	(0.0368)	(0.0368)
Black	-0.190***	-0.131**	-0.102**	-0.101**
	(0.0174)	(0.0515)	(0.0440)	(0.0440)
South	-0.125***	-0.105***	-0.0950***	-0.0954***
	(0.0154)	(0.0229)	(0.0217)	(0.0218)
Urban	0.161***	0.131***	0.116***	0.115***
	(0.0152)	(0.0298)	(0.0263)	(0.0263)
Constant	4.734***	3.753***	3.268***	3.262***
	(0.0702)	(0.817)	(0.682)	(0.683)
N	3010	3010	3010	3010
R^2	0.291	0.225	0.145	0.143

Standard errors in parentheses.

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6 Part III - Maximum Likelihood (ML)

6.1 Set 1: Maximum Likelihood (ML) Estimation and Specification Tests

6.1.1 Introduction

The ML estimation methodology is grounded on the assumption that the (conditional) distribution of an observed phenomenon (the endogenous variable) is known up to a finite number of unknown parameters.

These unknown parameters are estimated by taking those values that give the observed data the highest probability (likelihood) of being drawn, given the assumed conditional distribution.

6.1.2 Example 1: discrete setting

Consider a large pool filled with black and white balls. We are interested in the fraction of white balls, p, in this pool.

To obtain information on p we extract a random sample of n balls. Let us denote

$$y_i = \begin{cases} 1 & \text{if ball } i \text{ is white,} \\ 0 & \text{otherwise.} \end{cases}$$

By assumption,

$$\Pr(y_i = 1) = p. \tag{77}$$

Suppose our sample contains $n_1 = \sum_i y_i$ white balls and $n - n_1$ black balls. The probability of obtaining such a sample (in a given order) is

$$L(p) = p^{n_1}(1-p)^{n-n_1}. \tag{78}$$

The expression in (78), viewed as a function of the unknown parameter p, is the *likelihood function*. The maximum likelihood estimator of p chooses the value of p that maximizes (78), i.e. the probability of drawing the observed sample.

For computational reasons it is often more convenient to maximize the natural logarithm of the likelihood, which is a monotonic transformation. This is the *log-likelihood* function:

$$\log L(p) = n_1 \log p + (n - n_1) \log(1 - p). \tag{79}$$

Maximizing (79) with respect to p gives the first-order condition:

$$\frac{d \log L(p)}{dp} = \frac{n_1}{p} - \frac{n - n_1}{1 - p} = 0, \tag{80}$$

which, solving for p, yields the ML estimator

$$\hat{p}_{ML} = \frac{n_1}{n}. \tag{81}$$

To check that this solution corresponds to a maximum, we look at the second derivative:

$$\frac{d^2 \log L(p)}{dp^2} = -\frac{n_1}{p^2} - \frac{n - n_1}{(1 - p)^2} < 0, \tag{82}$$

which confirms that $L(\hat{p}_{ML})$ is a maximum.

6.1.3 Example 2: continuous setting

Consider a bivariate classical linear regression model with normally distributed errors:

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i, \tag{83}$$

with

$$\varepsilon_i \mid x \sim NID(0, \sigma^2), \tag{84}$$

where NID stands for *normally* (N) and *independently* (I) *distributed* (D) errors.

Given these assumptions,

$$y_i \mid x \sim NID(\beta_1 + \beta_2 x_i, \sigma^2). \tag{85}$$

Therefore, the contribution of observation i to the likelihood is the value of the normal density at y_i:

$$f(y_i \mid x_i; \beta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2} \frac{(y_i - \beta_1 - \beta_2 x_i)^2}{\sigma^2}\right]. \tag{86}$$

Because the errors are independent, the joint density of $y_1, y_2, \ldots, y_n$ conditional on x is

$$f(y_1, y_2, \ldots, y_n \mid x; \beta, \sigma^2) = \prod_i f(y_i \mid x_i; \beta, \sigma^2) \tag{87}$$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \prod_i \exp\left[-\frac{1}{2} \frac{(y_i - \beta_1 - \beta_2 x_i)^2}{\sigma^2}\right]. \tag{88}$$

The likelihood function is identical to this joint density, viewed as a function of (β, σ^2):

$$L(\beta, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \prod_i \exp\left[-\frac{1}{2} \frac{(y_i - \beta_1 - \beta_2 x_i)^2}{\sigma^2}\right]. \tag{89}$$

Taking logs,

$$\log L(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2} \sum_i \frac{(y_i - \beta_1 - \beta_2 x_i)^2}{\sigma^2}. \tag{90}$$

The first term in (90) does not depend on β, so maximizing with respect to β_1, β_2 is equivalent to minimizing the residual sum of squares:

$$S(\beta) = \sum_i (y_i - \beta_1 - \beta_2 x_i)^2.$$

Thus, the ML estimators for β_1 and β_2 coincide with the OLS estimators. In matrix notation,

$$\hat{\beta}_{ML} = \hat{\beta}_{OLS} = (X'X)^{-1} X'y. \tag{91}$$

Given (91), we can substitute $y_i - \beta_1 - \beta_2 x_i$ with the residuals e_i. The log-likelihood as a function of σ^2 becomes

$$\log L(\sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2}\sum_i \frac{e_i^2}{\sigma^2}. \tag{92}$$

Differentiating (92) with respect to σ^2 gives the first-order condition:

$$\frac{d\log L(\sigma^2)}{d\sigma^2} = -\frac{n}{2}\frac{1}{\sigma^2} + \frac{1}{2}\sum_i \frac{e_i^2}{\sigma^4} = 0. \tag{93}$$

Solving for σ^2 yields the ML estimator

$$\hat{\sigma}^2_{ML} = \frac{e'e}{n}. \tag{94}$$

This estimator is consistent but biased: it differs from the unbiased estimator

$$s^2 = \frac{e'e}{n-K},$$

that arises from OLS.

6.1.4 The ML estimator: general setup

Suppose that we are interested in the conditional distribution of y_i given x_i. The probability mass (in a discrete setting) or density function (in a continuous setting) can be written as

$$f(y_i \mid x_i; \theta), \tag{95}$$

where θ is the vector of unknown parameters.

Assume that observations are mutually independent. Then the joint (conditional) probability mass or density of the sample $(y_1, \ldots, y_n)$ given $(x_1, \ldots, x_n)$ is

$$f(y_1, \ldots, y_n \mid X; \theta) = \prod_i f(y_i \mid x_i; \theta). \tag{96}$$

The likelihood function is therefore

$$L(\theta) = \prod_i L_i(\theta) = \prod_i f(y_i \mid x_i; \theta), \tag{97}$$

where $L_i(\theta)$ is the contribution of observation i to the likelihood.

The ML estimator for θ solves the maximization problem

$$\max_\theta \log L(\theta) = \max_\theta \sum_i \log L_i(\theta). \tag{98}$$

6.1.5 First-order conditions and the score

The first-order conditions for (98) are

$$\left.\frac{\partial \log L(\theta)}{\partial \theta}\right|_{\theta=\hat{\theta}_{ML}} = \sum_i \left.\frac{\partial \log L_i(\theta)}{\partial \theta}\right|_{\theta=\hat{\theta}_{ML}} = 0. \tag{99}$$

If the log-likelihood is globally concave, there is a unique global maximum and the ML estimator is uniquely determined by (99). Only in special cases can $\hat{\theta}_{ML}$ be found analytically; more often numerical optimization methods are required.

For notational convenience, define the *score*:

$$s(\theta) = \frac{\partial \log L(\theta)}{\partial \theta} = \sum_i \frac{\partial \log L_i(\theta)}{\partial \theta} = \sum_i s_i(\theta), \tag{100}$$

where $s(\theta)$ is the score vector and $s_i(\theta)$ is the score contribution of observation i.

The first-order conditions state that the K sample averages of the score contributions, evaluated at $\hat{\theta}_{ML}$, should be zero:

$$s(\theta)\big|_{\theta=\hat{\theta}_{ML}} = \sum_i s_i(\theta)\big|_{\theta=\hat{\theta}_{ML}} = 0. \tag{101}$$

6.1.6 General properties of the ML estimator

Provided that the likelihood function is correctly specified, it can be shown under weak regularity conditions that:

(a) **Consistency**:

$$\hat{\theta}_{ML} \xrightarrow{p} \theta. \tag{102}$$

(b) **Asymptotic normality**:

$$\sqrt{n}\,(\hat{\theta}_{ML} - \theta) \xrightarrow{d} N(0, V), \tag{103}$$

where V is the asymptotic covariance matrix.

(c) **Asymptotic efficiency**: the ML estimator is asymptotically efficient and achieves the Cramér–Rao lower bound among all consistent estimators (linear and non-linear). In this case, the asymptotic covariance matrix is

$$V = [I(\theta)]^{-1},$$

where $I(\theta)$ is the (Fisher) information matrix.

6.1.7 The information matrix

If the model is correctly specified, the covariance matrix V is determined by the inverse of the Fisher information matrix. The Fisher information can be defined in terms of the second derivatives (Hessian) of the log-likelihood.

For observation i, define the information in i as

$$I_i(\theta) = -E\left[\frac{\partial^2 \log L_i(\theta)}{\partial\theta\partial\theta'}\right]. \tag{104}$$

This $K \times K$ matrix summarizes the expected amount of information about θ contained in observation i.

The average information for a sample of size n is

$$I_n(\theta) = \frac{1}{n}\sum_i I_i(\theta) = -E\left[\frac{1}{n}\sum_i \frac{\partial^2 \log L_i(\theta)}{\partial\theta\partial\theta'}\right] = -E\left[\frac{1}{n}\frac{\partial^2 \log L(\theta)}{\partial\theta\partial\theta'}\right]. \tag{105}$$

The limiting information matrix is

$$I(\theta) = \lim_{n\to\infty} I_n(\theta). \tag{106}$$

In the special case where observations are identically and independently distributed,

$$I(\theta) = I_n(\theta) = I_i(\theta). \tag{107}$$

Under appropriate regularity conditions, the asymptotic covariance matrix of the ML estimator is equal to the inverse of the information matrix:

$$V = [I(\theta)]^{-1}.$$

6.1.8 Information, curvature and precision

- The Fisher Information matrix $I(\theta)$ is the negative expected Hessian of the log-likelihood.
- It measures how sharply the log-likelihood bends around its maximum.
- More curvature ⇒ larger second derivatives ⇒ more information, greater precision, smaller asymptotic variance.
- Less curvature ⇒ smaller second derivatives ⇒ less information, lower precision, larger asymptotic variance.

6.1.9 Cramér–Rao bound

For an unbiased estimator $\hat{\theta}$:

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}. \tag{108}$$

- No unbiased estimator can have variance below $1/I(\theta)$.
- The data fundamentally limit how precisely θ can be estimated.

Connection to MLE.

- The MLE achieves the Cramér–Rao bound asymptotically.
- Therefore it is the most efficient regular estimator in large samples.

6.1.10 Alternative expression of the information matrix

An alternative expression for the information matrix comes from the result that

$$J_i(\theta) = E[s_i(\theta)s_i(\theta)'] \tag{109}$$

is identical to $I_i(\theta)$, provided that the likelihood function is correctly specified.

6.1.11 Estimating the covariance matrix

The covariance matrix can be consistently estimated by replacing expectations with sample averages and unknown parameters with ML estimates.
The estimator based on the Hessian form (105) is

$$\hat{V}_H = \left[-\frac{1}{n} \sum_i \frac{\partial^2 \log L_i(\theta)}{\partial\theta\partial\theta'} \Big|_{\theta=\hat{\theta}_{ML}} \right]^{-1}. \tag{110}$$

The estimator based on the outer product of gradients (OPG) form (109) is

$$\hat{V}_G = \left[\frac{1}{n} \sum_i s_i(\hat{\theta}_{ML})\, s_i(\hat{\theta}_{ML})' \right]^{-1}. \tag{111}$$

6.1.12 Two forms of Fisher information

There are two equivalent definitions of Fisher information (under correct specification):

- **Curvature form:**
$$I(\theta) = -E[H_i(\theta)] = -E\left[\frac{\partial^2 \ell_i(\theta)}{\partial\theta\partial\theta'} \right],$$
where $H_i(\theta)$ is the Hessian and $\ell_i(\theta) = \log L_i(\theta)$.
- **Score-variance form:**
$$I(\theta) = E[s_i(\theta)s_i(\theta)'].$$

These two expressions coincide only when the likelihood is correctly specified and regularity conditions hold.

Practical meaning.

- When the model coincides with the true data-generating process, curvature and score variation convey the same information.
- In that case, the asymptotic variance of the MLE simplifies to $\text{Avar}(\hat{\theta}_{ML}) = I(\theta_0)^{-1}$.

6.1.13 When to use each expression in practice

Use the Hessian form $I(\theta) = -E[H_i(\theta)]$:

- when the second derivative is easy to compute analytically;
- when the model is correctly specified (e.g. normal, logit, Poisson);
- when the likelihood is well-behaved and concave.

Use the OPG form $I(\theta) = E[s_i(\theta)s_i(\theta)']$:

- when the Hessian is difficult or unstable to compute;
- when score computation is simpler or more numerically stable;
- useful in nonlinear or latent-variable models.

When they are not equal.

- Under likelihood misspecification: functional form errors, heteroskedasticity, omitted variables, etc., the information equality fails and
$$E[s_i(\theta)s_i(\theta)'] \neq -E[H_i(\theta)].$$

6.1.14 Misspecification and robust variance (sandwich form)

Note: misspecified likelihood. If the likelihood function is not correctly specified:

- Properties (a) and (b) (consistency and asymptotic normality) still hold under mild conditions: the ML estimator remains consistent and asymptotically normal.
- However, the covariance matrix is not given by the inverse of the Fisher information; the information equality fails:

$$E[s_i(\theta)s_i(\theta)'] \neq -E[H_i(\theta)].$$

In general, the asymptotic variance of the ML estimator is

$$\text{Avar}(\hat{\theta}_{ML}) = A^{-1}BA^{-1}, \tag{112}$$

where

$$A = -E[H_i(\theta_0)], \qquad B = E[s_i(\theta_0)s_i(\theta_0)'].$$

- A captures the curvature of the (possibly misspecified) log-likelihood.
- B captures the variability of the score under the true data-generating process.
- Only when the model is correctly specified do we have $A = B$, recovering the usual ML variance.

This "sandwich" form is the basis of robust standard errors.

Sandwich estimator in practice. A practical robust (Huber–White) sandwich estimator is

$$\widehat{\text{Var}}(\hat{\theta}) = \big[-H_n(\hat{\theta})\big]^{-1}\left(\sum_{i=1}^{n} s_i(\hat{\theta})s_i(\hat{\theta})'\right)\big[-H_n(\hat{\theta})\big]^{-1},$$

where $H_n(\hat{\theta})$ is the observed average Hessian.

- This combines curvature (Hessian) and score variability.
- It is valid under heteroskedasticity or model misspecification.
- Software (e.g. Stata) uses this structure with options like `robust`, `cluster`, or `vce(robust)`.

Key practical takeaway.

- Use $I(\theta)^{-1}$ only when the model is correctly specified.
- Use the sandwich estimator whenever robustness is needed.

6.1.15 Example 1: discrete setting – information matrix

Recall the second derivative:

$$\frac{d^2 \log L(p)}{dp^2} = -\frac{n_1}{p^2} - \frac{n - n_1}{(1-p)^2}. \tag{113}$$

The information for p is

$$I(\theta) = -E\left[\frac{1}{n}\frac{d^2 \log L(p)}{dp^2}\right] \tag{114}$$

$$= -E\left[\frac{1}{n}\left(-\frac{n_1}{p^2} - \frac{n - n_1}{(1-p)^2}\right)\right] \tag{115}$$

$$= \frac{p}{p^2} + \frac{1-p}{(1-p)^2} = \frac{1}{p(1-p)}. \tag{116}$$

Therefore,

$$\sqrt{n}(\hat{p}_{ML} - p) \xrightarrow{d} N\big(0,\, p(1-p)\big). \tag{117}$$

6.1.16 Example 2: continuous setting – information matrix

The log-likelihood function is

$$\log L(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2}\sum_i \frac{(y_i - x_i'\beta)^2}{\sigma^2}. \tag{118}$$

The score contribution for observation i is

$$s_i(\beta, \sigma^2) = \begin{bmatrix} \dfrac{\partial \log L_i(\beta, \sigma^2)}{\partial \beta} \\ \dfrac{\partial \log L_i(\beta, \sigma^2)}{\partial \sigma^2} \end{bmatrix} = \begin{bmatrix} \dfrac{y_i - x_i'\beta}{\sigma^2} x_i \\ -\dfrac{1}{2\sigma^2} + \dfrac{1}{2}\dfrac{(y_i - x_i'\beta)^2}{\sigma^4} \end{bmatrix}. \tag{119}$$

Using $\varepsilon_i = y_i - x_i'\beta$, this can be written as

$$s_i(\beta, \sigma^2) = \begin{bmatrix} \dfrac{\varepsilon_i}{\sigma^2} x_i \\ -\dfrac{1}{2\sigma^2} + \dfrac{1}{2}\dfrac{\varepsilon_i^2}{\sigma^4} \end{bmatrix}. \tag{120}$$

The outer product is

$$s_i(\beta, \sigma^2) s_i(\beta, \sigma^2)' = \begin{bmatrix} \dfrac{\varepsilon_i^2}{\sigma^4} x_i x_i' & \dfrac{\varepsilon_i}{\sigma^2}\left(-\dfrac{1}{2\sigma^2} + \dfrac{1}{2}\dfrac{\varepsilon_i^2}{\sigma^4}\right) x_i \\ \dfrac{\varepsilon_i}{\sigma^2}\left(-\dfrac{1}{2\sigma^2} + \dfrac{1}{2}\dfrac{\varepsilon_i^2}{\sigma^4}\right) x_i' & \left(-\dfrac{1}{2\sigma^2} + \dfrac{1}{2}\dfrac{\varepsilon_i^2}{\sigma^4}\right)^2 \end{bmatrix}. \tag{121}$$

Using the normality of ε_i:

$$E(\varepsilon_i) = 0, \quad E(\varepsilon_i^2) = \sigma^2, \quad E(\varepsilon_i^3) = 0, \quad E(\varepsilon_i^4) = 3\sigma^4,$$

we obtain

$$J_i(\theta) = E[s_i(\theta) s_i(\theta)'] = \begin{bmatrix} \dfrac{1}{\sigma^2} x_i x_i' & 0 \\ 0' & \dfrac{1}{2\sigma^4} \end{bmatrix}. \tag{122}$$

Hence, the asymptotic covariance matrix is

$$I(\beta, \sigma^2)^{-1} = \begin{bmatrix} \sigma^2 \Sigma_{XX}^{-1} & 0 \\ 0' & 2\sigma^4 \end{bmatrix}, \tag{123}$$

where

$$\Sigma_{XX} = \lim_{n \to \infty} \frac{1}{n} \sum_i x_i x_i'.$$

From this we finally obtain the limiting distributions

$$\sqrt{n}(\hat{\beta}_{ML} - \beta) \xrightarrow{d} N\big(0, \sigma^2 \Sigma_{XX}^{-1}\big), \tag{124}$$

$$\sqrt{n}(\hat{\sigma}^2_{ML} - \sigma^2) \xrightarrow{d} N(0, 2\sigma^4). \tag{125}$$

6.2 Set 2: Specification Tests

6.2.1 Hypothesis Testing

Consider the problem where we estimate the K-dimensional parameter vector θ by maximizing the likelihood function

$$\max_\theta \log L(\theta) = \max_\theta \sum_i \log L_i(\theta). \tag{1}$$

Suppose that we are interested in testing a set of J linear restrictions on the same parameter vector:

$$H_0 : R\theta = r, \tag{2}$$

where R is a $J \times K$ matrix and r is a $J \times 1$ vector.

6.2.2 Three Asymptotically Equivalent Test Procedures

The null hypothesis in (2) can be tested by applying three different test principles. In general, the corresponding tests are asymptotically equivalent (that is, the test statistics share the same asymptotic distribution).

In a few special cases they even give the same numerical outcome (the computed statistics are exactly the same). Each procedure has its own merits and advantages; in practice, the test that is simplest to compute from available results is often chosen.

- **Wald test**
- **Likelihood ratio (LR) test**
- **Lagrange multiplier (LM) or score test**

6.2.3 The Wald Test

The Wald test starts from the asymptotic distribution

$$\sqrt{n}\,(\hat{\theta}_{ML} - \theta) \xrightarrow{d} N[0, V], \tag{3}$$

from which it follows that the J-dimensional vector $R\hat{\theta}_{ML}$ also has an asymptotic normal distribution:

$$\sqrt{n}\,\left(R\hat{\theta}_{ML} - R\theta\right) \xrightarrow{d} N[0, RVR']. \tag{4}$$

Under the null hypothesis $R\theta = r$, we can construct the Wald statistic:

$$\xi_W = n\big(R\hat{\theta}_{ML} - r\big)'\big(R\hat{V}R'\big)^{-1}\big(R\hat{\theta}_{ML} - r\big), \tag{5}$$

where $\hat{V}$ is a consistent estimator of V.

Under H_0, ξ_W has an asymptotic χ^2 distribution with J degrees of freedom.

6.2.4 The Likelihood Ratio Test

The likelihood ratio (LR) test compares the maximized log-likelihoods of the unrestricted and restricted models.

Estimate the model twice:

- without restrictions: $\hat{\theta}_{ML}$;
- with the null hypothesis imposed: $\tilde{\theta}_{ML}$, with $R\tilde{\theta}_{ML} = r$.

Maximizing a function subject to constraints cannot yield a larger value than maximizing it without constraints, so:

$$\log L(\hat{\theta}_{ML}) - \log L(\tilde{\theta}_{ML}) \geq 0.$$

If this difference is small, imposing the restrictions has little effect and the restrictions are plausible.

The LR test statistic is

$$\xi_{LR} = 2\big[\log L(\hat{\theta}_{ML}) - \log L(\tilde{\theta}_{ML})\big], \tag{6}$$

which, under the null, has an asymptotic χ^2 distribution with J degrees of freedom.

The LR test is only appropriate if the two models are *nested* (the restricted model is a special case of the unrestricted model). An attractive feature of this test is that it is not sensitive to how non-linear restrictions are written. The Wald test can handle non-linear restrictions, but its result *does* depend on the particular formulation of those restrictions.

6.2.5 The Lagrange Multiplier (LM) or Score Test

To introduce the general idea of an LM test, suppose that the null hypothesis restricts some components of the parameter vector $\theta = (\theta_1', \theta_2')'$ to fixed values, i.e. $\theta_2 = r$.

The name "Lagrange multiplier" comes from the fact that the test is implicitly based on the Lagrange multipliers in the constrained maximization problem.

Consider the constrained problem

$$\max_{\theta} \log L(\theta) = \max_{\theta} \sum_i \log L_i(\theta) \quad \text{with } \theta_2 = r, \tag{7}$$

where $\theta = (\theta_1', \theta_2')'$.

The associated Lagrangian is

$$\max_{\theta,\lambda} \log H(\theta, \lambda) = \max_{\theta,\lambda} \left\{ \sum_i \log L_i(\theta) - \lambda'(\theta_2 - r) \right\}, \tag{8}$$

where λ is the vector of Lagrange multipliers.

Forming the first-order conditions of (8) with respect to θ, and evaluating at the constrained ML estimate $\tilde{\theta}_{ML}$, we obtain:

$$\sum_i \frac{\partial \log H_i(\theta, \lambda)}{\partial \theta_1}\bigg|_{\tilde{\theta}_{ML}} = \sum_i s_{i1}(\tilde{\theta}_{ML}) = 0, \tag{9a}$$

$$\sum_i \frac{\partial \log H_i(\theta, \lambda)}{\partial \theta_2}\bigg|_{\tilde{\theta}_{ML}} = \sum_i s_{i2}(\tilde{\theta}_{ML}) - \hat{\lambda}_{ML} = 0, \tag{9b}$$

where the score contribution $s_i(\theta)$ is partitioned as $s_i(\theta) = \left(s_{i1}(\theta)', s_{i2}(\theta)'\right)'$, corresponding to θ_1 and θ_2, respectively.

From (9b) we see that the vector of Lagrange multipliers $\hat{\lambda}_{ML}$ equals the sum of the first derivatives with respect to the restricted parameters θ_2, evaluated at the constrained estimator $\tilde{\theta}_{ML} = (\tilde{\theta}_{1,ML}', r')'$. Thus, $\hat{\lambda}_{ML}$ measures the extent to which the unrestricted first-order conditions with respect to θ_2 are violated at $\tilde{\theta}_{ML}$.

It can be shown that, under the null,

$$\frac{1}{n}\hat{\lambda}_{ML} \xrightarrow{d} N(0, V_\lambda), \tag{10}$$

where

$$V_\lambda = \left[I_{22}(\theta) - I_{21}(\theta) I_{11}(\theta)^{-1} I_{12}(\theta)\right]^{-1}, \tag{126}$$

and V_λ is the lower-right $J \times J$ block of the inverse of the information matrix $I(\theta)$.

In detail, the information matrix can be written as

$$I(\theta) = \begin{bmatrix} I_{11}(\theta) & I_{12}(\theta) \\ I_{21}(\theta) & I_{22}(\theta) \end{bmatrix}. \tag{11}$$

Computationally, we use the fact that V_λ is the lower-right $J \times J$ block of $I(\theta)^{-1}$. Denote this block by $I^{22}(\theta)$; then

$$I(\theta)^{-1} = \begin{bmatrix} I^{11}(\theta) & I^{12}(\theta) \\ I^{21}(\theta) & I^{22}(\theta) \end{bmatrix}, \tag{12}$$

with

$$I^{22}(\theta) = \left[I_{22}(\theta) - I_{21}(\theta) I_{11}(\theta)^{-1} I_{12}(\theta)\right]^{-1}. \tag{13}$$

The LM test statistic is then

$$\xi_{LM} = \frac{1}{n}\hat{\lambda}_{ML}' \, \hat{I}^{22}(\tilde{\theta}_{ML}) \, \hat{\lambda}_{ML}, \tag{14}$$

where $\hat{I}^{22}(\tilde{\theta}_{ML})$ is a consistent estimator of $I^{22}(\theta)$ based on the constrained estimator $\tilde{\theta}_{ML}$.

Under the null hypothesis, ξ_{LM} has an asymptotic χ^2 distribution with J degrees of freedom.

6.3 Set 3: Models with Limited Dependent Variables

6.3.1 Introduction

In many empirical applications, the variables of interest are of a discrete or mixed discrete–continuous nature. When such variables are to be explained, the standard linear regression model is generally inappropriate.

It is therefore important to consider alternative frameworks that can be used to model discrete or discrete/continuous dependent variables.

6.3.2 Binary choice (binary response) models

Binary choice (or binary response) models are applied to discrete variables that can assume only two values (typically 0 and 1).

Examples:

- $y_i = 1$ if family i owns a car, $y_i = 0$ otherwise.
- $y_i = 1$ if individual i goes to work by car, $y_i = 0$ if individual i goes to work by tube.

Note that conceptually the two examples are different. In the first example there is a natural ordering (owning vs. not owning), while in the second (mode of transport) there is no natural ordering of the alternatives.

6.3.3 Multi-response models

Multi-response models are applied to discrete variables that can assume a finite number of values.

Examples:

- $y_i = 2$ if a family owns more than one car, $y_i = 1$ if it owns only one car, $y_i = 0$ if it does not own a car. Here there is a logical ordering in the alternatives.
- $y_i = 2$ if an individual buys product C, $y_i = 1$ if they buy product B, $y_i = 0$ if they buy product A. In this case there is no natural ordering among the alternatives.

6.3.4 Binary models: the linear regression approach

Suppose we are interested in explaining whether a family owns a car or not. The phenomenon is described by a binary variable y_i, which takes the value 1 if the family owns a car and 0 otherwise. Explanatory variables for observation i are collected in the vector x_i.

The standard linear model is

$$y_i = x_i'\beta + \varepsilon_i. \tag{127}$$

Under the usual exogeneity assumption $E[\varepsilon_i \mid x_i] = 0$, we have

$$E[y_i \mid x_i] = x_i'\beta.$$

But, since y_i is binary,

$$E[y_i \mid x_i] = 1 \cdot P(y_i = 1 \mid x_i) + 0 \cdot P(y_i = 0 \mid x_i) \tag{128}$$

$$= P(y_i = 1 \mid x_i). \tag{129}$$

Therefore,

$$P(y_i = 1 \mid x_i) = x_i'\beta. \tag{130}$$

Thus, the linear model implies that $x_i'\beta$ is a probability and must therefore lie between 0 and 1. This is only possible if the values of x_i are bounded and if the parameters β satisfy certain restrictions. In most applications this is hard to guarantee.

6.3.5 Problems with the linear probability model

In addition, the error term in (127) is highly non-normal and heteroskedastic.

Because y_i can only be 0 or 1, the error term $\varepsilon_i = y_i - x_i'\beta$ also takes only two possible values, $1 - x_i'\beta$ and $-x_i'\beta$. The conditional distribution of ε_i is

$$P(\varepsilon_i = 1 - x_i'\beta \mid x_i) = P(y_i = 1 \mid x_i) = x_i'\beta, \tag{131}$$

$$P(\varepsilon_i = -x_i'\beta \mid x_i) = P(y_i = 0 \mid x_i) = 1 - x_i'\beta. \tag{132}$$

This implies that

$$\mathrm{Var}(\varepsilon_i \mid x_i) = x_i'\beta\,(1 - x_i'\beta), \tag{133}$$

so the variance of the error term is not constant and depends on x_i (and β): a clear violation of homoskedasticity.

6.3.6 Introduction to binary choice models

To overcome these limitations, we can replace (130) with

$$P(y_i = 1 \mid x_i) = F(x_i'\beta), \tag{134}$$

where $F(\cdot)$ takes values only in $[0, 1]$.

A natural choice is to let $F(\cdot)$ be a cumulative distribution function (CDF). Common choices are:

- the standard normal CDF $\Phi(w)$,
- the standard logistic CDF $\Lambda(w)$.

6.3.7 The Probit model

If we adopt the standard normal distribution,

$$F(w) = \Phi(w) = \int_{-\infty}^{w} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}t^2\right)\, dt, \tag{135}$$

then

$$P(y_i = 1 \mid x_i) = \Phi(x_i'\beta). \tag{136}$$

This is the main assumption of the *Probit* model.

6.3.8 The Logit model

If we instead use the standard logistic distribution,

$$F(w) = \Lambda(w) = \frac{\exp(w)}{1 + \exp(w)}, \tag{137}$$

the probability becomes

$$P(y_i = 1 \mid x_i) = \Lambda(x_i'\beta) = \frac{\exp(x_i'\beta)}{1 + \exp(x_i'\beta)}. \tag{138}$$

An equivalent formulation is

$$\Lambda(w) = \frac{\exp(w)}{1 + \exp(w)} = \frac{1}{1 + \exp(-w)}. \tag{139}$$

This is the basic assumption of the *Logit* model.

6.3.9 Interpreting the parameters

The parameters β in binary choice models are not straightforward to interpret directly, so β is generally not the ultimate object of interest. A common approach is to look at the *marginal effects* of the explanatory variables on the probability $P(y_i = 1 \mid x_i)$.

For a continuous variable x_{ki}, the marginal effect is

$$\frac{\partial P(y_i = 1 \mid x_i)}{\partial x_{ki}} = f(x_i'\beta)\,\beta_k, \tag{140}$$

where $f(\cdot)$ is the probability density function corresponding to $F(\cdot)$.

These marginal effects are observation-specific (they depend on x_i). In practice, one often reports:

- *average marginal effects* (average over all i), or
- *marginal effects at the mean* of the explanatory variables.

6.3.10 Marginal effects: Probit

Under the Probit model, the marginal effect is

$$\frac{\partial P(y_i = 1 \mid x_i)}{\partial x_{ki}} = \varphi(x_i'\beta)\,\beta_k, \tag{141}$$

where the standard normal PDF is

$$\varphi(x_i'\beta) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(x_i'\beta)^2\right]. \tag{142}$$

6.3.11 Marginal effects: Logit

Under the Logit model, the marginal effect is

$$\frac{\partial P(y_i = 1 \mid x_i)}{\partial x_{ki}} = \lambda(x_i'\beta)\,\beta_k, \tag{143}$$

where the logistic PDF is

$$\lambda(x_i'\beta) = \frac{\exp(x_i'\beta)}{[1 + \exp(x_i'\beta)]^2}. \tag{144}$$

6.3.12 Alternative interpretation of the Logit model

For the Logit model, we can write

$$\ln\left(\frac{p_i}{1 - p_i}\right) = x_i'\beta, \tag{145}$$

where $p_i = P(y_i = 1 \mid x_i)$.

The left-hand side is the *log-odds ratio* (the logarithm of the odds of outcome 1 versus outcome 0). Thus, the coefficients β describe the effect of the explanatory variables on the log-odds of $y_i = 1$.

6.3.13 Binary models in latent form

Binary choice models can also be derived from an underlying behavioral model, leading to a latent variable representation. This is useful even when explicit behavioral assumptions are not spelled out.

Let y_i^* be an unobservable (latent) variable, often interpreted as the utility difference between the two states:

- $y_i = 1$: e.g. working,
- $y_i = 0$: e.g. not working.

Assume a linear additive relationship:

$$y_i^* = x_i'\beta + \varepsilon_i. \tag{146}$$

Define the observed outcome as

$$y_i = \begin{cases} 1 & \text{if } y_i^* > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$\begin{aligned} P(y_i = 1 \mid x_i) &= P(y_i^* > 0 \mid x_i) && (147)\\ &= P(x_i'\beta + \varepsilon_i > 0) && (148)\\ &= P(\varepsilon_i > -x_i'\beta) && (149)\\ &= 1 - F(-x_i'\beta). && (150) \end{aligned}$$

If the distribution of ε_i is symmetric, this becomes

$$1 - F(-x_i'\beta) = F(x_i'\beta).$$

6.3.14 Latent form: identification and normalization

The scale of the latent utility y_i^* is not identified, so a normalization on the distribution of ε_i is required. Typically, this means fixing the variance of ε_i at a given value.

Probit (latent form).

$$\begin{aligned} y_i^* &= x_i'\beta + \varepsilon_i, \\ \varepsilon_i \mid x_i &\sim NID(0,1), \\ y_i &= 1 \quad \text{if } y_i^* > 0, \\ y_i &= 0 \quad \text{if } y_i^* \le 0. \end{aligned}$$

Logit (latent form).

$$\begin{aligned} y_i^* &= x_i'\beta + \varepsilon_i, \\ \varepsilon_i \mid x_i &\sim LID\left(0, \frac{\pi^2}{3}\right), \\ y_i &= 1 \quad \text{if } y_i^* > 0, \\ y_i &= 0 \quad \text{if } y_i^* \le 0, \end{aligned}$$

where LID denotes a logistic distribution, and $\pi^2/3$ is the variance of the standard logistic.

6.3.15 ML estimation of binary models

Given the choice of a specific distribution $F(\cdot)$, these non-linear models can be estimated by Maximum Likelihood. The likelihood function is

$$L(\theta) = \prod_i L_i(\theta) = \prod_i f(y_i \mid x_i; \theta), \tag{151}$$

and for a binary model we can write

$$L(\beta) = \prod_i P(y_i = 1 \mid x_i; \beta)^{y_i} P(y_i = 0 \mid x_i; \beta)^{1-y_i} \tag{152}$$

$$= \prod_i F(x_i'\beta)^{y_i} [1 - F(x_i'\beta)]^{1-y_i}. \tag{153}$$

The log-likelihood is then

$$\log L(\beta) = \sum_i y_i \log F(x_i'\beta) + \sum_i (1 - y_i) \log[1 - F(x_i'\beta)]. \tag{154}$$

Differentiating with respect to β yields the first-order conditions. In general, this system of K equations is non-linear in the parameters and must be solved using numerical methods.

6.3.16 Goodness-of-fit

A goodness-of-fit measure summarizes how well the model approximates the observed data, playing a role analogous to R^2 in linear regression.

For binary models, fit can be assessed:

- by comparing fitted probabilities with observed response frequencies;
- by evaluating the model's ability to forecast the observed outcomes.

6.3.17 Pseudo (McFadden) R^2

Let $\log L_1$ denote the maximized log-likelihood of the model of interest (unrestricted model), and $\log L_0$ the maximized log-likelihood of a restricted model where all coefficients except the constant are set to zero.

Clearly, $\log L_1 \geq \log L_0$. Intuitively, the larger the difference, the more the unrestricted model improves upon the restricted one.

The McFadden pseudo-R^2 is defined as

$$\text{Pseudo } R^2 = 1 - \frac{\log L_1}{\log L_0}. \tag{155}$$

Because the log-likelihood is a sum of log probabilities, we have

$$\log L_0 \leq \log L_1 \leq 0.$$

If all parameters except the constant are zero, then $\log L_0 = \log L_1$ and the statistic equals 0. If the model were able to reproduce the outcomes perfectly ($\hat{p}_i = y_i$), then $\log L_1 = 0$ and the pseudo-R^2 would be 1. Hence the upper bound of the McFadden pseudo-R^2 is 1.

6.3.18 Measures based on correct and incorrect predictions

Another way to evaluate fit is to compare correct and incorrect predictions.

Typically, one predicts

$$\hat{y}_i = 1 \quad \text{if } F(x_i'\hat{\beta}_{ML}) > \tfrac{1}{2}, \tag{156}$$

$$\hat{y}_i = 0 \quad \text{if } F(x_i'\hat{\beta}_{ML}) \leq \tfrac{1}{2}. \tag{157}$$

We can then construct a 2×2 table of predictions $\hat{y}_i$ versus actual values y_i:

	$\hat{y}_i = 0$	$\hat{y}_i = 1$	Total
$y_i = 0$	n_{00}	n_{01}	n_0
$y_i = 1$	n_{10}	n_{11}	n_1
Total	N_0	N_1	n

Several goodness-of-fit measures can be derived from this table. The overall proportion of incorrect predictions is

$$w_{r1} = \frac{n_{01} + n_{10}}{n}, \tag{158}$$

which can be compared to the proportion of incorrect predictions from a restricted model with an intercept only. For the restricted model, the predicted probability is $\hat{p}_{ML} = n_1/n$. If $\hat{p}_{ML} > 1/2$, then

$$w_{r0} = 1 - \frac{n_1}{n}; \tag{159}$$

if $\hat{p}_{ML} \leq 1/2$, then

$$w_{r0} = \frac{n_1}{n}. \tag{160}$$

A prediction-based goodness-of-fit statistic is

$$R_p^2 = 1 - \frac{w_{r1}}{w_{r0}}. \tag{161}$$

Because it is possible that the unrestricted model predicts worse than the restricted model (i.e. $w_{r1} > w_{r0}$), R_p^2 can be negative. This is, of course, an unfavourable sign for the unrestricted model.

6.3.19 Appendix: Newton–Raphson method (basic idea)

The Newton–Raphson method is a numerical procedure for solving non-linear equations $f(x) = 0$.
Basic idea:

- Start from an initial guess x_0, hopefully close to a root r.
- At each step, approximate $f(x)$ around the current guess by its tangent line.
- Use the intersection of this tangent with the x-axis as the next approximation.

The iteration formula is

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}, \tag{162}$$

where:

- $f(x_{n-1})$ is the function value at the current guess,
- $f'(x_{n-1})$ is the derivative (slope of the tangent) at that point,
- x_n is the updated guess.

6.3.20 Newton–Raphson: geometric derivation

Geometric interpretation:

- Near a current guess x_0, approximate $f(x)$ by its first-order Taylor expansion:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

- At the root r, we have $f(r) = 0$. Using the approximation:

$$0 \approx f(x_0) + f'(x_0)(r - x_0).$$

- Solving for r, we get

$$r \approx x_0 - \frac{f(x_0)}{f'(x_0)}.$$

This leads to the iterative scheme

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}.$$

If f is smooth and the initial guess is sufficiently close, convergence is typically *quadratic*: the approximation error is roughly squared at each step.

6.3.21 Newton–Raphson for Maximum Likelihood

In Maximum Likelihood estimation we want to maximize the log-likelihood $\log L(\beta)$.

The first-order condition for a maximum is

$$\frac{\partial \log L(\beta)}{\partial \beta} = 0,$$

i.e. the score is zero. This is a system of non-linear equations

$$s(\beta) \equiv \frac{\partial \log L(\beta)}{\partial \beta} = 0.$$

We can apply Newton–Raphson to $f(\beta) = s(\beta)$. The update step is

$$\beta_{t+1} = \beta_t - \left[\frac{\partial^2 \log L(\beta_t)}{\partial \beta \partial \beta'}\right]^{-1} \frac{\partial \log L(\beta_t)}{\partial \beta}, \tag{163}$$

where:

- $\frac{\partial \log L(\beta_t)}{\partial \beta}$ is the score at β_t,
- $\frac{\partial^2 \log L(\beta_t)}{\partial \beta \partial \beta'}$ is the Hessian (curvature).

At each iteration we approximate the log-likelihood by a quadratic (second-order Taylor expansion) around β_t and move to the maximum of this local quadratic approximation.

7 Part IV - Panel Data

7.1 Set 1: Panel Data

Panel data (or longitudinal data) consist of observations on entities or "individuals" (persons, households, firms, states, regions, ...) for multiple time periods. The distinguishing feature relative to pure cross-sectional data is the presence of multiple observations over time for each entity.

The econometric interest in panel data has mainly two motivations:

- The desire to exploit panel data to control for unobserved time-invariant heterogeneity in cross-sectional models.
- The use of panel data to disentangle components of variance and estimate transition probabilities, and more generally to study the dynamics of cross-sectional populations.

These two motivations are loosely associated with two strands of the panel data literature:

- fixed effects models,
- random effects models.

7.1.1 Motivation: Endogeneity and Unobserved Heterogeneity

Regression lines from economic data often cannot be given a causal interpretation. One reason is that we may expect correlation between explanatory variables and the error term in the relation of interest.

Examples:

- The classical supply–demand simultaneity problem due to market equilibrium.
- Measurement error: if the explanatory variable we observe is not the exact variable to which agents respond, but a noisy measure of it, then the unobservable term in the equation of interest will contain the measurement error, which will be correlated with the regressor.
- Unobserved heterogeneity: if characteristics that have a direct effect on both the left-hand-side and right-hand-side variables are omitted, the explanatory variables will generally be correlated with the error, and regression coefficients will be biased measures of structural effects.

Thus, researchers are often confronted with large cross-sectional data sets from which precise correlations can be determined, but that nevertheless carry little information about parameters of genuine policy interest.

The traditional responses to these problems have been:

- multiple regression (adding controls),
- instrumental variables models.

In practice, we often lack data on the necessary conditioning variables or suitable instruments to achieve identification in these ways.

7.1.2 Motivation: Using Panel Data

A major motivation for using panel data is the ability to control for possibly correlated, time-invariant heterogeneity without observing it.

Consider a cross-sectional regression of the form

$$y_{i1} = \beta x_{i1} + \alpha_i + \varepsilon_{i1}, \tag{1}$$

with

$$E(\varepsilon_{i1} \mid x_{i1}, \alpha_i) = 0.$$

Here α_i captures individual-specific, time-invariant unobserved heterogeneity.

If α_i is observed, β can be identified from a multiple regression of y on x and α.

(We omit an overall intercept term β_0 in (1) because we have not restricted the mean of α_i to be zero. An equivalent formulation would include a constant and impose $E(\alpha_i) = 0$.)

If α_i is *not* observed, identification of β requires either:

- no correlation between x_{i1} and α_i:

$$\text{Cov}(x_{i1}, \alpha_i) = 0 \quad \Rightarrow \quad \beta = \frac{\text{Cov}(x_{i1}, y_{i1})}{\text{Var}(x_{i1})}, \tag{2}$$

- or an external instrument z_i uncorrelated with both ε_{i1} and α_i, but correlated with x_{i1}:

$$\text{Cov}(z_i, \alpha_i) = 0 \quad \Rightarrow \quad \beta = \frac{\text{Cov}(z_i, y_{i1})}{\text{Cov}(z_i, x_{i1})}. \tag{3}$$

If neither of these options is available, but we observe a second period $t = 2$ for the same individuals, we can use panel data.

Suppose:

$$y_{i2} = \beta x_{i2} + \alpha_i + \varepsilon_{i2}, \tag{4}$$

and

$$E(\varepsilon_{it} \mid x_{i1}, x_{i2}, \alpha_i) = 0 \quad \text{for } t = 1, 2.$$

Then, differencing eliminates α_i:

$$y_{i2} - y_{i1} = \beta(x_{i2} - x_{i1}) + (\varepsilon_{i2} - \varepsilon_{i1}), \tag{5}$$

and we can identify β from

$$\beta = \frac{\text{Cov}(\Delta x_{i1}, \Delta y_{i1})}{\text{Var}(\Delta x_{i1})}. \tag{6}$$

7.1.3 The Fixed Effects Model

The fixed effects (FE) model is a linear regression model in which the intercept is individual-specific:

$$y_{it} = x_{it}'\beta + \alpha_i + \varepsilon_{it}, \tag{7}$$

where:

- y_{it} is the outcome for individual i at time t,
- x_{it} is a vector of regressors,
- α_i is the individual-specific effect (time-invariant),
- ε_{it} is the idiosyncratic error term.

Usually one assumes $\varepsilon_{it} \sim \text{IID}(0, \sigma_\varepsilon^2)$ and that all x_{it} are independent of all ε_{it}.

7.1.4 The LSDV Estimator

An alternative way to write the fixed effects model is:

$$y_{it} = x'_{it}\beta + \sum_{j=1}^{N} \alpha_j d_{ij} + \varepsilon_{it}, \tag{8}$$

where $d_{ij} = 1$ if $i = j$ and 0 otherwise. That is, we include a dummy for each individual.

If x_{it} already includes a constant, then we include only $N - 1$ dummies to avoid perfect multicollinearity.

In both cases, all parameters can be estimated by OLS. The resulting estimator is the *least squares dummy variable* (LSDV) estimator.

7.1.5 Individual Effects and Time Effects

The LSDV formulation introduces individual-specific intercepts (a full set of individual dummies). These capture all time-invariant characteristics of each entity, absorbing unobserved heterogeneity that does not change over time.

Panel data models can also include a full set of time dummies:

$$\sum_{t=2}^{T} \lambda_t D_t,$$

where $D_t = 1$ for period t, and one dummy is omitted to avoid multicollinearity.

Time dummies control for common shocks or factors that vary over time but not across individuals, such as:

- macroeconomic trends,
- policy reforms,
- inflation,
- technological progress.

Thus, with panel data we can separately identify:

- individual fixed effects (time-invariant heterogeneity),
- time fixed effects (period-specific shocks),
- effects of regressors varying across both individuals and time.

7.1.6 The Within (Fixed Effects) Estimator

The LSDV estimator may be unattractive if there are many entities (large N). An alternative is to use the "within" transformation, working with deviations from individual means.

Start from:

$$\bar{y}_i = \bar{x}'_i\beta + \alpha_i + \bar{\varepsilon}_i, \tag{9}$$

where $\bar{y}_i = T^{-1}\sum_t y_{it}$, and $\bar{x}_i$, $\bar{\varepsilon}_i$ are the corresponding averages.

Subtracting (9) from (7) gives:

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)'\beta + (\varepsilon_{it} - \bar{\varepsilon}_i), \tag{10}$$

for all i, t.

Equation (10) is called the within transformation: a regression in deviations from individual means, which no longer includes α_i.

The OLS estimator of β from (10) is the *within estimator* or *fixed effects estimator*. It is numerically identical to the LSDV estimator.

In matrix notation, the within estimator can be written as:

$$\hat{\beta}_{FE} = \left[\sum_{i=1}^{N}\sum_{t=1}^{T}(x_{it} - \bar{x}_i)(x_{it} - \bar{x}_i)'\right]^{-1}\left[\sum_{i=1}^{N}\sum_{t=1}^{T}(x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i)\right]. \tag{11}$$

7.1.7 Properties of the Within Estimator

Unbiasedness. If all x_{it} are independent of all ε_{it} (for all leads and lags), the fixed effects estimator can be shown to be unbiased for β.

Consistency. If we assume mean independence

$$E[(x_{it} - \bar{x}_i)\varepsilon_{it}] = 0,$$

the FE estimator is consistent (this is weaker than full independence). Importantly, this requires that x_i is mean independent of ε_{it}, which in turn requires strict exogeneity:

$$E[x_{it}\varepsilon_{is}] = 0 \quad \text{for all } t, s.$$

Strict exogeneity means that the regressors do not depend on past, present or future values of the error term. Predetermined variables, such as lagged dependent variables, are not strictly exogenous and would violate the condition for consistency of the FE estimator.

7.1.8 Inference for the Fixed Effects Estimator

Assuming ε_{it} is i.i.d. across individuals and time with variance σ_ε^2, the covariance matrix of the FE estimator is

$$V(\hat{\beta}_{FE}) = \sigma_\varepsilon^2 \left[\sum_{i=1}^{N} \sum_{t=1}^{T} (x_{it} - \bar{x}_i)(x_{it} - \bar{x}_i)' \right]^{-1}. \tag{12}$$

A consistent estimator for σ_ε^2 is:

$$\hat{\sigma}_\varepsilon^2 = \frac{1}{N(T-1)} \sum_{i=1}^{N} \sum_{t=1}^{T} \hat{\varepsilon}_{it}^2, \tag{13}$$

where $\hat{\varepsilon}_{it}$ are the FE residuals (from the within regression).

7.1.9 Robust Inference (Cluster-Robust Standard Errors)

To allow for general heteroskedasticity and arbitrary autocorrelation within each individual, we can use a robust covariance matrix:

$$\hat{V}^R(\hat{\beta}_{FE}) = \left[\sum_{i=1}^{N} \sum_{t=1}^{T} \tilde{x}_{it}\tilde{x}_{it}' \right]^{-1} \left[\sum_{i=1}^{N} \sum_{t=1}^{T} \sum_{s=1}^{T} \hat{\varepsilon}_{it}\hat{\varepsilon}_{is}\tilde{x}_{it}\tilde{x}_{is}' \right] \left[\sum_{i=1}^{N} \sum_{t=1}^{T} \tilde{x}_{it}\tilde{x}_{it}' \right]^{-1}, \tag{14}$$

where $\tilde{x}_{it} = x_{it} - \bar{x}_i$.

Standard errors from $\hat{V}^R$ are called panel-robust, cluster-robust, or clustered standard errors, because they are robust not only to heteroskedasticity but also to autocorrelation within each "cluster" (entity).

7.1.10 The Two–Way Fixed Effects (TWFE) Model

A natural extension of the FE framework allows for both individual and time-specific effects:

$$y_{it} = x_{it}'\beta + \alpha_i + \lambda_t + \varepsilon_{it}, \tag{164}$$

where:

- α_i captures all time-invariant characteristics of individual i,
- λ_t captures time-specific shocks that affect all individuals (business cycles, policy changes, macro trends, ...).

Estimation can be done by:

- including N individual dummies and $T-1$ time dummies (LSDV),
- applying the within transformation that subtracts individual means and including $T-1$ time dummies,
- applying a within transformation that subtracts both individual and time means.

The TWFE model controls for unobserved heterogeneity in both dimensions, so identification comes from variation over time within individuals, after removing common time shocks.

7.1.11 The First-Difference Estimator

An alternative way to eliminate α_i in the simple FE model is by first differencing. Starting from:

$$y_{it} = x'_{it}\beta + \alpha_i + \varepsilon_{it},$$

we consider

$$y_{it} - y_{i,t-1} = (x_{it} - x_{i,t-1})'\beta + (\varepsilon_{it} - \varepsilon_{i,t-1}), \tag{15}$$

or, in compact notation,

$$\Delta y_{it} = \Delta x'_{it}\beta + \Delta\varepsilon_{it}. \tag{16}$$

Applying OLS to (16) yields the first-difference estimator:

$$\hat{\beta}_{FD} = \left[\sum_{i=1}^{N}\sum_{t=1}^{T}\Delta x_{it}\Delta x'_{it}\right]^{-1}\left[\sum_{i=1}^{N}\sum_{t=1}^{T}\Delta x_{it}\Delta y_{it}\right]. \tag{17}$$

In a model with individual and time fixed effects, after first differencing we can include time dummies in the differenced regression to capture changes in the time effects $(\lambda_t - \lambda_{t-1})$. This is equivalent (for β) to estimating a TWFE model with individual and time dummies.

7.1.12 Properties of the First-Difference Estimator

Consistency of the FD estimator requires:

$$E[(x_{it} - x_{i,t-1})(\varepsilon_{it} - \varepsilon_{i,t-1})] = 0,$$

which is weaker than the strict exogeneity condition required for FE. It allows for some correlation between x_{it} and $\varepsilon_{i,t-s}$ for $|s| \geq 2$.

However, because the FD error $\Delta\varepsilon_{it}$ is serially correlated, the variance–covariance matrix of $\hat{\beta}_{FD}$ should account for this.

In general, the FD estimator is less efficient than the FE estimator, except when $T = 2$, in which case FE and FD coincide.

7.1.13 The Difference-in-Differences (DiD) Estimator

A simple but powerful application of differencing is the difference-in-differences (DiD) estimator. It applies when we want to estimate the impact of a "treatment" on an outcome (e.g. effect of a drug on blood pressure, effect of an increase in the minimum wage on earnings, ...).

In this framework, the treatment effect is inferred by comparing changes in outcomes over time between:

- a population enrolled in the program (treatment group),
- a population not enrolled (control group).

Let r_{it} be a binary regressor equal to 1 if individual i receives the treatment in period t and 0 otherwise. Consider the fixed effects model:

$$y_{it} = \delta r_{it} + \mu_t + \alpha_i + \varepsilon_{it}, \tag{18}$$

where μ_t is a time-specific fixed effect.

First differencing eliminates α_i:

$$\Delta y_{it} = \delta\Delta r_{it} + \Delta\mu_t + \Delta\varepsilon_{it}, \tag{19}$$

which can be estimated by OLS under the assumption $E(\Delta r_{it}\Delta\varepsilon_{it}) = 0$.

7.1.14 Difference-in-Differences with Two Periods

Consider $T = 2$. In the first period nobody is treated ($r_{i1} = 0$ for all i). In the second period, only some individuals are treated; thus $r_{i2} = 1$ for a subset, and 0 for the others.

Then (19) becomes:

$$y_{i2} - y_{i1} = \delta(r_{i2} - r_{i1}) + \mu + (\varepsilon_{i2} - \varepsilon_{i1}), \tag{20}$$

where $\mu = \mu_2 - \mu_1$ is a constant.

OLS estimation of (20) yields:

- $\Delta\bar{y}_2^{\text{untreated}} = \hat{\mu}$: average change in y for the untreated group ($r_{i2} = 0$),

- $\Delta\bar{y}_2^{\text{treated}} = \hat{\mu} + \hat{\delta}$: average change in y for the treated group ($r_{i2} = 1$).

Thus,

$$\hat{\delta} = \Delta\bar{y}_2^{\text{treated}} - \Delta\bar{y}_2^{\text{untreated}}. \tag{21}$$

The estimator (21) is called the difference-in-differences estimator, because it is the difference between the time differences in outcomes for treated and untreated groups.

- The first difference (over time) removes unobservable fixed effects and controls for unobserved time-invariant individual characteristics.
- The second difference (between treated and untreated) isolates the treatment effect, under the assumption that time trends are common between the two groups.

7.1.15 Error Components and the Random Effects Model

Another major motivation for using panel data is the possibility of separating permanent from transitory components of variation.

Consider the model:

$$y_{it} = \beta_0 + x_{it}'\beta + \alpha_i + u_{it}, \tag{22}$$

where:

- $u_{it} \sim \text{IID}(0, \sigma_u^2)$,
- $\alpha_i \sim \text{IID}(0, \sigma_\alpha^2)$,
- all x_{it} are independent of all u_{it} and α_i.

In (22), the composite error $\alpha_i + u_{it}$ has two components:

- an individual-specific component α_i, permanent and constant over time,
- a transitory component u_{it}, uncorrelated over time.

Hence, conditional on x_{it}, a fraction

$$\frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_u^2}$$

of the total variance of y_{it} corresponds to differences that remain constant over time.

Because α_i and u_{it} are assumed independent of all x_{it}, the pooled OLS estimator for β_0 and β is unbiased and consistent.

7.1.16 The Random Effects GLS Estimator

However, the composite error $\alpha_i + u_{it}$ exhibits a particular form of autocorrelation. The usual OLS standard errors are not valid, and a more efficient GLS estimator can be obtained by exploiting the error covariance structure (the Balestra–Nerlove estimator).

Let y_i and X_i be the $T \times 1$ and $T \times K$ stacks of observations for individual i. The GLS estimator is:

$$\hat{\beta}_{GLS} = \left[\sum_{i=1}^{N} X_i'\Omega^{-1}X_i\right]^{-1}\left[\sum_{i=1}^{N} X_i'\Omega^{-1}y_i\right], \tag{23}$$

where the $T \times T$ covariance matrix Ω for each individual is:

$$\Omega = \sigma_u^2 I_T + \sigma_\alpha^2 \iota_T \iota_T' = \begin{bmatrix} \sigma_u^2 + \sigma_\alpha^2 & \sigma_\alpha^2 & \dots & \sigma_\alpha^2 \\ \sigma_\alpha^2 & \sigma_u^2 + \sigma_\alpha^2 & \dots & \sigma_\alpha^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_\alpha^2 & \dots & \sigma_\alpha^2 & \sigma_u^2 + \sigma_\alpha^2 \end{bmatrix}, \tag{24}$$

with I_T the $T \times T$ identity matrix and ι_T a $T \times 1$ vector of ones.

This GLS estimator is unfeasible because it depends on the unknown variances σ_α^2 and σ_u^2. A feasible GLS (FGLS) estimator replaces them by consistent estimates.

Common choices are:

$$\hat{\sigma}_u^2 = \frac{1}{N(T-1) - k}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\tilde{y}_{it} - \tilde{x}_{it}'\hat{\beta}_{FE}\right)^2, \tag{25}$$

$$\hat{\sigma}_\alpha^2 = \frac{1}{N}\sum_{i=1}^{N}\left(\bar{y}_i - \bar{x}_i'\hat{\beta}_{BG}\right)^2 - \frac{\hat{\sigma}_u^2}{T}, \tag{26}$$

where:

- $\hat{\beta}_{FE}$ is the within (fixed effects) estimator,
- $\hat{\beta}_{BG}$ is the between estimator (OLS regression of $\bar{y}_i$ on $\bar{x}_i$),
- $\tilde{y}_{it}$, $\tilde{x}_{it}$ denote demeaned values (for FE).

7.1.17 The Between Estimator and the RE Estimator

More precisely, the between estimator is the OLS estimator in the model for the individual means:

$$\bar{y}_i = \beta_0 + \bar{x}_i'\beta + \alpha_i + \bar{u}_i, \tag{27}$$

where the error variance is $\sigma_\alpha^2 + \frac{1}{T}\sigma_u^2$, leading to (26).

Having obtained an estimate $\hat{\Omega}$, the random effects estimator is the feasible GLS estimator:

$$\hat{\beta}_{RE} = \left[\sum_{i=1}^{N} X_i'\hat{\Omega}^{-1}X_i\right]^{-1}\left[\sum_{i=1}^{N} X_i'\hat{\Omega}^{-1}y_i\right]. \tag{28}$$

7.1.18 Fixed or Random Effects? The Hausman Test

Hausman (1978) proposed a test for correlated effects by comparing the FE (within) and RE estimators.

He suggested the statistic:

$$h = (\hat{\beta}_{FE} - \hat{\beta}_{RE})'\left(\hat{V}_{FE} - \hat{V}_{RE}\right)^{-1}(\hat{\beta}_{FE} - \hat{\beta}_{RE}), \tag{29}$$

where $\hat{V}_{FE}$ and $\hat{V}_{RE}$ are consistent estimators of the covariance matrices of $\hat{\beta}_{FE}$ and $\hat{\beta}_{RE}$, respectively.

- Under H_0 (uncorrelated effects), both FE and RE are consistent, so we expect $\hat{\beta}_{FE} - \hat{\beta}_{RE}$ to be small.
- Under H_1 (correlated effects), FE remains consistent while RE is not, so their difference tends to be large.

The Hausman statistic h has a χ^2 distribution with K degrees of freedom, where K is the number of elements in β.